# V5A10 Analytic Number Theory

So Murata

WiSe 25/26, University of Bonn

## 1 Classical Number Theory

**Theorem 1.1** (Euclid)**.** *There are infinitely many prime numbers.*

**Definition 1.1.** $\pi : \mathbb{N} \to \mathbb{N}$ *be a function such that*

$$\pi(n) = \{\textit{prime numbers less than } n\}.$$

**Remark 1.1.**

$$\frac{\pi}{n\ln(n)} \approx 1.$$

**Definition 1.2.**

$$\mathrm{Li}(x) = \int_0^x \frac{1}{\ln(t)} dt.$$

**Notation 1.1.** *Given* $f, g : \mathbb{R} \to \mathbb{C}$*,*

$$f(x) = O(g(x))$$

*means that*

$$\exists K \in (0, \infty), x_0 \in \mathbb{R}, \text{ s. t. } \forall x > x_0, |f(x)| \leq K|f(x)|.$$

**Notation 1.2.** *Let* $f, g : \mathbb{R} \to \mathbb{C}$ *be functions.* $f \sim g$ *denotes that*

$$\lim_{n\to\infty} \frac{f(x)}{g(x)} = c,$$

*for some constant* $c$*.*

**Notation 1.3.** *Let* $f : \mathbb{R} \to \mathbb{C}$*,*

$$\mathrm{Li}(x) \sim \frac{x}{\ln x} \sum_{k=0}^{\infty} \frac{k!}{(\ln x)^k},$$

*denotes that*

$$\mathrm{Li}(x) = \frac{x}{\ln x} \sum_{k=1}^{N-1} \frac{k!}{(\ln x)^k} + O\left(\frac{x}{(\ln x)^{N+1}}\right).$$

*and as* $x \to \infty$*, this holds for any* $N \geq 1$*.*

**Remark 1.2.** *By the integration by parts, we see that it's asymptotic expansion is*

$$\mathrm{Li}(x) \approx \frac{x}{\ln(x)} \sum_{k=0}^{\infty} \frac{k!}{(ln(x))^k}.$$

**Theorem 1.2** (Prime Number Theorem)**.**

$$\lim_{x\to\infty} \frac{\pi(x)}{\mathrm{Li}(x)} = 1.$$

**Definition 1.3** (First Chebyschev Function)**.**

$$\vartheta(x) = \sum_{p\leq x} \ln p.$$

**Definition 1.4** (Second Chebyschev Function)**.**

$$\psi(x) = \sum_{\substack{m,p \\ p^m \leq x}} \ln p.$$

**Remark 1.3.** *We can rewrite the second Chebyschev function as follows.*

$$\psi(x) = \sum_{\substack{p\leq x \\ m=\max\{n\in\mathbb{N} \mid p^n \leq x\}}} m \ln p.$$

**Definition 1.5** (Möbius Function)**.**

$$\mu(n) = \begin{cases} 1 \quad (n=1) \\ (-1)^k \quad (n = p_1 \cdots p_k, p_i = p_j \Rightarrow i = j) \\ 0 \quad (\exists p \text{ s. t. } p^2 | n). \end{cases}$$

**Remark 1.4.** *The prime number theorem is equivalent to the following statements.*

*1).* $\psi(x) \sim x$.

*2).* $\theta(x) \sim x$.

*3).* $\lim_{x\to\infty} \frac{\sum_{n\leq x}}{\frac{\mu(n)}{n}} x = 0$.

**Conjecture 1.1** (Twin Prime Conjecture)**.** *There exists infinitely many primes $p$ such that $p+2$ is also prime.*

**Conjecture 1.2** (Goldbach's Conjecture)**.** *Let $n \in \mathbb{N}$ be an even number greater than 2, then there exists two primes $p, q$ such that $n = p + q$.*

**Conjecture 1.3** (Hardy-Littlewood Conjecture)**.**

$$\#\left\{ \begin{matrix} \textit{prime numbers } p \textit{ such that} \\ 2p+1 \textit{ is also a prime and } p < x \end{matrix} \right\} \sim \frac{x}{(\log x)^2}.$$

**Definition 1.6** (Riemann-Zeta Function)**.** *We define* $\zeta : \mathbb{C} \to \mathbb{C}$ *such that*

$$\zeta(s) = \sum_{n \in \mathbb{N}} \frac{1}{n^s}.$$

**Remark 1.5.** *Supposer* $\mathrm{Re}(s) > 1$*, then we have*

$$\begin{aligned} |\zeta(s)| &= \sum_{n \in \mathbb{N}} \frac{1}{|n|^s} \\ &= \sum_{n \in \mathbb{N}} \frac{1}{n^{\mathrm{Re}(s)}} \end{aligned}$$

*By multiplying* $\frac{1}{2^s}$*, we obtain*

$$\frac{1}{2^s}\zeta(s) = \sum_{n \in \mathbb{N}} \frac{1}{(2n)^s}.$$

*We get*

$$\left(1 - \frac{1}{2^s}\right)\zeta(s) = \sum_{n \in \mathbb{N}} \frac{1}{(2n+1)^s}.$$

*Continuing this procedure, we get the following proposition.*

**Proposition 1.1.**

$$\zeta(s) = \prod_p \left(1 - \frac{1}{p^s}\right)$$

**Theorem 1.3** (Weierstrass)**.** *Let* $A \subseteq \mathbb{C}$ *and consider a sequence of functions* $(f_n : A \to \mathbb{C})_{n \in \mathbb{N}}$ *such that there exists a sequence of non-negative numbers* $(M_n)_{n \in \mathbb{N}}$ *such that*

*i).* $\forall x \in A, |f_n(x)| \leq M_n$.

*ii).* $\sum_{n \in \mathbb{N}} M_n$ *converges.*

*Then the sequence converges uniformly.*

**Theorem 1.4.** *Suppose the conditions in the previous theorem. If each function is analytic on a compact subset of* $A$*, then the limit is also analytic.*

**Corollary 1.1.** *Let* $A$ *be a compact subset of a complex plane where* $\mathrm{Re}(s) > 1$*. Then there exists* $\delta > 0$ *such that* $\mathrm{Re}(s) > 1 + \delta$ *and*

$$\sum_{n \in \mathbb{N}} \left| \frac{1}{n^s} \right| \leq \sum_{n \in \mathbb{N}} \frac{1}{n^{1+\delta}} < \infty.$$

**Fact 1.1.** *The Riemann zeta function can be analytically continued to the whole plane except for* $s = 1$*.*

**Definition 1.7** (Gamma function)**.** *We define the Gamma function* $\Gamma : \mathbb{C} \to \mathbb{C}$ *as*

$$\Gamma(s) := \int_0^\infty t^{s-1} e^{-t} dt, \mathrm{Re}(s) > 0.$$

**Proposition 1.2.**

$$\zeta(s) = \frac{1}{\Gamma(s)} \int_0^\infty \frac{x^{s-1}}{e^x - 1} dx.$$

**Remark 1.6.** $\zeta(1+it) \neq 0$ *if* $t \in \mathbb{R}, t \neq 0$. $\zeta(s) \neq 0$ *for* $0 < s < 1$.

**Remark 1.7** (Functional Equation)**.**

$$\zeta(1-s) = 2(2\pi)^{-s} \cos\left(\frac{\pi s}{2}\right) \Gamma\left(\frac{s}{2}\right) \zeta(s)$$

**Remark 1.8.** $\Gamma(s)$ *is defined for* $\mathrm{Re}(s) > 0$ *and can be analytically continued to the whole place except for* $\{-n \mid n \in \mathbb{Z}_{\geq 0}\}$.

**Remark 1.9.** *For* $s = -2m$ *where* $m \in \mathbb{N}$, *we see* $\zeta(s) = 0$.

$$\begin{aligned}
\zeta(0) &= \frac{2}{2\pi} \lim_{s \to 1} \cos\left(\frac{\pi s}{2}\right) \zeta(s) \\
&= \frac{1}{\pi} \lim_{s \to 1} \frac{\cos(\frac{\pi s}{2})}{s-1} \lim_{s \to 1} (s-1)\zeta(s) \\
&= \frac{1}{\pi} \times \frac{-\pi}{2} \times 1 \\
&= -\frac{1}{2}.
\end{aligned}$$

**Definition 1.8.** *The critical strip is the subset of the complex plane with its real part between* 0 *and* 1. *The critical line is the line where* $\mathrm{Re}(s) = \frac{1}{2}$.

**Conjecture 1.4** (Riemann Hypothesis)**.** *Let* $s$ *be an element of the critical strip. If* $\zeta(s) = 0$ *then* $\mathrm{Re}(s) = \frac{1}{2}$ *(i.e. it lies on the critical line).*

**Notation 1.4.** *Let* $T > 0$. *We denote* $N(T)$ *the number of zeros of* $\zeta$ *in the critical strip whose coefficient of the imaginary part is in* $(0, T)$. *That is*

$$N(T) = |\{\sigma + it \in \mathbb{C} \mid 0 < \sigma < 1, 0 < t < T\}|.$$

**Proposition 1.3.**

$$\lim_{T \to \infty} \frac{N(T) 2\pi}{T \log(T)} = 1.$$

*Sketch of Proof (needs refinement).*

$$\psi(x) = \frac{1}{2\pi i} \int_l \frac{-\zeta'(s)}{\zeta(s)} \frac{x^s}{s} ds$$

where $l$ is the line $l = a$ for some $a > 1$.

$$\psi(s) = x - \sum_{\rho \text{ non-trivial zeros}} \frac{x^\rho}{\rho} - \frac{\zeta'(0)}{\zeta(0)} - \log(1 - x^{-2}).$$

□

**Definition 1.9.** *Let $q \in \mathbb{N}$ and $a$ be a natural number coprime to $q$. We define*

$\pi(x; q, a) = |\{$*prime numbers $p$ less than or equal to $x$ such that* $p \equiv a \mod q\}|$

**Proposition 1.4.**

$$\pi(x; , q, a) \sim \frac{x}{\varphi(q)\log(x)}$$

*where $\varphi$ is a Euler phi-function.*

**Theorem 1.5** (Brun–Titchmarsh)**.** *For any $q < x$, we have*

$$\pi(x; q, a) < \frac{2x}{\varphi(q)\log(\frac{x}{q})}.$$

**Remark 1.10.**

$$\mathrm{Li}(x) \sim \frac{x}{\ln x} \sum_{k=0}^{\infty} \frac{k!}{(\ln x)^k}.$$

*Indeed we have*

$$\mathrm{Li}(x) = . \int_2^x \frac{dt}{\ln t}.$$

*Observe that*

$$\int_2^t \frac{1}{(\ln t)^N} \sim \frac{x}{(\ln x)^N}$$

*for all $N \geq 1$. Thus $\mathrm{Li}(x)$ can be expressed in terms of polynomials in $\frac{x}{\ln(x)}$, by keep replacing the greatest temr with the above approximation.*

# 2 Arithmetic Functions

## 2.1 Multiplicative Functions

**Definition 2.1.** *A function $f : \mathbb{N} \to \mathbb{C}$ is said to be*

*1). multiplicative if for any $(m, n) = 1$, we have $f(mn) = f(m)f(n)$,*

*2). completely multiplicative if for any natural numbers $m, n$, we have $f(mn) = f(m)f(n)$.*

**Example 2.1.** *Möbius function $\mu$ is multiplicative.*

**Definition 2.2** (Von-Mangoldt Function)**.** *The Von-Mangoldt function* $\Lambda : \mathbb{N} \to \mathbb{C}$ *is defined as*

$$\Lambda(n) = \begin{cases} \log(p) & (n = p^k \text{ for some } k \geq 1), \\ 0 & (\text{otherwise}). \end{cases}$$

**Definition 2.3** (Euler Phi Function)**.** *The Euler phi funtion is* $\varphi : \mathbb{N} \to \mathbb{C}$ *such that*

$$\varphi(n) = \{1 \leq a \leq n \mid (a, n) = 1\}.$$

**Example 2.2.** $\varphi$ *is multiplicative but* $\Lambda$ *is not.*

**Definition 2.4** (Dirichlet Characters Modulo $q$)**.** *Let* $q \in \mathbb{N}$ *be a natural number and* $q \geq 2$*.*

$$\chi_1 : (\mathbb{Z}/q\mathbb{Z})^\times \to \mathbb{C}^\times$$

*be a group homomorphism. The Dirichlet character function modulo* $q$ *with respect to* $\chi_1$ *is such that*

$$\chi(n) = \begin{cases} \chi_1(\overline{n}) & ((n, q) = 1), \\ 0 & (\text{otherwise}). \end{cases}$$

**Example 2.3.** *For* $q = 3$*, we have* $(\mathbb{Z}/3\mathbb{Z})^\times = \{\pm 1\}$*. The only possible character is* $\pm 1 \mapsto \pm 1$*. Therefore, we have*

$$\chi(1) = 1, \chi(2) = -1, \chi(0) = 0.$$

**Theorem 2.1.**

$$\sum_{d|n} \mu(d) = \begin{cases} 1 & n = 1, \\ 0 & (\text{otherwise}). \end{cases}$$

*Proof.* When $n = 1$, this is trivial. Suppose $n \neq 1$. We factorize $n$ by

$$n = \prod_{i=1}^{k} p_i^{\alpha_i}$$

where $p_i$ is a prime and $\alpha_i \in \mathbb{N}$ for each $i = 1, \cdots, n$.

Observe that

$$\sum_{d|n} \mu(d) = \sum_{d| \prod_{i=1}^{k} p_i} \mu(d).$$

Now we see

$$\sum_{d| \prod_{i=1}^{k} p_i} \mu(d) = \sum_{j=0}^{k} \binom{k}{j} (-1)^j = \sum_{j=0}^{k} \binom{k}{j} (1)^{k-j} (-1)^j = (1-1)^k = 0.$$

□

**Proposition 2.1** (Möbius Inversion Formula)**.** *Let $f, g : \mathbb{N} \to \mathbb{C}$ be functions (we do not assume them to be multiplicative). If*

$$\sum_{d|n} g(d) = f(n),$$

*holds if and only if*

$$\sum_{d|n} \mu(d) f\left(\frac{n}{d}\right) = g(n).$$

*Proof.*

$$\sum_{d|n} \mu(d) f\left(\frac{n}{d}\right) = \sum_{d|n} \mu(d) \sum_{e|\frac{n}{d}} g(e).$$

$e|\frac{n}{d}$ if and only if $de|n$ thus obtain,

$$\sum_{d|n} \mu(d) f\left(\frac{n}{d}\right) = \sum_{d|n} \mu(d) \sum_{de|n} g(e).$$

In particular, we get the expression

$$= \sum_{de|n} \mu(d) g(e).$$

By reordering, we get

$$= \sum_{e|n} g(e) \sum_{d|\frac{n}{e}} \mu(d).$$

By Proposition 2.1, we get

$$\sum_{d|\frac{n}{e}} \mu(d) = 0$$

unless $e = n$. □

**Proposition 2.2.**

$$\sum_{d|n} \varphi(d) = n.$$

*Proof.* Consider $(\mathbb{Z}/n\mathbb{Z})^\times$. We know that

$$|(\mathbb{Z}/n\mathbb{Z})^\times| = \varphi(n).$$

Later □

**Theorem 2.2.**

$$\varphi(n) = n \prod_{p|n} \left(1 - \frac{1}{p}\right).$$

*Proof.* Using Proposition 2.2, we have,

$$\sum_{d|n} \mu(d)\frac{n}{d} = \varphi(n).$$

Dividing both sides by $n$ and observe that $\mu(d) \neq 0$ if and only if $d$ is a prime factor of $n$.

$$\begin{aligned}\frac{\varphi(n)}{n} &= \sum_{d|n} \frac{\mu(d)}{d}, \\ &= 1\sum_{p|n}\frac{1}{p} + \sum_{p_1,p_2|n}\frac{1}{p_1p_2} - \cdots .\end{aligned}$$

By the induction on the number of prime divisors of $n$, we get the statement. □

**Proposition 2.3.** *We have the following properties of $\varphi$.*

*1).* $n|m \Rightarrow \varphi(n)|\varphi(m)$.

*2).* $\varphi(n)$ *is even for* $n \geq 3$.

*3).* $\varphi(2n) = \begin{cases} 2\varphi(n), & (2|n) \\ \varphi(n), & (2 \nmid n). \end{cases}$

*4).* $\varphi$ *is multiplicative.*

*5).* $\varphi(mn) = \varphi(m)\frac{\varphi(n)d}{\varphi(d)}$ *where* $d = (m,n)$.

*6).* $\varphi(n^m) = n^{m-1}\varphi(n)$.

*Proof.* Exercise. □

**Theorem 2.3.** *The following statements are equivalent.*

*1).* $\sum_{d|n} \Lambda(d) = \log n$

*2).* $\sum_{d|n} \mu(d) \log \frac{n}{d} = \Lambda(n)$.

*And in particular* $\sum_{d|n} \Lambda(d) = \log n$ *holds.*

*Proof.* The equivalence is a direct corollary of Möbius inversion formula (ie. Proposition 2.1). For the latter, Write

$$n = \prod_{i=1}^{k} p_i^{\alpha_i}.$$

We have

$$\sum_{d|n} \Lambda(d) = \sum_{i=1}^{k} \alpha_i \log p_i = \log(n).$$

□

**Notation 2.1.** *Let $n \in \mathbb{N}$, suppose a prime $p$ divides $n$. Then we denote $\alpha(p)$ to be the highest prime power factor of $n$.*

**Theorem 2.4.** *Let $f : \mathbb{N} \to \mathbb{C}$ be a multiplicative function. Then*

$$\sum_{d|n} f(d) = \prod_{p|n} \left( \sum_{i=0}^{\alpha(p)} f(p^i) \right).$$

*In particular $\sum_{d|n} f(d)$ is also multiplicative.*

*Proof.* Let $d|n$, then we have $d = \prod_{i=1}^{k} p_i^{\beta_i}$ for some $0 \leq \beta_i \leq \alpha(p_i)$. Since $f$ is multipliative we have

$$f(d) = \prod_{i=1}^{k} f(p_i^{\beta_i}).$$

The second part is a direct result of the first part. □

**Remark 2.1.** *The Second Chebyschev Function $\psi$ can be written as*

$$\psi(x) = \sum_{d \leq x} \Lambda(d).$$

*This follows directly from the definition.*

**Definition 2.5** (Dirichlet Series)**.** *Let $f : \mathbb{N} \to \mathbb{C}$ be a function and $s \in \mathbb{C}$. We define*

$$\sum_{n \in \mathbb{N}} \frac{f(n)}{n^s}.$$

*For another arithmetic function $g : \mathbb{N} \to \mathbb{C}$, we define*

$$\sum_{n \in \mathbb{N}} \frac{f(n)}{n^s} + \sum_{n \in \mathbb{N}} \frac{g(n)}{n^s} = \sum_{n \in \mathbb{N}} \frac{(f(n) + g(n))}{n^s}.$$

*and*

$$\left( \sum_{n \in \mathbb{N}} \frac{f(n)}{n^s} \right) \times \left( \sum_{n \in \mathbb{N}} \frac{g(n)}{n^s} \right) = \sum_{n,m \in \mathbb{N}} \frac{(f(n)g(m))}{(nm)^s}.$$

$$= \sum_{t \in \mathbb{N}} \sum_{n|t} \frac{\left(f(n)g\left(\frac{n}{t}\right)\right)}{t^s}.$$

Recall the taylor expansion of $\ln x$ we get

$$\ln 2 = \sum_{n \in \mathbb{N}} \frac{(-1)^{n+1}}{n}.$$

Rearranging the following way

$$\left(1 - \frac{1}{2}\right) - \frac{1}{4} + \left(\frac{1}{3} - \frac{1}{6}\right) - \frac{1}{8} + \cdots$$

we get this equals to $\frac{1}{2} \ln 2$.

**Theorem 2.5.** *Let $s \in \mathbb{C}$ be $\mathrm{Re}(s) > 1$, we have*

$$\frac{1}{\zeta(s)} = \sum_{n \geq 1} \frac{\mu(n)}{n^s}.$$

*Proof.*

$$\begin{aligned}
\zeta(s) \sum_{n \geq 1} \frac{\mu(n)}{n^s} &= \left( \sum_{n \in \mathbb{N}} \frac{1}{n^s} \right) \left( \sum_{n \in \mathbb{N}} \frac{\mu(n)}{n^s} \right) \\
&= \sum_{t \in \mathbb{N}} \frac{1}{t^s} \sum_{n | t} \mu(n) \\
&= 1.
\end{aligned}$$

□

**Theorem 2.6.** *For $\mathrm{Re}(s) > 1$, we have*

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{n \in \mathbb{N}} \frac{\Lambda(n)}{n^s}.$$

*From this we derive*

$$\lim_{n \to \infty} \frac{\log n}{n^\varepsilon} = 0.$$

*Proof.*

$$\begin{aligned}
\zeta(s) \sum_{n \in \mathbb{N}} \frac{\Lambda(n)}{n^s} &= \left( \sum_{m \in \mathbb{N}} \frac{1}{m^s} \right) \left( \sum_{n \in \mathbb{N}} \frac{\Lambda(n)}{n^s} \right), \\
&= \sum_{t \in \mathbb{N}} \frac{1}{t^s} \left( \sum_{n | t} \Lambda\left( \frac{t}{n} \right) \right), \\
&= \sum_{t \in \mathbb{N}} \frac{\log(t)}{t^s}, \\
&= -\zeta'(s).
\end{aligned}$$

□

**Remark 2.2.**

$$\begin{aligned}
\sum_{n \in \mathbb{N}} \left| \frac{\Lambda(s)}{n^s} \right| &\leq \sum_{n \in \mathbb{N}} \frac{\log(n)}{n^\sigma}, \\
&\ll \sum_{n \in \mathbb{N}} \frac{n^\varepsilon}{n^\sigma}, \\
&= \sum_{n \in \mathbb{N}} \frac{1}{n^{\sigma - \varepsilon}}.
\end{aligned}$$

*We have* $\lim_{n\to\infty} \frac{\log(n)}{n^\varepsilon} = 1$ *and the last equation is convergent if and only if* $\sigma - \varepsilon > 1$. *Thus we have* $\sigma > 1 + \varepsilon$.

**Definition 2.6.** *Let* $D \subseteq \mathbb{C}$ *be open. A meromorphic function is* $f : D \to \mathbb{C}$ *which is analytic on* $D$ *except for a discrete set of poles of* $f$.

**Remark 2.3.** $\frac{\zeta'(s)}{\zeta(s)}$ *is a meromorphic functions with a pole at* $s = 1$ *and where* $\zeta(s)$ *vanishes. Indeed, For general* $\frac{f}{g}$, *it is analytic if* $f, g$ *are analytic and* $g \neq 0$.

*1).* $\zeta(s)$ *is analytic except* $s = 1$.

*2).* $\zeta'(s)$ *has a pole of order* $2$ *at* $s = 1$.

*3).* $\zeta(s)$ *has a pole of order* $1$ *at* $s = 1$.

Recall that for $|z| \geq 1$, we have,

1. $|z| \geq 1 \Rightarrow \sum_{n\in\mathbb{Z}_{\geq 0}} z^n = \frac{1}{1-z}$,
2. $\prod_{n\in\mathbb{N}}(1 + a_n)$ is convergent if $\sum_n a_n$ is absolutely convergent,
3. therefore $\prod_{n\in\mathbb{N}}(1 + a_n)$ is convergent if and only if $\prod_{n\in\mathbb{N}}(1 + |a_n|)$ is convergent.

**Theorem 2.7.** *Let* $f : \mathbb{N} \to \mathbb{C}$ *be a map.*
*If* $f$ *is multiplicative and for* $\mathrm{Re}(s) > r_0, r_0 \in \mathbb{R}$ *then we have,*

$$\sum_{n\in\mathbb{N}} \frac{f(n)}{n^s} = \prod_p \left( \sum_{\nu\geq 0} f(p^\nu) p^{-\nu s} \right).$$

*If* $f$ *is completely multiplicative, then*

$$\sum_{n\in\mathbb{N}} \frac{f(n)}{n^s} = \prod_p (1 - f(p)p^{-s})^{-1}.$$

*Proof.* Let $A(x) = \{n \in \mathbb{N} \mid \text{primes factors of } n \text{ are } \leq x\}$, then

$$\prod_{p\leq x} \sum_{\nu=0}^{\infty} f(p^\nu) p^{-\nu s} = \sum_{n\in A} \frac{f(n)}{n^s}.$$

Therefore,

$$\begin{aligned}\left|\prod_{p\leq x}\sum_{\nu=0}^{\infty} f(p^{\nu})p^{-\nu s} - \sum_{n\in\mathbb{N}}\frac{f(n)}{n^s}\right| &= \left|\sum_{n\in A(x)}\frac{f(n)}{n^s} - \sum_{n\in\mathbb{N}}\frac{f(n)}{n^s}\right|,\\ &= \left|\sum_{n\in\mathbb{N}\setminus A(x)}\frac{f(n)}{n^s}\right|,\\ &\leq \sum_{n\notin A(x)}\frac{|f(n)|}{n^{\mathrm{Re}(s)}},\\ &\leq \sum_{n>x}\frac{|f(n)|}{n^{\mathrm{Re}(s)}} \to 0.\end{aligned}$$

The last limit is due to that it is a tail of a an absolutely convergent series. Since $f$ is completely multipliative, we have

$$f(p^{\nu}) = (f(p))^{\nu}.$$

Therefore, we get,

$$\begin{aligned}\prod_p\left(\sum_{\nu\in\mathbb{Z}_{\geq 0}}(f(p^{\nu})p^{-\nu s})\right) &= \prod_p\left(\sum_{\nu\in\mathbb{Z}_{\geq 0}}(f(p)p^{-s})^{-\nu}\right),\\ &= \prod_p\left(\frac{1}{1-f(p)p^{-s}}\right).\end{aligned}$$

□

**Example 2.4.** *Take $f(n) = 1$ as above we get,*

$$\sum_{n\in\mathbb{N}}\frac{1}{n^s} = \zeta(s) = \prod_p\left(1-\frac{1}{p^s}\right)^{-1}, \mathrm{Re}(s) > 1.$$

**Example 2.5.**

$$\sum_{n\in\mathbb{N}}\frac{\chi(n)}{n^s} = \prod_p\left(1-\frac{\chi(p)}{p^s}\right)^{-1}, \mathrm{Re}(s) > 1.$$

**Example 2.6.**

$$\begin{aligned}\sum_{n\in\mathbb{N}}\frac{\mu(n)}{n^s} &= \prod_p\left(1+\frac{\mu(p)}{p^s}\right)^{-1},\\ &= \prod_p\left(1-\frac{1}{p^s}\right)^{-1},\\ &= \frac{1}{\zeta(s)}.\end{aligned}$$

**Example 2.7.** *Note that $\phi(n) \leq n$. Thus for $\mathrm{Re}(s) > 2$, we have,*

$$\sum_{n\in\mathbb{N}} \frac{\phi(n)}{n^s} = \frac{\zeta(s-1)}{\zeta(s)}.$$

## 2.2 Order of arithmetic functions

**Definition 2.7.** *Let $f, g : \mathbb{N} \to \mathbb{C}$, we denote,*

$$f(n) = O(g(n)),$$

*if there is $K > 0$ and $n_0 \in \mathbb{N}$ such that*

$$n \geq n_0 \Rightarrow |f(n)| \leq K|g(n)|.$$

*An alternative notation for this is $f(n) = O(g(n))$.*

**Definition 2.8.** *We define following arithmetic functions,*

*1). $\nu(n) := \sum_{p|n} 1$, a number of primer divisors of $n$,*

*2). $d(n) := \sum_{d|n} 1$, the number of divisors of $n$,*

*3). $\sigma(n) := \sum_{d|n} d$, the sum of all the divisors of $n$.*

**Lemma 2.1.**

$$\nu(n) \ll \log(n).$$

*Proof.* Let $n = \prod_{i=1}^{k} p_i^{\alpha(p_i)}$. Then $\nu(n) = k$. Since $p_i \geq 2$, we have,

$$\begin{aligned}\log(n) &= \sum_{i=1}^{k} \alpha(p_i)\log(p_i),\\ &\geq k\log(2).\end{aligned}$$

Therefore $\nu(n) \leq \frac{\log(n)}{\log 2}$. □

**Lemma 2.2.**

$$\sum_{k=2}^{n} \frac{1}{k} \leq \log(n+1).$$

*Proof.* We know that

$$\int_1^n \frac{1}{t}dt = \log(n).$$

For $1 \leq k \leq t \leq k+1 \leq n$, we have,

$$\int_k^{k+1} \frac{1}{k+1}dt \leq \int_k^{k+1} \frac{1}{t}dt \leq \int_k^{k+1} \frac{1}{k}dt.$$

Thus we have,

$$\frac{1}{k+1} \leq \log(k+1) - \log(k) \leq \frac{1}{k}.$$

By telescopoing sum we get

$$\sum_{k=2}^{n} \frac{1}{k} \leq \log(n+1).$$

□

**Lemma 2.3.**

$$\sigma(n) \ll n(1+\log(n)) \sim n\log(n).$$

*Proof.*

$$\begin{aligned}
\sigma(n) = \sum_{d|n} \frac{n}{d}, \\
&= n \sum_{d|n} \frac{1}{d}, \\
&\leq n \left(1 + \sum_{d=2}^{n} \frac{1}{d}\right), \\
&\leq (1+\log(n)).
\end{aligned}$$

The last inequality is due to Lemma 2.2. □

**Exercise 2.1.** *Show that*

$$\sum_{k=1}^{n} \frac{1}{k} = \log(n) + O(1).$$

*That is*

$$\left| \sum_{k=1}^{n} \frac{1}{k} - \log(n) \right| \ll 1.$$

*Hint: Replace $\frac{1}{t}$ by an increasing function and derive the similar inequality to Lemma 2.2.*

*Proof.* By Lemma 2.2, we have,

$$\left| \sum_{k=1}^{n} \frac{1}{k} - \log(n) \right| \leq |1 + \log(n) - \log(n)| \leq 1.$$

□

**Lemma 2.4.**

$$d(n) \leq 2\sqrt{n}.$$

*Proof.* If $n = d_1 d_2$ then one of them must be less than or equal to $\sqrt{n}$. $\square$

We have an improved inequality,

**Proposition 2.4.** *for $\varepsilon > 0$, we have,*

$$d(n) \ll n^{\varepsilon}.$$

*Proof.* Recall that for $n = \prod_{i=1}^{k} p_i^{\alpha_i}$, we have $d(n) = \prod_{i=1}^{k}(\alpha_i+1)$. In particular, we have,

$$\frac{d(n)}{n^{\varepsilon}} = \prod_{i=1}^{k} \frac{(\alpha_i+1)}{p_i^{\varepsilon\alpha_i}}.$$

Let $A = \{i \mid p_i^{\varepsilon} \geq 2\}$. Rcall that for $x \geq 1$, $x+1 \leq 2^x$ that is

$$\frac{x+1}{2^x} \leq 1.$$

Then,

$$\prod_{i\in A} \frac{\alpha_i+1}{p_i^{\varepsilon\alpha_i}} \leq \prod_{i\in A} \frac{\alpha_i+1}{2^{\alpha_i}} \leq 1.$$

For $p_i^{\varepsilon} < 2$, we observe,

$$p_i^{\varepsilon\alpha_i} = e^{\varepsilon\alpha_i \log(p_i)} \geq \varepsilon\alpha_i \log(p_i).$$

Therefore,

$$\begin{aligned}
\prod_{i\notin A} \frac{\alpha_i+1}{p_i^{\varepsilon\alpha_i}} &\leq \prod_{i\notin A} \left( \frac{\alpha_i}{p_i^{\varepsilon\alpha_i}} + 1 \right), \\
&\leq \prod_{i\notin A} \left( \frac{\alpha_i}{\varepsilon\alpha_i \log(p_i)} + 1 \right), \\
&\leq \prod_{i\notin A} \left( \frac{1}{\varepsilon \log(2^{\frac{1}{\varepsilon}})} + 1 \right), \\
&\leq \prod_{i\notin A} \left( \frac{1}{\log(2)} + 1 \right).
\end{aligned}$$

Combining two cases, we obtain the statement. $\square$

**Notation 2.2.** *Let $x \in \mathbb{R}$. We denote*

1. *the integer part $[x] \in \mathbb{Z}$ which is the greatest integer not exceeding $x$,*
2. *the fraction part $\{x\} = x - [x]$.*

**Proposition 2.5.**

$$\frac{\sum_{n\leq x} d(n)}{x} \sim \log(x).$$

*Proof.* By definition, we have,

$$\sum_{n\le x} d(n) = \sum_{n\le x}\sum_{d|n} 1.$$

$d|n$ if and only if there is $e$ such that $de = n$. Using this we obtain,

$$\sum_{n\le x} d(n) = \sum_{\substack{(e,d)\\ de\le x}} 1 = \sum_{d\le x}\sum_{e\le \frac{x}{d}} 1 = \sum_{d\le x}\left[\frac{x}{d}\right].$$

Using the definition of $[x]$, we have,

$$\sum_{n\le x} d(n) = \sum_{d\le x}\frac{x}{d} - \left\{\frac{x}{d}\right\} = x\sum_{d\le x}\frac{1}{d} - \sum_{d\le x}\left\{\frac{x}{d}\right\} = x(\log(x) + O(1)) + O(x).$$

Thus we obtain the statement. □

**Definition 2.9.** *Let $f : \mathbb{N} \to \mathbb{C}$, we say the average order of $f$ is $g : \mathbb{R} \to \mathbb{C}$ if*

$$\frac{\sum_{n\le x} f(n)}{x} \sim g(x).$$

Proposition 2.5 can be restated as follows.

**Proposition 2.6.** *The average order of $d$ is $\log(x)$.*

**Exercise 2.2.** *Examine the following statements.*

1. *Is it true that $d(n) \ll \log n$?*
2. *Do we have $d(n) = O(n^{\varepsilon})$ for any $\varepsilon > 0$?*
3. *What is the optimal bound for $d(n)$?*

**Theorem 2.8.** *There exists $c_1, c_2 > 0$ such that*

$$c_1 \le \frac{\varphi(n)\sigma(n)}{n^2} \le c_2.$$

*Proof.* Recall that

$$\varphi(n) = n\prod_{p|n}\left(1 - \frac{1}{p}\right),$$

and

$$\sigma(n) = \prod_{p|n}\frac{p^{\alpha(p)+1} - 1}{p - 1}.$$

Thus we obtain,

$$\frac{\sigma(n)}{n} = \frac{\prod_{p|n}(1 + p + \cdots + p^{\alpha(p)})}{\prod_{p|n} p^{\alpha(p)}} = \prod_{p|n}\left(\frac{\frac{1}{p^{\alpha(p)+1}} - 1}{\frac{1}{p} - 1}\right)$$

By multiplying two we get,

$$\frac{\varphi(n)\sigma(n)}{n^2} = \prod_{p|n}\left(1 - \frac{1}{p^{\alpha(p)+1}}\right) \leq 1.$$

On the other hand,

$$\prod_{p}\left(1 - \frac{1}{p^2}\right) \leq \frac{\varphi(n)\sigma(n)}{n^2}.$$

The left hand side is equal to $\frac{1}{\zeta(2)}$ which is $\frac{6}{\pi^2}$. □

**Lemma 2.5.** *For $x \geq 1$ and $d$ an integer such that $d \leq x$, we have,*

$$\left[\frac{x}{d}\right]\left(\left[\frac{x}{d}\right] + 1\right) = \frac{x^2}{d^2} + o\left(\frac{x}{d}\right).$$

*Proof.*

$$\begin{aligned}
\left[\frac{x}{d}\right]\left(\left[\frac{x}{d}\right] + 1\right) &= \left(\frac{x}{d} - \left\{\frac{x}{d}\right\}\right)\left(\left(\frac{x}{d} - \left\{\frac{x}{d}\right\}\right) + 1\right), \\
&= \frac{x^2}{d^2} - 2\left\{\frac{x}{d}\right\}\frac{x}{d} + \left(\left\{\frac{x}{d}\right\}\right)^2 + \frac{x}{d} + \left\{\frac{x}{d}\right\}, \\
&= \frac{x^2}{d^2} + O(1)\cdot\frac{x}{d} + O(1).
\end{aligned}$$

□

**Theorem 2.9.** *The average order of $\varphi$ is $\frac{3x}{\pi^2}$.*

*Proof.*

$$\begin{aligned}
\sum_{n\leq x}\varphi(n) &= \sum_{n\leq x} n \sum_{d|n}\frac{\mu(d)}{d}, \\
&= \sum_{\substack{d,e \\ de\leq x}} e\mu(d), \\
&= \sum_{d\leq x}\mu(d)\left(\sum_{e\leq\frac{x}{d}} e\right),
\end{aligned}$$

Since $\sum_{e\leq\frac{x}{d}} e$ is a sum of an arithmetic progression, we have,

$$\sum_{e\leq\frac{x}{d}} e = \frac{1}{2}\left[\frac{x}{d}\right]\left(\left[\frac{x}{d}\right] + 1\right).$$

Substituting this, we obtain,

$$\sum_{n\leq x}\varphi(n)=\frac{1}{2}\sum_{d\leq x}\left(\mu(d)\left[\frac{x}{d}\right]\left(\left[\frac{x}{d}\right]+1\right)\right).$$

Note that

$$\frac{x}{d}=\left[\frac{x}{d}\right]+\left\{\frac{x}{d}\right\}=\left[\frac{x}{d}\right]+O(1).$$

Using Lemma 2.5, that is,

$$\left[\frac{x}{d}\right]\left(\left[\frac{x}{d}\right]+1\right)=\frac{x^2}{d^2}+o\left(\frac{x}{d}\right).$$

We then have,

$$\begin{aligned}\sum_{n\leq x}\varphi(n)&=\frac{1}{2}\sum_{d\leq x}\mu(d)\left(\frac{x^2}{d^x}+O\left(\frac{x}{d}\right)\right),\\&=\frac{x^2}{2}\sum_{d\leq x}\frac{\mu(d)}{d^2}+O\left(x\sum_{d\leq x}\frac{\mu(d)}{d}\right),\\&=\frac{x^2}{2}\left(\sum_{d\geq 1}\frac{\mu(d)}{d^2}-\sum_{d\geq x}\frac{\mu(d)}{d^2}\right)+O\left(x\sum_{d\geq x}\frac{\mu(d)}{d}\right),\\&=\frac{x^2}{2}\frac{1}{\zeta(2)}-\frac{x^2}{2}\sum_{d\geq x}\frac{\mu(d)}{d}^2+O\left(x\sum_{d\leq x}\frac{\mu(d)}{d}\right).\end{aligned}$$

We have,

$$\begin{aligned}\left|\sum_{d\geq x}\frac{\mu(d)}{d^2}\right|&\leq\sum_{d\geq x}\frac{1}{d^2}\\&\ll\int_x^\infty\frac{dt}{t^2},\\&\ll\frac{1}{x}.\end{aligned}$$

and,

$$\left|\sum_{d\leq x}\frac{\mu(d)}{d}\right|\ll\ln x.$$

Using these we have,

$$\begin{aligned}\sum_{n\leq x}\varphi(n)&=\frac{x^2}{2}\frac{1}{\zeta(2)}+O(x)+O(x\ln x),\\&=\frac{x^2}{2\zeta(2)}+O(x\ln x).\end{aligned}$$

We conclude that

$$\frac{\sum_{n\leq x}\varphi(n)}{x^2} \overset{x\to\infty}{\to} \frac{1}{2\zeta(2)}.$$

In particular,

$$\frac{\sum_{n\leq x}\varphi(n)}{x} \sim \frac{x}{2\zeta(2)} = \frac{x\cdot 6}{2\pi^2}.$$

□

## 2.3 Abel's Summation Formula

Recall the harmonic series $\sum_{n\in\mathbb{N}}\frac{1}{n}$ is divergent. Our next goal is to find such $A_x$ that

$$\lim_{n\to\infty}\left(\sum_{n\leq x}\frac{1}{n} - A_x\right)$$

exists.

**Remark 2.4** (Euler-Mascheroni constant)**.** *By taking $A_x = \log(x)$, we have*

$$\lim_{n\to\infty}\left(\sum_{n\leq x}\frac{1}{n} - \log(x)\right) = \gamma,$$

*exists. Such $\gamma$ is called Euler-Mascheroni constant.*

**Remark 2.5** (Euler Kronecer constant)**.** *Take $A_x = \log(x)$, we have*

We can show that

$$\psi = \lim_{s\to 1^+}\left(\frac{\zeta'(s)}{\zeta(s)} - \frac{1}{s-1}\right).$$

Hint first show that

$$\zeta(s) = \frac{1}{s-1} + \psi + o(s-1).$$

**Proposition 2.7** (Abels' summation formula)**.** *Given $(a_n)_{n\in\mathbb{N}}$ in $\mathbb{C}$ and $f(n)$ is continuously differentiable in $[1,x]$. Set*

$$A(x) := \sum_{n\leq x} a_n.$$

*Then we have,*

$$\sum_{n\leq x} a_n f(n) = A(x)f(x) - \int_1^x A(t)f'(t)dt.$$

*Proof.* Observe that

$$a_n = A(n) - A(n-1).$$

Assume $x \in \mathbb{N}$. We substitute this to $\sum_{n \leq x} a_n f(n)$, we get,

$$\begin{aligned}
\sum_{n \leq x} a_n f(n) &= \sum_{n \leq x} \left(A(n) - A(n-1)\right) f(n), \\
&= \sum_{n \leq x} A(n) f(n) - \sum_{n \leq x} A(n-1) f(n), \\
&= \sum_{n \leq x} A(n) f(n) - \sum_{n \leq x-1} A(n) f(n+1), \\
&= A(x) f(x) - \sum_{n \leq x-1} A(n)(f(n+1) - f(n)), \\
&= A(x) f(x) - \sum_{n \leq x-1} A(n) \int_n^{n+1} f'(t) dt, \\
&= A(x) f(x) - \sum_{n \leq x-1} \int_n^{n+1} A(t) f'(t) dt, \\
&= A(x) f(x) - \int_1^x A(t) f'(t) dt.
\end{aligned}$$

For the case when $n \notin \mathbb{N}$ and $n > 1$,

$$\sum_{n \leq x} a_n f(n) = \sum_{n \leq [x]} a_n f(n).$$

Using the previous case, we get,

$$\sum_{n \leq x} a_n f(n) = A([x]) f([x]) - \int_1^{[x]} A(t) f'(t) dt.$$

Remains to show that we can remove the brackets. To do so,

$$\begin{aligned}
\sum_{n \leq x} a_n f(n) &= A([x]) f([x]) - \int_1^x A(t) f'(t) dt + \int_{[x]}^x A(t) f'(t) dt, \\
&= A([x]) f([x]) - \int_1^x A(t) f'(t) dt + A([x]) \int_{[x]}^x f'(t) dt, \\
&= A([x]) f([x]) - \int_1^x A(t) f'(t) dt + A([x]) f(x) - A([x])(f[x]), \\
&= A([x]) f(x) - \int_1^x A(t) f'(t) dt.
\end{aligned}$$

□

**Corollary 2.1.** *Under the settings of Proposition 2.7, we have,*

$$\sum_{p\leq x} a_p f(p) = A(x)f(x) - \int_2^x A(t)f'(t)dt.$$

*Proof.* Consider $b_n = \pi(n)a_n$ and apply Proposition 2.7. □

**Corollary 2.2.**

1. $\sum_{n\leq x} \frac{1}{n} = \ln x + \psi + o\left(\frac{1}{x}\right)$.
2. $\sum_{n\leq x} \frac{1}{n^s} = \frac{x^{1-s}}{1-s} + \zeta(s) + o\left(\frac{1}{x^s}\right)$, *where* $\mathrm{Re}(s) > 0, s \neq 1$.

*We also have the following equivalent forms of prime number theorem when* $x \to \infty$.

$$\begin{aligned}\sum_{n\leq x} \Lambda(n) \sim x \Leftrightarrow \pi(x) &\sim \frac{x}{\ln x},\\ \Leftrightarrow \sum_{p\leq x} \ln p &\sim x,\\ \Leftrightarrow \sum_{n\leq x} \mu(n) &= o(x).\end{aligned}$$

*Proof.* Consider $f(t) = \frac{1}{t}$ and $a_n = 1$ for all $n \in \mathbb{N}$.

$$\begin{aligned}\sum_{n\leq x} \frac{1}{n} &= \frac{[x]}{x} + \int_1^x \frac{[t]}{t^2} dt,\\ &= \frac{x - \{x\}}{x} + s\int_1^x \frac{t - \{t\}}{t^2} dt,\\ &= 1 + \log x - \frac{\{x\}}{x} - \int_1^\infty \frac{\{t\}}{t^2} dt.\end{aligned}$$

For general $s, \mathrm{Re}(s) > 0, s \neq 1$, we have

$$\sum_{n\leq x} \frac{1}{n^s} = x^{1-s} + o\left(\frac{1}{x^s}\right) + \frac{sx^{1-s}}{1-s} - \frac{s}{1-s} - s\int \frac{\{t\}}{t^{s+1}} dt.$$

Recall that

$$\left[\int \frac{1}{t^s} = \frac{t^{-s+1}}{1-s}\right]_1^x = \frac{x^{1-s}}{1-s} - \frac{1}{1-s}.$$

Using this we obtain,

$$\begin{aligned}
\sum_{n\le x}\frac{1}{n^s} &= \frac{x^{1-s}}{1-s}-\frac{s}{1-s}+o\left(\frac{1}{x^s}\right)-x\int_1^x\frac{\{t\}}{t^{s+1}}dt,\\
x^{1-s}\left[1+\frac{1}{1-s}\right] &= \frac{x^{1-s}}{1-s}-\frac{s}{1-s}+o\left(\frac{1}{x^s}\right)-s\int_1^x\frac{\{t\}}{t^{s+1}}dt\\
\int_1^\infty\frac{\{t\}}{t^{s+1}} &< \infty,\\
&\le \int_1^\infty\frac{1}{t^{\mathrm{Re}(s)+1}}<\infty.
\end{aligned}$$

As $x\to\infty$, the left hand side goes to $\zeta(s)$, for the right hand side, we get $=\frac{-s}{1-s}-s\int_1^\infty\frac{\{t\}}{t^{s+1}}dt$, where $\mathrm{Re}(s)>1$,

$$\zeta(s)=\frac{-s}{1-s}-s\int\frac{\{t\}}{t^{s+1}}dt.$$

Identity theorem for analytic function tells us the analytic continuation of Riemann zeta function is unique. □

**Remark 2.6.** *It is an exercise that*

$$\int_1^\infty\frac{\{t\}}{t^{s+1}}dt$$

*where* $\mathrm{Re}(x)>0$. *From Stein-Schakarchi 5.2, 5.3, we have*

$$\sum f_n(z)\overset{unif.}{\to}f(z)$$

*is analytic where* $\mathrm{Re}(x)>0$ *and* $s\neq 1$, *also in this case,*

$$\zeta(s)=\frac{-s}{1-s}-s\int\frac{\{t\}}{t^{s+1}}dt.$$

*holds.*

**Remark 2.7** (Exercise)**.** *Let* $M\in\mathbb{N}$ *and*

$$\lim_{x\to\infty}\left(\sum_{\substack{n\le x\\(n,M)=1}}\frac{1}{n}-\frac{\phi(M)}{M}\ln x\right)$$

*exists.*

Suppose we have, $\pi(x)\sim\frac{x}{\ln x}$. We want to show

$$\theta(x):=\sum_{p\le x}\ln p\sim x,$$

Consider the following sequence

$$a_n = \begin{cases} 1, & (n \text{ is prime}), \\ 0, & (\text{otherwise}). \end{cases}$$

and

$$f(t) = \ln t.$$

Using Abel summation formula,

$$\sum_{p \le x} \ln p = \pi(x) \ln(x) - \int_1^\infty \frac{\pi(t)}{t} dt.$$

Dividing both sides by $x$, we obtain,

$$\frac{\theta(x)}{x} = \frac{\pi(x) \ln x}{x} - \frac{1}{x} \int_1^x \frac{\pi(t)}{t} dt.$$

**Remark 2.8** (Exercise)**.** *Use* $\frac{\pi(t)}{t} \sim \frac{1}{\ln t}$ *and*

$$\int_1^x \frac{dt}{\ln t} = o(x),$$

*prove that*

$$\lim_{x \to \infty} \frac{1}{x} \int_1^x \frac{\pi(t)}{t} dt = 0.$$

$$\begin{aligned} \int_2^{\sqrt{x}} \frac{dt}{\ln t} = \int_{\sqrt{x}}^x \frac{dt}{\ln t} &\le \frac{1}{\ln 2} \int_2^{\sqrt{x}} dt + \frac{1}{\ln \sqrt{x}} \int_{\sqrt{x}}^x dt, \\ &= \frac{\sqrt{x} - 2}{\ln 2} + \frac{x - \sqrt{x}}{\ln \sqrt{x}}, \\ &= o(x). \end{aligned}$$

**Remark 2.9.**

$$\psi(x) = \sum_{n \le x} \Lambda(n) = \sum_{\substack{1 \le k \\ p^k \le x}} \ln p.$$

*Thus we see,*

$$\psi(x) - \theta(x) = \sum_{\substack{2 \le k \\ p, p^k \le x}} \ln p.$$

*Also,*

$$p^k \le x \Rightarrow k \le \frac{\ln x}{\ln p}.$$

*Using* $k \ge 2$*,*

$$p \le x^{\frac{1}{k}} \le \sqrt{x}.$$

*Thus we have the inequalities,*

$$\begin{aligned}
\psi(x) - \theta(x) &\leq \sum_{p \leq \sqrt{x}} \ln p \left( \sum_{2 \leq k \leq \frac{\ln x}{\ln p}} 1 \right), \\
&\leq \sum_{p \leq \sqrt{x}} \ln x, \\
&\leq \ln(x) \sum_{n \leq \sqrt{x}} 1, \\
&\leq \sqrt{x} \ln x. \\
\Rightarrow \frac{\psi(x)}{x} &= \frac{\theta(x)}{x} + O\left( \frac{\sqrt{x} \ln x}{x} \right).
\end{aligned}$$

*Therefore we obtain,*

$$\psi(x) \sim x \Leftrightarrow \theta(x) \sim x.$$

**Remark 2.10.** *As exercises, find the closed expressions for the following summations.*

$$\sum_{n \in \mathbb{N}} \frac{\mu(n)}{n}, \sum_{p \leq x} \frac{1}{p} = \ln \ln x + o(1).$$

## 2.4 Characters

**Definition 2.10.** *Let $G$ be a finite group. A character is a group homomorphism $f : G \to \mathbb{C}^\times$.*

**Remark 2.11.** *Let us denote*

$$\hat{G} := \{ f : G \to \mathbb{C}^\times \mid \textit{characters} \}.$$

*If $G$ is finite abelian then $|\hat{G}| = |G|$. Furthermore, such characters are linearly independent over $\mathbb{C}$.*

**Definition 2.11.** *Let $q \in \mathbb{N}$ and $q \geq 3$. A character is a group homomorphism modulo $q$ is a group homomorphism*

$$\chi' : (\mathbb{Z}/q\mathbb{Z})^\times \to \mathbb{C}^\times.$$

*Given a character $\chi' : (\mathbb{Z}/q\mathbb{Z})^\times \to \mathbb{C}^\times$. We can define a Dirichlet character modulo $q$, $\chi : \mathbb{Z} \to \mathbb{C}^\times$ as follows.*

$$\chi(a) := \begin{cases} \chi'(\overline{a}), & (a, q) = 1, \\ 0, & (a, q) \neq 1. \end{cases}$$

**Remark 2.12.** *From Remark 2.11, there are exactly $\varphi(q)$ many Dirichlet characters modulo $q$. Furthermore,*

$$\chi(a)^{\varphi(q)} = \chi'(\overline{a})^{\varphi(q)} = \chi'(\overline{a}^{\varphi(q)}) = \chi'(\overline{1}) = 1.$$

*In particular, images of $\chi$ are $\varphi(q)$-th roots of unity.*

**Example 2.8.** *For $q = 3$,*

$$(\mathbb{Z}/3\mathbb{Z})^\times = \{\overline{1}, \overline{2}\} \to \mathbb{C}^\times.$$

*We only have two characters, a trivial one and $\overline{2} \mapsto -1$.*

**Example 2.9.** *For $q = 5$,*

$$(\mathbb{Z}/5\mathbb{Z})^\times = \{\overline{1}, \overline{2}, \overline{3}, \overline{4}\}$$

| | $\overline{1}$ | $\overline{2}$ | $\overline{3}$ | $\overline{4}$ |
|---|---|---|---|---|
| $\chi_{1,5}(n)$ | 1 | 1 | 1 | 1 |
| $\chi_{2,5}(n)$ | 1 | $-1$ | $-1$ | 1 |
| $\chi_{3,5}(n)$ | 1 | $i$ | $-i$ | $-1$ |
| $\chi_{4,5}(n)$ | 1 | $-i$ | $i$ | $-1$ |

*$\chi_{1,5}$ is called a principle/trivial character.*

**Definition 2.12.** *A character $\chi : \mathbb{Z} \to \mathbb{C}^\times$ is called*

*1). trivial if $\chi(g) = 1$ for all $g \in G$,*

*2). even if $\chi(-1) = 1$,*

*3). odd if $\chi(-1) = -1$.*

*We also define these notions for characters $\chi_0 : (\mathbb{Z}/q\mathbb{Z})^\times \to \mathbb{C}^\times$ accordingly if the character $\chi : \mathbb{Z} \to \mathbb{C}^\times$ induced by it has these properties. Trivial characters are often denoted by $\chi_0$.*

**Theorem 2.10.**

$$\sum_{a \bmod q} \chi(a) = \begin{cases} \phi(q) & (\chi = \chi_0), \\ 0 & (\textit{otherwise}). \end{cases}$$

*We also have,*

$$\sum_{\chi \bmod q} \chi(a) = \begin{cases} \phi(a) & (\overline{a} = 1), \\ 0 & (\textit{otherwise}). \end{cases}$$

*Proof.* If $\chi$ is principle then the first assertion is clear. Suppose $\chi$ is not principle then there is $b \in \{1, \cdots, q\}$ such that $\chi(b) \neq 1$ and $(b, q) = 1$. Let

$$s = \sum_{a \bmod q} \chi(a).$$

Then by the definiion of group homomorphisms, we have,

$$\chi(b)s = s.$$

But $\chi(b) \in \mathbb{C}$, this means $s = 0$ as $\mathbb{C}$ is an integral domain.

For the second assertion, let $\overline{a} \neq 1$, then

$$\exists \chi' \bmod q, \text{ s. t. } \chi'(a) \neq 1.$$

Thus we get,

$$s = \sum_{\chi \bmod q} \chi(a), s \cdot \chi(a) = \sum_{\chi \bmod q} \chi\chi'(a) = s \Rightarrow s = 0.$$

The statement when $\overline{a} = 1$ follows from Remark 2.11. $\square$

**Remark 2.13.** *One can check in the table of Example 2.9 that Theorem 2.10 indeed holds.*

**Exercise 2.3.**

$$\sum_{\substack{\chi \bmod q, \\ \chi(-1)=1}} \chi(a) = \begin{cases} \frac{\phi(a)}{2} & (\overline{a} = 1, -1), \\ 0 & (\textit{otherwise}). \end{cases},$$

$$\sum_{\substack{\chi \bmod q, \\ \chi(-1)=-1}} \chi(a) = \begin{cases} \frac{\phi(a)}{2} & (\overline{a} = 1), \\ -\frac{\phi(a)}{2} & (\overline{a} = -1), \\ 0 & (\textit{otherwise}). \end{cases},$$

Obviously we have the following equalities.

$$\sum_{n \leq x} \chi(n) = \sum_{\substack{n \leq x \\ (n,q)=1}} \chi(n) = \sum_{n \leq kq} \chi(n) + \sum_{n=kq+1}^{x} \chi(n),$$

where $k$ is the largest integer such that $kq \leq x$. Then we observe from Theorem 2.10

$$\sum_{n \leq kq} \chi(a) = k \left( \sum_{n=1}^{q} \chi(n) \right) = 0,$$

unless $\chi$ is trivial. Also we have,

$$\left| \sum_{n \leq x} \chi(n) \right| = \left| \sum_{\substack{kq+1 \leq n \leq x \\ (n,q)=1}} \chi(n) \right| \leq \sum_{\substack{kq+1 \leq n \leq x \\ (n,q)=1}} 1 \leq \sum_{\substack{kq+1 \leq n \leq kq+q \\ (n,q)=1}} 1 = \phi(q).$$

Thus we conclude,

$$\sum_{n \leq x} \chi(n) \leq \varphi(q),$$

**Exercise 2.4.**

$$\sum_{n \leq x} \chi_0(n) = ?.$$

**Theorem 2.11** (Pólya–Vinogradov)**.**

$$\sum_{n\le x}\chi(n)\ll\sqrt{q}\ln q,(\chi\neq\chi_0 \mod q).$$

*notice that the above expression is bounded by* $\sqrt{q}\ln\ln q$*. Furthermore, this is uniform in* $q$ *that is the constant does not depend on* $q$*.*

$$\sum_{n\ge1}\frac{1}{n^s}=\frac{x^{1-s}}{1-s}-\frac{s}{1-s}+O(x^{-s})-s\int_1^x\frac{\{t\}}{t^{s+1}}dt.$$

For $\mathrm{Re}(s)>1$, as $x\to\infty$ we have,

$$\sum_{n\ge1}^{\infty}\frac{1}{n^s}=\frac{-s}{1-s}-s\int_1^\infty\frac{\{t\}}{t^{s+1}}dt.$$

The last expression is analytic since,

$$\sum\int_n^{n+1}\frac{\{t\}}{t}^{s+1}dt\overset{\text{uniformly}}{\to}\int_1^\infty\frac{\{t\}}{t^{s+1}},\text{when }\mathrm{Re}(s)>0.$$

Suppose $\zeta(s)\neq0$, where $\mathrm{Re}(s)>0$, then Euler product exists.

**Theorem 2.12.** *Set*

$$A(n):=\sum_{n\in\mathbb{N}}a_n.$$

*Assume* $A(x):=\sum_{n\le x}a_n=O(x^\delta)$*, then we have, for* $\mathrm{Re}(s)>\delta$*,*

$$\sum_{n=1}^{\infty}\frac{a_n}{n^s}=s\int_1^\infty\frac{A(t)}{t^{s+1}}dt.$$

*Hence the Dirichlet series converges for* $\mathrm{Re}(s)>\delta$*.*

*Proof.*

$$\sum_{n\in\mathbb{N}}\frac{a_n}{n^s}=\frac{A(x)}{x^s}+s\int_1^x\frac{A(t)}{t^{s+1}}dt.$$

As $A(x)=O(x^\delta)$ and $\mathrm{Re}(s)>\delta$, $\frac{A(x)}{x^s}=O(x^{\delta-\mathrm{Re}(s)})$. Therefore, as $x\to\infty$, we have,

$$\sum_{n\in\mathbb{N}}\frac{a_n}{n^s}=s\int_1^\infty\frac{A(t)}{t^{s+1}}dt.$$

Again using the assumption, we have,

$$\int_1^\infty\left|\frac{A(t)}{t^{s+1}}\right|dt\ll\int_1^\infty t^{\delta-\mathrm{Re}(s)-1}dt=\frac{t^{\delta-\mathrm{Re}(s)}}{\delta-\mathrm{Re}(s)}\bigg|_0^\infty=\frac{1}{\delta-\mathrm{Re}(s)}.$$

Thus the integral is convergent. □

**Definition 2.13.** *For* $\mathrm{Re}(s) > 1$*, we define*

$$L(s,\chi) := \sum_{n\in\mathbb{N}} \frac{\chi(n)}{n^s}.$$

**Remark 2.14.** *Since* $\mathrm{Re}(s) > 1$ *and for any character* $\chi : \mathbb{Z} \to \mathbb{C}^\times$ *we have* $|\chi(n)| \le 1$*,* $L(s,\chi)$ *is uniformly absolutely convergent.*

**Example 2.10.** *Let* $\chi : \mathbb{Z} \to \mathbb{C}^\times$ *be a non-principal character modulo* $q$ *and set* $A(n) = \chi(n)$*. Recall that*

$$\sum_{n\le x} \chi(n) \le q.$$

*Taking* $A(n) = \chi(n)$ *and apply Theorem 2.12, we obtain, for* $\mathrm{Re}(s) > 0$*,*

$$L(s,\chi) = \sum_{n\in\mathbb{N}} \frac{\chi(n)}{n^s} = s\int_1^\infty \frac{\sum_{n\le t}\chi(n)}{t^{s+1}}dt.$$

*Since for* $\mathrm{Re}(s) > 1, L(s,\chi)$ *is absolutely uniformly convergent. By Theorem 2.7, we have,*

$$L(s,\chi) = \prod_p \left(1 - \frac{\chi(p)}{p^s}\right)^{-1}.$$

*If* $\chi = \chi_0$*, a principal character, we have,*

$$L(s,\chi_0) = \prod_{(p,q)=1} \left(1 - \frac{1}{p^s}\right)^{-1} = \prod_p \left(1 - \frac{1}{p^s}\right)^{-1} \prod_{p|q} \left(1 - \frac{1}{p^s}\right).$$

*Using* $\zeta$ *function, we have,*

$$L(s,\chi_0) = \zeta(s) \prod_{p|q} \left(1 - \frac{1}{p^s}\right).$$

**Theorem 2.13.** $\zeta$ *has an analytic continuation for* $\mathrm{Re}(s) > 0$ *besides* $s = 1$*. For* $s = 1$ *we have a simple pole of residue* $1$*.*

*Proof.* Recall from the proof of Corollary 2.2. We have for $\mathrm{Re}(s) > 1$,

$$\zeta(s) = \frac{-s}{1-s} - s\int_1^\infty \frac{\{t\}}{t^{s+1}}dt.$$

The right hand side of the equation is analytic when $\mathrm{Re}(s) > 0, s \ne 1$. When $s = 1$, we have,

$$\lim_{s\to 1^+} (s-1)\zeta(s) = \lim_{s\to 1^+} \left(s - s(s-1)\int_1^\infty \frac{\{t\}}{t^{s+1}}dt\right) = 1.$$

□

**Corollary 2.3.** *For* $\mathrm{Re}(s) > 0$ *and* $s \neq 1$*, we have an analytic continuation of* $L(s, \chi_0)$ *where* $\chi_0$ *is a principal character modulo* $q$*, which is*

$$L(s, \chi_0) = \zeta(s) \prod_{p|q} \left(1 - \frac{1}{p^s}\right).$$

*Obviously at* $s = 1$*, it has a simple pole of residue* $\prod_{p|q} \left(1 - \frac{1}{p^s}\right)$*, which we can write as*

$$\mathrm{Res}(L(s, \chi_0), 1) = \frac{\varphi(q)}{q}.$$

*Proof.* A direct corollary of Theorem 2.13. For the residuce, write,

$$q = \prod_{p|q} p^{\alpha(p)},$$

then

$$\varphi(q) = \prod_{p|q} (p-1) p^{\alpha(p)-1} = q \prod_{p|q} (p-1) p^{-1}.$$

□

**Theorem 2.14.** *Let* $\chi$ *be a non-principal character, then there is an analytic continuation of* $L(s, \chi)$ *for* $\mathrm{Re}(s) > 0$*.*

*Proof.* From Example 2.10, we have,

$$L(s, \chi) = \sum_{n \in \mathbb{N}} \frac{\chi(n)}{n^s} = s \int_0^\infty \frac{\sum_{n \leq t} \chi(n)}{t^{s+1}} dt.$$

The right hand side is analytic. □

**Theorem 2.15.**

$$\sum_{n \in \mathbb{N}} \frac{a_n}{n^s}$$

*is analytic in its range of convergence.*

**Remark 2.15.** *We have the following conjecture.*

$$L\left(\frac{1}{2}, \chi\right) = 0, \chi \neq \chi_0.?$$

$$\zeta\left(\frac{1}{2}\right) = \frac{1}{1 - \sqrt{2}} \sum_{n \in \mathbb{N}} \frac{(-1)^{n-1}}{\sqrt{n}} \approx -1.46 \cdots$$

**Definition 2.14.** *A character is said to be quadratic if its values are either* $\pm 1$*.*

**Remark 2.16.** *Suppose $\chi \neq \chi_0$. We have,*

$$L(s,\chi) = 0 \quad \text{if } s = 0, -2, -4, \text{ when } \chi \text{ is an even character.}$$
$$L(s,\chi) = 0 \quad \text{if } s = -1, -3, -5, \text{ when } \chi \text{ is an odd character.}$$

**Lemma 2.6.** *For $\sigma > 1$ and $t \in \mathbb{R}$, we have,*

$$\mathrm{Re}(\ln(\zeta(\sigma + it))) = \sum \frac{\Lambda(n)}{n^\sigma \ln n} \ln(t \ln(n)).$$

*And also,*

$$\mathrm{Re}(3\ln(\zeta(\sigma)) + 4\ln(\zeta(\sigma + it)) + \ln(\zeta(\sigma + 2it))) \geq 0.$$

*Proof.*

$$\begin{aligned}
\zeta(s) &= \prod_p \left(1 - \frac{1}{p^s}\right)^{-1}, \sigma > 0, \\
\ln(\zeta(s)) &= -\sum_p \ln(1 - p^{-s}) \\
&= \sum_{p,n} \frac{1}{np^{ns}}, \sigma > 1.
\end{aligned}$$

We have,

$$\sum_{n \geq 2} \frac{\Lambda(n)}{n^s \ln n} = \sum_{p,k,k \geq 1} \frac{\ln p}{p^{ks} \ln p^k} = \sum_{p,k} \frac{1}{kp^{ks}} = \sum_{n \geq 2} \Lambda(n).$$

$$\mathrm{Re}(3\ln\zeta(\sigma) + 4\ln\zeta(\sigma + it) + \ln\zeta(\sigma + 2it)) = \sum_{n \geq 2} \frac{\Lambda(n)}{n^\sigma \ln n}(3 + 4\cos(t\ln n) + \cos(2t\ln n)) \geq 0,$$

since

$$3 + 4\cos\theta + \cos 2\theta = 2(\cos\theta + 1)^2 \geq 0.$$

$$= \ln|\zeta(\sigma)^3 \zeta(\sigma + it)^4 \zeta(\sigma + 2it)| \geq 0.$$

Thus we have,

$$\mathrm{Re}(\ln(z)) \leq$$

□

**Theorem 2.16.** *For $t \in \mathbb{R}\backslash\{0\}$, we have*

$$\zeta(1 + it) \neq 0.$$

*Proof.* Using Lemma 2.6, we have, Thus we get,

$$|\zeta(\sigma)^3\zeta(\sigma+it)^4\zeta(\sigma+2it)| \geq 1.$$

Suppose $\zeta(1+it_0)=0$, for $t_0 \in \mathbb{R}\backslash\{0\}$. Suppose further that the order of zero is $m \in \mathbb{N}$. Then by looking at, and taking $\sigma \to 1+$

$$\underbrace{((\sigma-1)\zeta(\sigma))^3}_{\to\text{finite}} \underbrace{\left(\frac{\zeta(\sigma+it_0)}{(\sigma-1)^m}\right)^4}_{\to\text{non-zero}} \underbrace{((\sigma-1)^{4m-3}\zeta(\sigma+2it_0))}_{\to 0} \to 0.$$

Contradicts to that the abosolute value of above expression is at least 1. □

**Theorem 2.17.** *$\frac{\zeta'(s)}{\zeta(s)}$ has an analytic continuation to $\mathrm{Re}(s)=1, s\neq 1$. And for $s=1$, we have a simple pole of residue $-1$.*

*Proof.* We have,

$$(s-1)\zeta(s) = s - s(s-1)\int_1^\infty \frac{\{t\}}{t^{s+1}}dt.$$

Set

$$f(s) := 1-(s-1)\int_1^\infty \frac{\{t\}}{t^{s+1}}dt,$$

so that $(s-1)\zeta(s)=sf(s)$. We already have $f(s)$ is analytic when $\mathrm{Re}(s)>0$. Differentiating both sides we get,

$$(s-1)\zeta'(s)+\zeta(s) = sf'(s)+f(s).$$

Dividing both sides by $(s-1)\zeta(s)$ we get,

$$\frac{\zeta'(s)}{\zeta(s)}+\frac{1}{s-1} = \frac{sf'(s)+f(s)}{(s-1)\zeta(s)}.$$

The right hand side is analytic when $\zeta$ does not vanish. From Theorem 2.16, by letting $s=1+it$ for some $t\in\mathbb{R}\backslash\{0\}$, we get the desired analytic continuation. For $s=1$, we have

$$(s-1)\frac{\zeta'(s)}{\zeta(s)} = \frac{sf'(s)+f(s)}{\zeta(s)}-1.$$

Recall that $\zeta(s)$ has a pole at 1 and observe that $f(1)=1, f'(s)$ is finite thus the last statement follows. □

**Theorem 2.18.** *Let $\chi$ be a non-real Dirichlet character of modulo $q$, then*

$$L(1,\chi)\neq 0.$$

*Proof.* Recall that

$$L(s,\chi) = \prod_p \left(1 - \frac{\chi(p)}{p^s}\right)^{-1},$$

where $\mathrm{Re}(s) > 1$. Now consider,

$$\log L(s,\chi) = \sum_{n,p} \frac{\chi(p^s)}{np^{ns}} = \sum_{n\geq 2} \frac{\Lambda(n)\chi(n)}{n^s \log n}.$$

Set $\zeta_{\varphi(q)} = e^{\frac{2\pi i}{\varphi(q)}}$. Then for any $n \in (\mathbb{Z}/q\mathbb{Z})^\times$, there is $n' = n(n,q,\chi)$ (ie $n'$ depends on $n, q$, and $\chi$) such that

$$\chi(n) = \zeta_{\varphi(q)}^{n'}$$

Note that for a cyclic group $G = \langle g \rangle$, and a character $\chi : G \to \mathbb{C}^\times \in \hat{G}$, there is $a \in \{1, \cdots, \varphi(q)\}$ such that

$$\chi(g) = \zeta_{\varphi(q)}^{a}.$$

We have, for $\sigma > 1$

$$\mathrm{Re}(\log L(\sigma,\chi)) = \sum_{n\geq 2} \frac{\Lambda(n)\cos\left(\frac{2\pi i}{\varphi(q)}\right)}{n^\sigma \log(n)}.$$

By Lemma 2.6, we have,

$$\mathrm{Re}(3\ln\zeta(\sigma) + 4\ln L(\sigma,\chi) + \ln L(\sigma,\chi^2)) \geq 0,$$

therefore,

$$|\zeta(\sigma)^3 L(\sigma,\chi)^4 L(\sigma,\chi^2)| \geq 1.$$

Note that $\chi^2 \neq \chi$ as it is non-real and $L(\sigma,\chi^2)$ is analytic by Theorem 2.14. If $L(\sigma,\chi) = 0$ then

$$\zeta(\sigma)^3 L(\sigma,\chi)^4$$

has a zero of order at least 1. Thus this is a contradiction. □

**Lemma 2.7.** *For $k \in \mathbb{Z}_{\geq 0}$,*

$$\left(\frac{\sin\left(k+\frac{1}{2}\right)\theta}{\sin\frac{\theta}{2}}\right)^2 = 2k + 1\sum_{j=1}^{2k} 2(2k+1-j)\cos j\theta.$$

**Theorem 2.19.** *Let $f : \mathbb{C} \to \mathbb{C}$ be a fnction such that*

*i). $f$ is analytic,*

*ii). $f \not\equiv 0$,*

*iii).* $\log f(s) = \sum_{n\in\mathbb{N}} \frac{a_n}{n^s}$ *for* $a_n \geq 0$ *and* $\mathrm{Re}(s) > 1$,

*iv).* $f$ *is analytic on* $\mathrm{Re}(s) = 1, s \neq 1$, *and it has a pole at* $s = 1$ *of order* $e$.

*If* $f(s) = 0$ *on the line* $\mathrm{Re}(s) = 1$, *then the order of zero is at most* $\frac{e}{2}$.

*Proof.* For $e \leq 2k-1$, suppose $f$ has a zero at $f(1+it_0)$ of order $k$. Let us define,

$$g(s) := f(s)^{2k+1} \prod_{j=1}^{2k} f(s+ijt_0)^{2(2k+1-j)} = f(s)^{2k+1} f(s+it_0)^{4k} \cdots .$$

Then at $s = 1$, $f(s)^{2k+1}$ has a pole of order $e(2k+1)$ and $\prod_{j=1}^{2k} f(s+ijt_0)^{2(2k+1-j)}$ has a zero of order $4k$ Note that

$$4k^2 - e(2k+1) \geq 1.$$

Thus $g$ has a zero at 1 of order at least 1, in particular, $g(1) = 0$.

Consider,

$$\begin{aligned}
\log g(\sigma) &= (2k+1)\log f(\sigma) + \sum_{j=1}^{2k} 2(2k+1-j)\log(f(\sigma+ijt_0)), \\
&= (2k+1)\sum_{n\in\mathbb{N}} \frac{a_n}{n^\sigma} + \sum_{j=1}^{2k} 2(2k+1-j)\left(\sum_{n\in\mathbb{N}} \frac{a_n}{n^{\sigma+ijt_0}}\right), \\
&= \sum_{n\in\mathbb{N}} \frac{a_n}{n^\sigma}\left(2k+1+\sum_{j=1}^{2k} 2(2k+i-j)e^{-ijt_0\log(n)}\right). \\
\mathrm{Re}(\log(g(\sigma))) &= \sum_{n\in\mathbb{N}} \frac{a_n}{n^\sigma}\left(2k+1+\sum_{j=1}^{2k} 2(2k+i-j)\cos(jt_0\log(n))\right).
\end{aligned}$$

Using Lemma 2.7, we get,

$$\mathrm{Re}\log(g(\sigma)) = \sum_{n\geq 1} \frac{a_n}{n^\sigma}\left(\frac{\sin\left(k+\frac{1}{2}\right)t_0\log(n)}{\sin\frac{t_0\log(n)}{2}}\right)^2 \geq 0.$$

As $\mathrm{Re}(g(\sigma)) \geq 1$, $|g(\sigma)| \geq 1$. This contradicts that $g$ has a zero of order at least 1 at $1+it_0$. □

**Corollary 2.4.** *For any Dirichlet character* $\chi$ *modulo* $q$ *we have,* $L(s,\chi)$ *is analytic over* $\mathrm{Re}(s) > 1$. *Furthermore* $L(s,\chi) \neq 0$ *on* $\mathrm{Re}(s) = 1, s \neq 1$.

*Proof.* The first assertion is due to Theorem 2.14. Let

$$f(s) := \prod_{\chi \bmod q} L(s, \chi).$$

Then,

$$\log(f(s)) = \sum_{\chi \bmod q} \log L(s, \chi) = \sum_{n,p} \frac{\sum_{\chi \bmod q} \chi(p^n)}{np^{ns}}, \mathrm{Re}(s) > 1.$$

Using Theorem 2.10, we get,

$$\log(f(s)) = \sum_{\substack{n,p \\ p^n \equiv 1 \bmod q}} \frac{\varphi(q)}{np^{ns}}.$$

Note $f$ has a pole of order at most 1 at $\mathrm{Re}(s) = 1$ and its residue is

$$\mathrm{Res}(f(s), 1) = \left(\lim_{s \to 1} (s-1) L(s, \chi_0)\right) \prod_{\substack{\chi \neq \chi_0 \\ \chi \bmod q}} L(1, \chi) = \frac{\varphi(q)}{q} \prod_{\substack{\chi \neq \chi_0 \\ \chi \bmod q}} L(1, \chi).$$

The right hand side is not 0 by Theorem 2.18. □

**Theorem 2.20.** *There are infinitely many primes.*

*Proof.* We have,

$$\zeta(s) = \prod_p \left(1 - \frac{1}{p^s}\right)^{-1}.$$

$$\log \zeta(s) = \sum_{n,p} \frac{1}{np^{ns}} = \sum_p \frac{1}{p^s} + \sum_{n \geq 2} \frac{1}{np^{ns}}.$$

Since $\zeta$ has a pole at 1, so does its log. Observe that at $s = 1$,

$$\sum_p \frac{1}{p^s} + \sum_{n \geq 2} \frac{1}{np^{ns}} \leq \sum_p \sum_{n \geq 2} \frac{1}{p^n} = \sum_p \frac{1}{p^2} \frac{1}{1 - \frac{1}{p}} = \sum_p \frac{1}{p(p-1)} \ll \sum_{n \in \mathbb{N}} \frac{1}{n^2}.$$

Thus $\lim_{s \to 1^+} \sum_p \frac{1}{p^s}$ must be infinity. □

**Lemma 2.8.** *Let $p$ be a prime and $a, q$ be coprimes such that $p \equiv a \bmod q$. Then we have,*

$$\sum_{\chi \bmod q} \chi(p^n) \overline{\chi(a)} = \begin{cases} \varphi(q), & (p^n \equiv a \bmod q), \\ 0, & (\text{otherwise}). \end{cases}$$

*Proof.* Since $\chi(a)$ is a $\varphi(q)$-th root of unity, we have, $\overline{\chi(a)} = \chi(a^{-1})$. Therefore,

$$\begin{aligned}\sum_{\chi \bmod q} \chi(p^n)\overline{\chi(a)} &= \sum_{\chi \bmod q} \chi(p^n)\chi(a^{-1}),\\ &= \sum_{\chi \bmod q} \chi(p^n a^{-1}),\\ &= \begin{cases}\varphi(q), & (p^n a^{-1} \equiv 1 \bmod q),\\ 0, & (\text{otherwise}).\end{cases}\end{aligned}$$

□

**Theorem 2.21** (Dirichlet's Theorem)**.** *Let $a, q$ be coprime. Then $(a + nq)_{n\in\mathbb{N}}$ contains infinitely many primes.*

*Proof.* Motivated by the alternative proof of the existence of infinitely many primes, examine,

$$\sum_{\chi \bmod q} \log L(s,\chi) = \sum_{n,p} \left( \frac{\sum_{\chi \bmod q} \chi(p^n)}{np^{ns}} \right),$$

where $\mathrm{Re}(s) > 1$. Using Theorem 2.10, we have,

$$\sum_{\chi \bmod q} \log L(s,\chi) = \sum_{\substack{n,p\\ p^n \equiv 1 \bmod q}} \frac{\varphi(q)}{np^{ns}} = \sum_{p^n \equiv 1 \bmod q} \frac{\varphi(q)}{p^s} + \sum_{\substack{p,n\geq 2\\ p^n \equiv 1 \bmod q}} \frac{\varphi(q)}{np^{ns}}.$$

Taking the log out we have,

$$\prod_{\chi \bmod q} L(s,\chi) = \exp\left( \varphi(q) \left( \sum_{p^n \equiv 1 \bmod q} \frac{1}{p^s} + \sum_{\substack{p,n\geq 2\\ p^n \equiv 1 \bmod q}} \frac{1}{np^{ns}} \right) \right).$$

Since for $\chi \neq \chi_0, L(1,\chi) \neq 0$ and by Corollary 2.3, we have,

$$\lim_{s\to 1^+} (s-1) \prod_{\chi \bmod q} L(s,\chi) = \frac{\varphi(q)}{q} \prod_{\substack{\chi \bmod q\\ \chi \neq \chi_0}} L(1,\chi).$$

By the same argument from Theorem 2.20, we see,

$$\sum_{\substack{p,n\geq 2\\ p^n \equiv \bmod q}} \frac{1}{np^{ns}} \leq \sum_{n\in\mathbb{N}} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

However,

$$\lim_{s\to 1^+} (s-1) \exp\left( \varphi(q) \left( \sum_{p^n \equiv 1 \bmod q} \frac{1}{p^s} + \sum_{\substack{p,n\geq 2\\ p^n \equiv 1 \bmod q}} \frac{1}{np^{ns}} \right) \right) \neq 0$$

which is only possible when

$$\lim_{s\to 1^+} \sum_{p^n \equiv 1 \bmod q} \frac{1}{p^s} = \infty.$$

Together with Lemma 2.8, we derived the general statement. □

**Theorem 2.22** (Bertrand's Postulate)**.** *For a sufficiently large $n \in \mathbb{N}$, there is a prime number inbetween $n$ and $2n$.*

*Proof.* Consider the second Chebyschev function, and by Remark 2.1, we have,

$$T(x) := \sum_{l\le x} \psi\left(\frac{x}{l}\right) = \sum_{l\le x}\sum_{n\le \frac{x}{l}} \Lambda(n) = \sum_{\substack{l,n\\ ln\le x}} \Lambda(n).$$

That is

$$\sum_{l\le x} \psi\left(\frac{x}{l}\right) = \sum_{m\le x}\sum_{d|m} \Lambda(d).$$

By Theorem 2.3,

$$T(x) = \sum_{m\le x} \log(m).$$

Using the definition,

$$\begin{aligned} T(x) - 2T\left(\frac{x}{2}\right) &= \sum_{l\le x} \psi\left(\frac{x}{l}\right) - 2\sum_{2l\le x} \psi\left(\frac{x}{2l}\right), \\ &= \sum_{l\le x} (-1)^{l+1} \psi\left(\frac{x}{l}\right), \\ &= \psi(x) - \psi\left(\frac{x}{2}\right) + \cdots . \end{aligned}$$

Again by Remark 2.1

$$x \le y \Rightarrow \psi(x) \le \psi(y).$$

In particular,

$$\psi(x) - \psi\left(\frac{x}{2}\right) \le \sum_{l\le x} (-1)^{l+1} \psi\left(\frac{x}{l}\right) \le \psi(x) - \psi\left(\frac{x}{2}\right) + \psi\left(\frac{x}{3}\right).$$

Recall Proposition 2.7, consider

$$A(x) = \sum_{n\le x} 1 = \lfloor x \rfloor,$$

and since $\log(x)$ is continuously differentiable on $[1, x]$, we have,

$$T(x) = \lfloor x \rfloor \log(x) - \int_1^x \frac{\lfloor x \rfloor}{x} dx = x\log(x) - x + O(\log(x)).$$

Therefore,

$$\begin{aligned} T(x) - 2T\left(\frac{x}{2}\right) &= x\log(x) - x + O(\log(x)) - 2\left(\frac{x}{2}\log\left(\frac{x}{2}\right) - \frac{x}{2} + O(\log(x))\right), \\ &= \log 2 \cdot x + O(\log(x)). \end{aligned}$$

Combining with the previous result, we get,

$$\psi(x) - \psi\left(\frac{x}{2}\right) \leq (\log 2)x + O(\log(x)) \leq \psi(x) - \psi\left(\frac{x}{2}\right) + \psi\left(\frac{x}{3}\right).$$

Generalizing this we obtain,

$$\psi\left(\frac{x}{2^{n-1}}\right) - \left(\frac{x}{2^n}\right) \leq \log(2)\frac{x}{2^{n-1}} + O(\log(x)).$$

By induction, we obtain,

$$\begin{aligned} \psi(x) - \psi\left(\frac{x}{2^n}\right) &\leq (\log 2)x\left(1 + \frac{1}{2} + \cdots + \frac{1}{2^{n-1}}\right) + O(n\log(x)), \\ &\leq (2\log 2)x + O(n\log(x)). \end{aligned} \tag{$\psi 1$}$$

Take $n$ to be the maximal such that $2^n \leq x$ that is $\lfloor \frac{x}{2^n} \rfloor = 1$. Then, $\psi\left(\frac{x}{2^n}\right) = 0$, thus,

$$\psi(x) \leq (2\log 2)x + O((\log(x))^2).$$

On the other hand,

$$\begin{aligned} \psi(x) - \psi\left(\frac{x}{2}\right) + \psi\left(\frac{x}{3}\right) &\geq (\log 2)x + O(\log(x)), \\ \Rightarrow \psi(x) - \psi\left(\frac{x}{2}\right) &\geq (\log(2))x + O(\log(x)) - \psi\left(\frac{x}{3}\right) \end{aligned}$$

Using Inequality ($\psi 1$), we get,

$$\begin{aligned} \psi(x) - \psi\left(\frac{x}{2}\right) &\geq (\log 2)x + O(\log(x)) - (2\log 2)\frac{x}{3} + O((\log(x))^2), \\ &= (\log 2)\frac{x}{3} + O((\log(x))^2). \end{aligned} \tag{$\psi 2$}$$

Using Remark 2.9, that is

$$\psi(x) - \theta(x) = O(\sqrt{x}\log(x)),$$

together with Inequalities ($\psi 1$) and ($\psi 2$), we obtain,

$$\theta(x) - \theta\left(\frac{x}{2}\right) \geq (\log 2)\frac{x}{3} + O(\sqrt{x}\log(x)).$$

Thus the right hand side is greater than 0 if $x$ is sufficiently large with coeffi of the inequality, therefore by definition of $\theta$, we have,

$$\theta(x) - \theta\left(\frac{x}{2}\right) = \sum_{\frac{x}{2} \leq p \leq x} \log(p) > 0.$$

□

**Theorem 2.23** (Chebyschev)**.**

*1).* $\theta(n) \leq 2n \log 2$,

*2).* $\psi(n) \geq 2n \log 2$,

*3). there exists* $A, B > 0$ *such that for sufficiently large* $n$*, we have*

$$\frac{Bx}{\log x} \leq \pi(x) \leq \frac{Ax}{\log x}.$$

*Proof.* □

## 2.5 Ikehara-Wiener Theorem and Its Applications

**Theorem 2.24** (Ikehara-Wiener)**.** *Let* $(b_n)_{n\in\mathbb{N}}$ *be a sequence of non-negative numbers and set,*

$$f(s) = \sum_{n\in\mathbb{N}} \frac{b_n}{n^s}.$$

*Suppose*

*i). the series converges abslutely for* $\mathrm{Re}(s) > 1$,

*ii).* $f$ *has an analytic continuation to* $\mathrm{Re}(s) = 1$ *except* $s = 1$,

*iii).* $f$ *has a simple pole at* $s = 1$ *with residue* $R \geq 0$.

*Then we have,*

$$\sum_{n\leq x} b_n = Rx + o(x).$$

*That is*

$$\lim_{x\to\infty} \frac{\sum_{n\leq x} b_n}{x} = R.$$

**Lemma 2.9.** *Let* $\chi$ *be a Dirichlet character modulo* $q$*, then*

$$L(s,\chi)^{-1} = \sum_{n\in\mathbb{N}} \frac{\chi(n)\mu(n)}{n^s}.$$

*Proof.* We immitate the proof of Theorem 2.5. Using Theorem 2.1,

$$L(s,\chi)\left(\sum_{n\in\mathbb{N}} \frac{\mu(n)\Lambda(n)}{n^s}\right) = \sum_{t\in\mathbb{N}}\sum_{n|t} \frac{\chi\left(\frac{t}{n}\right)\chi(n)\mu(n)}{t^s} = \sum_{t\in\mathbb{N}} \chi(t) \sum_{n|t} \mu(n) = 1.$$

□

**Theorem 2.25.** *Let $\chi$ be a character then, for* $\mathrm{Re}(s) > 1$,

$$-\frac{L'(s,\chi)}{L(s,\chi)} = \sum_{n\in\mathbb{N}} \frac{\chi(n)\Lambda(n)}{n^s}.$$

*Proof.* By Lemma 2.9 and Proposition 2.1

$$\begin{aligned}
-\frac{L'(s,\chi)}{L(s,\chi)} &= \left(\sum_{t\in\mathbb{N}} \frac{\chi(n)\log(t)}{t^s}\right)\left(\sum_{n\in\mathbb{N}} \frac{\mu(n)\chi(n)}{n^s}\right), \\
&= \sum_{t\in\mathbb{N}} \chi(t) \sum_{n|t} \frac{\mu\left(\frac{t}{n}\right)\log(n)}{t^s}, \\
&= \sum_{n\in\mathbb{N}} \frac{\chi(n)\Lambda(n)}{n^s}.
\end{aligned}$$

□

**Definition 2.15.**

$$\psi(x,q,a) = \sum_{\substack{n\le x \\ n\equiv a \bmod q}} \Lambda(n).$$

**Proposition 2.8.** *As $x \to \infty$, we have,*

$$\psi(x,q,a) \sim \frac{x}{\varphi(q)}.$$

*Proof.* Consider

$$\sum_{\chi \bmod q} \overline{\chi}(a) \left(\sum_{n\in\mathbb{N}} \frac{\chi(n)\Lambda(n)}{n^s}\right) = \sum_{n\in\mathbb{N}} \frac{\Lambda(n)}{n^s} \left(\sum_{\chi \bmod q} \chi(a^{-1}n)\right)$$

Using Lemma 2.8, we get,

$$-\sum_{\chi \bmod q} \overline{\chi}(a) \frac{L'(s,\chi)}{L(s,\chi)} = \varphi(q) \sum_{n\equiv a \bmod q} \frac{\Lambda(n)}{n^s}.$$

Recall from Corollary 2.3, we have,

$$\mathrm{Res}(L(s,\chi_0),1) = \frac{\varphi(q)}{q}.$$

And by Theorem 2.14, $L(s,\chi)$ is analytic at $s = 1$ for $\chi \neq \chi_0$. Also using Theorem 2.16, we have

$$\chi \neq \chi_0 \Rightarrow \frac{L'(s,\chi)}{L(s,\chi)} \text{ is analytic for } \mathrm{Re}(s) \ge 1.$$

Furthermore, we have Theorem 2.17, we have

$$\operatorname{Res}\left(\frac{\zeta'(s)}{\zeta(s)}, 1\right) = -1.$$

Combining these, we get,

$$\lim_{s\to 1}(s-1)\left(-\frac{1}{\varphi(q)}\sum_{\substack{\chi \bmod q\\ \chi\neq\chi_0}}\frac{L'(s,\chi)}{L(s,\chi)} - \frac{1}{\varphi(q)}\frac{L'(s,\chi_0)}{L(s,\chi_0)}\right) = \frac{1}{\varphi(q)}.$$

Now using Theorem 2.24, and apply $b_n = \Lambda(n)$ $f = \varphi(q)\sum_{n\equiv a \bmod q}\frac{\Lambda(n)}{n^s}$, we get,

$$\psi(x,q,a) = \sum_{n\leq x}\Lambda(n) = \frac{1}{\varphi(q)}x + o(x).$$

□

## 2.6 $L(s,\chi) \neq 0$ for Quadratic Characters

**Lemma 2.10.** *Let $f := \sum_{d|n}\chi(d)$, where $\chi$ is a character, then,*

$$\forall n \in \mathbb{N}, n \text{ is a perfect square} \Rightarrow f(n) \geq 0, f(n) \geq 1.$$

*Proof.* Recall $n = \prod_{p|n} p^{\alpha(p)}$. Using this we have,

$$\begin{aligned}\sum_{d|n}\chi(d) &= \prod_{p|n}\left(\sum_{k=0}^{\alpha(p)}\chi(p)^k\right),\\ &= \begin{cases}1 \quad \chi(p)=0,\\ \prod_{p|n}(1+\alpha(p)) \quad \chi(p)=1,\\ \prod_{p|n}\left(\frac{(1-(-1)^{\alpha(p)+1})}{2}\right) \quad \chi(p)=-1.\end{cases}\end{aligned}$$

Note that if $\alpha(p)$ are all even for $p|n$ (ie. $n$ is a perfect square), we have the last part of the cases equals to 1. Thus we have $f(n) \geq 1$. □

**Theorem 2.26.** *Let $f(n) = \sum_{d|n}\chi(d)$ for some character. Then we have,*

$$\sum_{n\leq x}\frac{f(n)}{\sqrt{n}} = 2\sqrt{x}L(1,\chi) + o(1).$$

*Proof.*

$$\begin{aligned}
\sum_{n\le x}\frac{f(n)}{\sqrt{n}} &= \sum_{n\le x}\left(\frac{\sum_{d|n}\chi(d)}{\sqrt{n}}\right),\\
&= \sum_{\substack{d,e\\ de\le x}}\frac{\chi(d)}{\sqrt{de}},\\
&= \sum_{\substack{d,e\le x\\ d\le\sqrt{x}}}\frac{\chi(d)}{\sqrt{de}}+\sum_{\substack{de\le x\\ d>\sqrt{x}}}\frac{\chi(d)}{\sqrt{de}},\\
&= \sum_{d\le\sqrt{x}}\frac{\chi(d)}{\sqrt{d}}\left(\sum_{e\le\frac{x}{d}}\frac{1}{\sqrt{e}}\right)+\sum_{e\le\sqrt{x}}\frac{1}{\sqrt{e}}\left(\sum_{\sqrt{x}<d\le\frac{x}{e}}\frac{\chi(d)}{\sqrt{(}d)}\right).
\end{aligned}$$

Recall that from Proposition 2.7,

$$\sum_{m\le x}\frac{1}{\sqrt{m}} = 2\sqrt{x}+B+o\left(\frac{1}{\sqrt{x}}\right),$$

where $B$ is some constant as $x\to\infty$.

Let $x,y\in\mathbb{R}$, such that $x<y$, we have,

$$\begin{aligned}
\sum_{x<d\le y}\frac{\chi(d)}{\sqrt{d}} &= \sum_{d\le y}\frac{\chi(d)}{\sqrt{d}}-\sum_{d\le x}\frac{\chi(d)}{\sqrt{(}d)},\\
\sum_{d\le x}\frac{\chi(d)}{\sqrt{d}} &= \frac{\sum_{d\le x}\chi(d)}{\sqrt{x}}+\frac{1}{2}\int_1^x\frac{\sum_{d\le t}\chi(d)}{t^{\frac{1}{2}}},\\
&= O\left(\frac{1}{\sqrt{x}}\right).
\end{aligned}$$

Using these equations, we have,

$$\begin{aligned}
\sum_{e\le\sqrt{x}}\frac{1}{\sqrt{e}}\left(\sum_{\sqrt{x}<d\le\frac{x}{e}}\frac{\chi(d)}{\sqrt{(}d)}\right) &= \sum_{e\le x}\frac{1}{\sqrt{e}}\left(O\left(\frac{1}{x^{\frac{1}{r}}}\right)\right),\\
&= \left(O\left(\frac{1}{x^{\frac{1}{r}}}\right)\right)\sum_{e\le x}\frac{1}{\sqrt{e}},\\
\sum_{e\le\sqrt{x}}\frac{1}{\sqrt{x}} &\ll \int_1^{\sqrt{x}}\frac{1}{\sqrt{t}}dt = x^{\frac{1}{4}}.
\end{aligned}$$

Thus we conclude,

$$\sum_{e\le\sqrt{x}}\frac{1}{\sqrt{e}}\left(\sum_{\sqrt{x}<d\le\frac{x}{e}}\frac{\chi(d)}{\sqrt{(}d)}\right) = O(1).$$

We also have,

$$\sum_{d\le\sqrt{x}}\frac{\chi(d)}{\sqrt{d}}\left(\sum_{e\le\frac{x}{d}}\frac{1}{\sqrt{e}}\right)=\sum_{d\le\sqrt{x}}\frac{\chi(d)}{\sqrt{d}}\left(2\sqrt{\frac{x}{d}}+B+o\left(\sqrt{\frac{d}{x}}\right)\right),$$

$$=2\sqrt{x}\sum_{d\le\sqrt{x}}\frac{\chi(d)}{d}+B\sum_{d\le\sqrt{x}}\frac{\chi(d)}{\sqrt{d}}+o\left(\frac{1}{\sqrt{x}}\sum_{d\le\sqrt{x}}\chi(d)\right).$$

$$2\sqrt{x}\sum_{d\le\sqrt{x}}\frac{\chi(d)}{d}=2\sqrt{x}\left(\sum_{d\ge1}\frac{\chi(d)}{d}-\sum_{d>\sqrt{x}}\frac{\chi(d)}{d}\right),$$

$$=2\sqrt{x}L(1,\chi)-2\sqrt{x}\sum_{d>\sqrt{x}}\frac{\chi(d)}{d},$$

$$=2\sqrt{x}L(1,\chi)-2\sqrt{x}o\left(\frac{1}{\sqrt{x}}\right),$$

$$=2\sqrt{x}L(1,\chi)+o(1).B\sum_{d\le\sqrt{x}}\frac{\chi(d)}{\sqrt{d}}\le o\left(\frac{B}{x^{\frac{1}{4}}}\right)=o(1).$$

$$\sum_{d\le\sqrt{x}}\chi(d)\le q?=o(1).$$

□

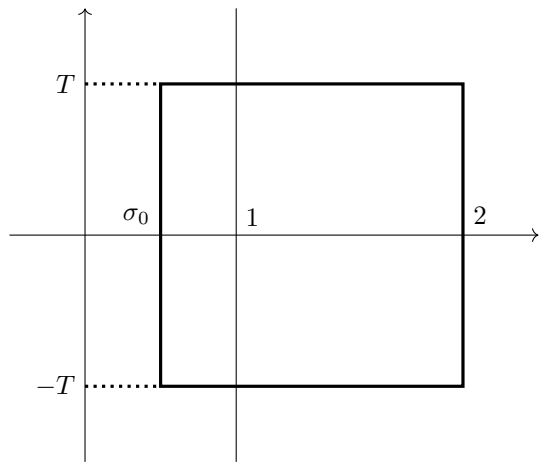
**Corollary 2.5.** *$L(1,\chi)\neq 0$ for a quadratic character $\chi$.*

*Proof.* To derive a contradiction, assume $L(1,\chi)=0$. Then, from Theorem 2.26 and Lemma 2.10

$$o(1)=\sum_{n\le x}\frac{f(n)}{\sqrt{n}}\ge\sum_{\substack{n\le x\\ n\text{ is a square}}}\frac{1}{\sqrt{n}}=\sum_{m\le\sqrt{x}}\frac{1}{m}=\log\sqrt{x}+o(1).$$

Thus we obtain $o(1)\ge\log\sqrt{x}+o(1)$ which is a cotradiction. □

**Lemma 2.11.** *Consider the following rectangle,*




*where $\sigma_0 = 1 - \log T$. On the boundary of the rectangle above, we have,*

1. $|\zeta(s)| = O(\log T), T \to \infty$,
2. $|\zeta'(s)| = O((\log T)^2)$.

*Proof.* Recall that

$$\zeta'(s) = \sum_{n=1}^{\infty} \frac{-\log n}{n^s}.$$

The first statement is due to the first derivative of $\zeta$. and it is assgined as an exercise. For the second part, for $\mathrm{Re}(s), \sigma > 1$, we have,

$$\begin{aligned}\zeta(s) &= \sum_{n\geq 1} \frac{1}{n^s}, \\ &= \sum_{n\leq T} \frac{1}{n^s} + \sum_{n>T} \frac{1}{n^s},\end{aligned}$$

Using Abel's summation formula by setting $a_n = 1, f(n) = \frac{1}{n^s}$, we have,

$$\begin{aligned}\zeta(s) = \sum_{n=1}^{\infty} &= -s \int_1^{\infty} \frac{[t]}{t^{s+1}} dt, \\ \sum_{n\leq T} \frac{1}{n^s} &= \frac{[T]}{T^s} - s \int_1^{T} \frac{[t]}{t^{s+1}} dt.\end{aligned}$$

Using this, we get,

$$\begin{aligned}\zeta(s) &= \sum_{n\leq T} \frac{1}{n^s} - \frac{\lfloor T \rfloor}{T^s} + s \int_1^{\infty} \frac{\lfloor t \rfloor}{t^{s+1}} dt, \\ &= \sum_{n\leq T} \frac{1}{n^s} + \frac{T^{1-s}}{s-1} + \frac{\{T\}}{T^s} - s \int_T^{\infty} \frac{\{u\}}{u^{s+1}} du.\end{aligned}$$

Note that the right hand side is analytic where $\mathrm{Re}(s) > 0$ and $s \neq 1$. Now we will estimate the last equation above on the boundary of the rectangle above.

$$\begin{aligned}
\left|\sum_{n\leq T}\frac{1}{n^s}\right| &\ll \int_1^T \frac{du}{u^{\mathrm{Re}(s)}},\\
&\leq \int_1^T \frac{du}{u^{\sigma_0}},\\
&= \frac{T^{1-\sigma_0}}{1-\sigma_0},\\
&\ll \log T.
\end{aligned}$$

Observe that ,

$$1-\sigma_0 = \frac{1}{\log T}, T^{1-\sigma_0} = T^{\frac{1}{\log T}} = \exp(\log T^{\frac{1}{\log T}}) = \exp(1) = e.$$

We also have,

$$\begin{aligned}
\left|\frac{T^{1-s}}{s-1}\right| &= \frac{T^{1-\mathrm{Re}(s)}}{|s-1|},\\
&\leq \frac{T^{1-\sigma_0}}{|s-1|},\\
&\leq \frac{e}{|s-1|},\\
&\leq \frac{e}{\sigma_0-1} = e\log T.
\end{aligned}$$

Also consider,

$$\left|\frac{\{T\}}{T}\right| \leq \frac{1}{T^{\mathrm{Re}(s)}} \leq 1.$$

Finally we have,

$$\begin{aligned}
\left|-s\int_T^\infty \frac{\{u\}}{u^{s+1}}du\right| &\leq |s|\int_T^\infty \frac{du}{u^{\mathrm{Re}(s)+1}},\\
&= -\frac{|s|}{\mathrm{Re}(s)u^{\mathrm{Re}(s)}}\Bigg|_T^\infty,\\
&= \frac{|s|}{\mathrm{Re}(s)T^{\mathrm{Re}(s)}},\\
&\leq \frac{|s|}{T^{\mathrm{Re}(s)}},\\
&\leq \frac{\sqrt{2^2+T^2}}{T^{\sigma_0}},\\
&\ll \frac{T}{T^{\sigma_0}} = T^{1-\sigma_0} = e.
\end{aligned}$$

For the second part, for $\mathrm{Re}(s) > 0, s \neq 1$,

$$\zeta'(s) = \sum_{n \leq T} \frac{-\log n}{n^s} + \frac{T^{1-s}(-\log T)}{s-1} + T^{1-s}\frac{-1}{(s-1)^2}$$
$$+ \frac{\{T\}\log T}{T^s} - \int_T^\infty \frac{\{u\}}{u^{s+1}}du + s\int_T^\infty \frac{\{u\}\log u}{u^{s+1}}du.$$

This is obtained simply differentiating the equation,

$$\zeta(s) = \sum_{n \leq T} \frac{1}{n^s} + \frac{T^{1-s}}{s-1} - \frac{\{T\}}{T^s} - s\int_T^\infty \frac{\{u\}}{u^{s+1}}du.$$

The statement can be shown using all the estimates obtained to show the first part. □

**Theorem 2.27** (Complex Mean Value Theorem)**.** *Let $f : \Omega \to \mathbb{C}$ be an analytic, where $\Omega$ is a convex open set. Let $a, b \in \Omega$, then there exists $z_1, z_2 \in (a, b)$ such that*

$$\mathrm{Re}(f'(z_1)) = \mathrm{Re}\left(\frac{f(b)-f(a)}{b-a}\right), \mathrm{Im}(f'(z_2)) = \mathrm{Im}\left(\frac{f(b)-f(a)}{b-a}\right).$$

**Theorem 2.28.** *There exist some constants $c_1, c_2$ such that*

$$1 - \frac{c_1}{(\log T)^9} \leq \sigma \leq 2 \Rightarrow |\zeta(s)| > \frac{c_2}{(\log T)^7},$$

*where $1 < |\mathrm{Im}(s)| \leq T$.*

*Proof.* We set $z = \sigma + it$ for $\sigma, t \in \mathbb{R}$. Recall Lemma 2.6,

$$|\zeta(\sigma)^3\zeta(\sigma+it)^4\zeta(\sigma+2it)| \geq 1, \sigma > 1.$$

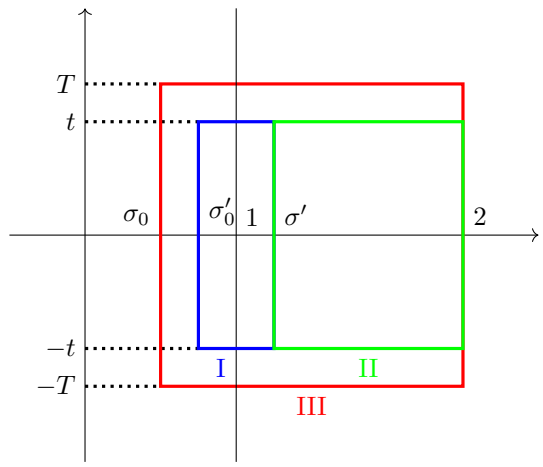
Thus we have,

$$|\zeta(\sigma+it)|^4 \geq |\zeta(\sigma)|^{-3}|\zeta(\sigma+2it)|^{-1}, \tag{$*$}$$

Consider

$$\sigma_0' = 1 - \frac{c_1}{(\log T)^9}, \sigma' = 1 + \frac{c_1}{(\log T)^9}$$

$c_1$ is a positive constant which will be adjusted later.




Fix the domain II where $\sigma' \leq \sigma \leq 2$, $1 \leq |t| \leq T$.

$$\zeta(s) = \frac{s}{s-1} - s\int_1^\infty \frac{\{u\}}{u^{s+1}}du, \mathrm{Re}(s) > 0, s \neq 1.$$
$$\zeta(\sigma) = \frac{\sigma - 1 + 1}{\sigma - 1} - \sigma\int_1^\infty \frac{\{u\}}{u^{\sigma+1}}du.$$
$$\ll \frac{1}{\sigma - 1} \text{ as } \sigma \to 1^+,$$
$$\Rightarrow \zeta(\sigma)^{-1} >> (\sigma - 1), \sigma \to 1^+$$

On the domain II, we have, $\sigma_0' < \sigma \leq 2, 1 \leq |t| \leq T$. Using Lemma 2.11,

$$|\zeta(\sigma + 2it)| = O(\log T).$$

Substituting these to Equation $(*)$, we have,

$$|\zeta(\sigma + it)|^4 >> (1-\sigma)^3(\log T)^{-1},$$
$$>> \frac{c_1^3}{(\log T)^{27}}(\log T)^{-1}.$$
$$\Rightarrow |\zeta(\sigma + it)| >> \frac{c_1^{\frac{3}{4}}}{(\log T)^7}, \tag{D1}$$

in the domain II.

On the domain I, similarly use Theorem 2.27 with

$$a = \sigma' + it, b = \sigma + it,$$

where,

$$1 - \frac{c_1}{(\log T)^9} \leq \sigma \leq \sigma'.$$

We have,

$$\frac{\zeta(\sigma' + it) - \zeta(\sigma + it)}{\sigma' - \sigma} = \mathrm{Re}(\zeta(z_1)) + \mathrm{Im}(\zeta(z_2)).$$

Furthermore,

$$\zeta(\sigma' + it) - \zeta(\sigma + it) = O((\sigma' - \sigma)(\log T)^2).$$

By the definition of $\sigma'$, we have,

$$\zeta(\sigma' + it) = \zeta(\sigma + it) + O\left(\frac{c_1}{(\log T)^7}\right). \tag{D2}$$

Combining Equations $(D1)$ and $(D2)$, there are some positive constants $A_1, A_2$ such that

$$\begin{aligned} |\zeta(\sigma + it)| &\geq |\zeta(\sigma' + it)| - \frac{A_1 c_1}{(\log T)^7}, \\ &\geq \frac{A_2 c_1^{\frac{3}{4}}}{(\log T)^7} - \frac{A_1 c_1}{(\log T)^7}. \end{aligned}$$

Now take $c_1$ sufficiently small so that in the region,

$$\sigma_0' \leq \sigma \leq \sigma', 1 \leq |t| \leq T,$$

we have,

$$|\zeta(\sigma + it)| \geq \frac{c_2}{(\log T)^7}.$$

Again combining this with Inequality $(D2)$, we obtain the statement. □

**Corollary 2.6.** *There exists some constant $c$ such that*

$$\frac{\zeta'(s)}{\zeta(s)} = O((\log T)^9),$$

*for* $1 - \frac{c}{(\log T)^9} \leq \mathrm{Re}(s) \leq 2$ *and* $1 \leq |\mathrm{Im}(s)| \leq T$.

**Notation 2.3.** *Let $c \in \mathbb{R}$ and $f : \mathbb{C} \supseteq \Omega \to \mathbb{C}$. Then we denote,*

$$\int_{(c)} f(s)ds = \lim_{T \to \infty} \int_{c-iT}^{c+iT} f(s)ds.$$

**Lemma 2.12.**

$$\frac{1}{2\pi i} \int_{(c)} y^s \frac{ds}{s} = \begin{cases} 0, & (0 < y < 1), \\ \frac{1}{2}, & (y = 1), \\ 1, & (y > 1). \end{cases}$$

**Lemma 2.13.** *Set,*

$$f(s) := \sum_{n=1}^{\infty} \frac{a_n}{n^s}.$$

*Suppose this is absolutely convergence in* $\mathrm{Re}(s) > c - \varepsilon$ *for some* $c \in \mathbb{R}, \varepsilon > 0$ *and $x$ is not an integer, then*

$$\sum_{n<x} a_n = \frac{1}{2\pi i} \int_{(c)} f(s) \frac{x^s}{s} ds.$$

*Proof.* Since $f(s)$ is a limit of a continuous function series which is absolutely convergent on $\mathrm{Re}(s) > c - \varepsilon$, thus this is uniformly convergent in a compact subset of its domain of convergence. Thus we can exchange the integral and the sum namely,

$$\frac{1}{2\pi i}\int_{(c)}\left(\sum_{n=1}^{\infty}\frac{a_n}{n^s}\right)\frac{x^s}{s}ds = \frac{1}{2\pi i}\sum_{n=1}^{\infty}a_n\int_{(c)}\left(\frac{x}{n}\right)^s\frac{ds}{s}.$$

By Lemma 2.12, and $\frac{x}{n} > 1 \Leftrightarrow n < x$, we derive the statement. □

**Corollary 2.7.** *Let $c > 1$ and $x$ be a non-integer, then we have,*

$$\psi(x) = \frac{1}{2\pi i}\int_{(c)} -\frac{\zeta'(s)}{\zeta(s)}\frac{x^s}{s}ds.$$

*Proof.* From Remark 2.1,

$$\psi(x) = \sum_{d\le x}\Lambda(d),$$

and by Theorem 2.6 set

$$f(s) = -\frac{\zeta'(s)}{\zeta(s)} = \sum_{n=1}^{\infty}\frac{\Lambda(n)}{n^s},$$

Thus applying Lemma 2.13, we get the statement. □

**Theorem 2.29.** *There is a positive constant $c' > 0$ such that*

$$\psi(x) = x + O(x\exp(-c'\log(x))).$$

*Proof.* Let $a = 1 + \frac{c}{\log T}$ for some $c$ where $T > 1$ is sufficiently large and will be chosen later. By Corollary 2.7, we have,

$$\psi(x) = \frac{1}{2\pi i}\int_{a-iT}^{a+iT} -\frac{\zeta'(s)}{\zeta(s)}\frac{x^s}{s}ds + \frac{1}{2\pi i}\int_{(a)} -\frac{\zeta'(s)}{\zeta(s)}\frac{x^s}{s}ds - \frac{1}{2\pi i}\int_{a-iT}^{a+iT} -\frac{\zeta'(s)}{\zeta(s)}\frac{x^s}{s}ds.$$

**Claim 1.**

$$\left|\frac{1}{2\pi i}\int_{(a)} -\frac{\zeta'(s)}{\zeta(s)}\frac{x^s}{s}ds - \frac{1}{2\pi i}\int_{a-iT}^{a+iT} -\frac{\zeta'(s)}{\zeta(s)}\frac{x^s}{s}ds\right| < \begin{cases} x^a\min\{1, T^{-1}, |\log x|^{-1}\}, & (x \ne 1), \\ \frac{a}{T}, \quad (x = 1). \end{cases}$$

<u>Proof:</u> Exercise. ■

Let

$$\psi(x) = \underbrace{\frac{1}{2\pi i}\int_{a-iT}^{a+iT} -\frac{\zeta'(s)}{\zeta(s)}\frac{x^s}{s}ds}_{=I_1} + \underbrace{\frac{1}{2\pi i}\int_{(a)} -\frac{\zeta'(s)}{\zeta(s)}\frac{x^s}{s}ds - \frac{1}{2\pi i}\int_{a-iT}^{a+iT} -\frac{\zeta'(s)}{\zeta(s)}\frac{x^s}{s}ds}_{=I_2}.$$

Again using Corollary 2.7 and the claim, we have,

$$\begin{aligned}
I_2 &= \frac{1}{2\pi i}\left(\int_{(a)} - \int_{a-iT}^{a+iT}\right)\left(\sum_{n=1}^{\infty}\frac{\Lambda(n)}{n^s}\right)\frac{x^s}{s}ds,\\
&= \frac{1}{2\pi i}\sum_{n=1}^{\infty}\Lambda(n)\left(\left(\int_{(a)} - \int_{a-iT}^{a+iT}\right)\left(\frac{x}{n}\right)^s\frac{ds}{s}\right),\\
&= \sum_{n=1}^{\infty}\Lambda(n)O\left(\left(\frac{x}{n}\right)^a\min\left(1, T^{-1}\left|\log\left(\frac{x}{n}\right)\right|^{-1}\right)\right),\\
&= O\left(\sum_{n=1}^{\infty}\Lambda(n)\left(\frac{x}{n}\right)^a\min\left(1, T^{-1}\left|\log\left(\frac{x}{n}\right)\right|^{-1}\right)\right).
\end{aligned}$$

Fix $x$ and let us consider two ranges $\left[\frac{x}{2}, \frac{3x}{2}\right]$ and $\left(-\infty, \frac{x}{2}\right) \cup \left(\frac{3x}{2}, \infty\right)$. If $n$ sits in the second range

$$\frac{\log\frac{n}{x}}{\frac{n}{x}-1} \geq 2\log\frac{3}{2}.$$

And if $n$ sits in the first interval, we have,

$$\left|\log\frac{x}{n}\right| \ll \left|\frac{n}{x}-1\right|^{-1}.$$

Using this we have,

$$\begin{aligned}
\sum_{n\in\left[\frac{x}{2},\frac{3x}{2}\right]}\Lambda(n)\left(\frac{x}{n}\right)^a\min\left(1, T^{-1}\left|\log\left(\frac{x}{n}\right)\right|^{-1}\right) &\ll \sum_{n\in\left[\frac{x}{2},\frac{3x}{2}\right]}\Lambda(n)\left(\frac{x}{n}\right)^a T^{-1}\left|\frac{n}{x}-1\right|^{-1}\\
&\ll \frac{\log x}{T}\sum_{n\in\left[\frac{x}{2},\frac{3x}{2}\right]}\frac{1}{\left|\frac{n}{x}-1\right|}.
\end{aligned}$$

Observe that

$$\begin{aligned}
\sum_{n\in\left[\frac{x}{2},x\right)}\frac{1}{\frac{n}{x}+1} &\ll \int_{\frac{x}{2}}^{\lfloor x\rfloor}\frac{dt}{1-\frac{t}{x}},\\
\sum_{n\in\left(x,\frac{3x}{2}\right]}\frac{1}{\frac{n}{x}-1} &\ll \int_{\lceil x\rceil}^{\frac{3x}{2}}\frac{dt}{\frac{t}{x}-1}.\\
\Rightarrow \sum_{n\in\left[\frac{x}{2},x\right)}\frac{1}{\frac{n}{x}+1} + \sum_{n\in\left(x,\frac{3x}{2}\right]}\frac{1}{\frac{n}{x}+1} &\ll x\log x.
\end{aligned}$$

Thus we conclude, as $x \to \infty$,

$$\sum_{n\in\left[\frac{x}{2},\frac{3x}{2}\right]}\frac{1}{\left|\frac{n}{x}-1\right|} \ll \frac{x(\log x)^2}{T}.$$

Let us turn our focus to the second range. We have $\left|\log \frac{x}{n}\right| > \log \frac{3}{2}$. Thus

$$\sum_{n<\frac{x}{2},\frac{3x}{2}<n} \Lambda(n)\left(\frac{x}{n}\right)^a \min\left(1, T^{-1}\left|\log\left(\frac{x}{n}\right)\right|^{-1}\right) \ll \frac{x^a}{T}\sum_{n<\frac{x}{2},\frac{3x}{2}<n}\frac{\Lambda(n)}{n^a},$$
$$\leq \frac{x^a}{T}\sum_{n=1}^{\infty}\frac{\Lambda(n)}{n^a}.$$

Recall that we have,

$$-\frac{\zeta'(a)}{\zeta(a)} = \sum_{n=1}^{\infty}\frac{\Lambda(n)}{n^a}.$$

Using Theorem 2.17, we see, as $a \to 1^+$,

$$\left|\frac{\zeta'(a)}{\zeta(a)}\right| \ll \frac{1}{a-1}.$$

Using Corollary 2.6, we see,

$$\frac{x^a}{T}\frac{1}{a-1} \ll \frac{x^a}{T}(\log T)^9.$$

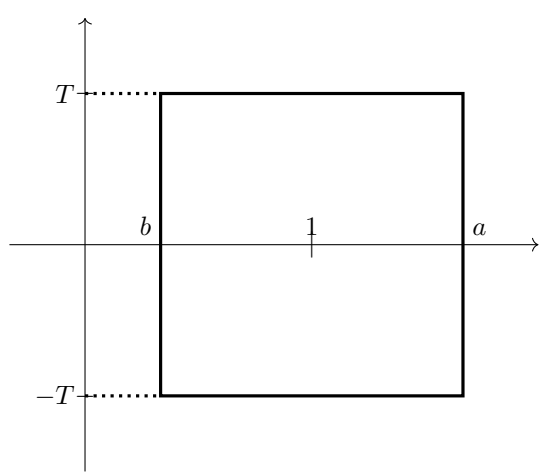
Combining all, we derive,

$$\psi(s) = \frac{1}{2\pi i}\int_{a-iT}^{a+iT}\frac{-\zeta'(s)}{\zeta(s)}\frac{x^s}{s}ds + O\left(\frac{x(\log x)^2}{T} + \frac{x^a}{T}(\log T)^9\right).$$

Recall Cauchy's Residue theorem, suppose $f : \Omega \to \mathbb{C}$ is meromorphic where $\Omega$ is simply connected with poles $a_i, 1 \leq i \leq n$ poles in $U \subseteq \Omega$ where the boundary of $U$ is a simple closed curve $\gamma$. Then,

$$\frac{1}{2\pi i}\int_{\gamma} f(s)ds = \sum_{i=1}^{n}\mathrm{Res}(f, a_i) + \text{constant}\,.$$

Consider the following closed path $R_T$ which we traverse counterclockwise.



where $a = 1 + \frac{c}{(\log(T))^9}, b = 1 - \frac{c}{(\log(T))^9}$ $\mathrm{Re}(s) > 0, s \neq 1$. We claim that $\frac{\zeta'(s)}{\zeta(s)}$ can be analytically continued to $\mathrm{Re}(s) \geq b$ except simple pole at $s = 1$ with residue $-1$. By Theorem 2.17 and Theorem 2.28, we obtain

$$\begin{aligned}\frac{1}{2\pi i}\int_{R_T} \frac{-\zeta'(s)}{\zeta(s)}\frac{x^s}{s}ds &= \mathrm{Res}_{s=1}\left(\frac{-\zeta'(s)}{\zeta(s)}\frac{x^s}{s}\right),\\ &= x.\end{aligned}$$

Now consider $R_T$ which is the closed path.

$$\int_{R_T} = \int_{a-iT}^{a+iT} + \int_{a+iT}^{b+iT} + \int_{b+iT}^{b-iT} + \int_{b-iT}^{a-iT}.$$

We then have,

$$\frac{1}{2\pi i}\int_{a-iT}^{a+iT} \frac{-\zeta'(s)}{\zeta(s)}\frac{x^s}{s}ds = x - \frac{1}{2\pi i}\left(\int_{a+iT}^{b+iT} + \int_{b+iT}^{b-iT} + \int_{b-iT}^{a-iT}\right)\frac{-\zeta'(s)}{\zeta(s)}\frac{x^s}{s}ds$$

$$\begin{aligned}\left|\frac{1}{2\pi i}\int_{a+iT}^{b+iT} \frac{-\zeta(s)}{\zeta(s)}\frac{x^s}{s}ds\right| &= \left|\frac{1}{2\pi i}\int_{a}^{b} \frac{-\zeta(u+iT)}{\zeta(u+iT)}\frac{x^{(u+iT)}}{(u+iT)}du\right|,\\ &\ll \int_a^b \left|\frac{-\zeta'(u+iT)}{\zeta(u+iT)}\right|\frac{x^u}{|u+iT|}du, \quad (|u+iT|^2 \geq 1)\\ &\overset{\text{Theorem 2.28}}{\ll} \frac{\log^9 T}{T}x^a\int_b^a du,\\ &\ll \frac{x^a}{T}.\end{aligned}$$

Now we have,

$$\begin{aligned}\left|\int_{b-iT}^{b+iT} \frac{-\zeta'(s)}{\zeta(s)}\frac{x^s}{s}ds\right| &= \left|\frac{1}{2\pi}\int_{-T}^{T} \frac{\zeta'(b+iu)}{\zeta(b+iu)}\frac{x^{b+iu}}{b+iu}du\right|,\\ &\ll \log^9 T\int_{-T}^{T}\frac{x^b}{\sqrt{b^2+u^2}}du,\\ &\ll x^b\log^9 T\int_0^T\frac{du}{\sqrt{b^2+u^2}}du,\\ &\ll x^b\log^9 T\int_1^T\frac{du}{u},\\ &= x^b\log^{10} T.\end{aligned}$$

Now back to the beginning,

$$\psi(x) = x + O\left(\frac{x^a}{T} + x^b\log^{10} T\right) + O\left(\frac{x\log^2 x}{T} + \frac{x^a}{T}\log^9 T\right).$$

Choose $T$ to be such that $2c\log x = \log^{10} T, x = e^{\frac{\log^{10} T}{2c}}$.

$$
\begin{aligned}
x^{\frac{c}{\log^9 T}} &= e^{\frac{\log T}{2}} = \sqrt{T}. \\
x^{1-\frac{c}{\log^9 T}} \cdot \log^{10} T + \frac{x\log^2 x}{T} &+ \frac{x^{1+\frac{c}{\log^9 T}}}{T}\log^9 T \\
&= x \cdot T^{-\frac{1}{2}} \log^{10} T + \frac{x\log^2 x}{T} \\
&+ x\frac{\sqrt{T}}{T}\log^{10} T\left(\frac{x}{\sqrt{T}}\right)(\log^{10} T + \frac{\log^2 x}{\sqrt{x}} + \log^9 T), \\
&\ll \frac{x}{T^s} = xe^{-c(\log x)^{\frac{1}{10}}}.
\end{aligned}
$$

□

**Remark 2.17.** *Suppose* $\mathrm{Re}(s) = \sigma$ *is bounded. We then have the following estimate.*

$$
\begin{aligned}
|\Gamma(s)| &\leq \int_0^\infty |t^{s-1}|e^{-t}dt, \\
&= \int_0^\infty t^{\sigma-1}e^{-t}dt, \\
&= \left(\int_0^1 + \int_1^\infty\right) t^{\sigma-1}e^{-t}dt, \\
&\leq \int_0^1 t^{\sigma-1}dt + \int_1^\infty t^{\sigma-1}e^{-t}dt, \left(\frac{t^n}{e^t} \overset{t\to\infty}{\to} 0\right). \\
&\ll \frac{1}{\sigma} + \int_1^\infty t^{\sigma-1}t^{-2\sigma}dt, \\
&\ll \frac{1}{\sigma}.
\end{aligned}
$$

**Theorem 2.30.**

$$F(s) = \int f(s,t)dt.$$

$F : \Omega \to \mathbb{C}$ *is analytic in* $\Omega$ *if*

*i).* $f(s,t)$ *is continuous in* $(s,t)$,

*ii).* $f(s,t)$ *is analytic in* $s$,

*iii).* $\int f(s,t)$ *is uniformly bounded on compact subsets of* $\Omega$.

In Remark 2.17, suppose $a \leq \mathrm{Re}(s) \leq b$ then the last two inequalities will be,

$$\begin{aligned}\int_0^1 t^{\sigma-1}dt + \int_1^\infty t^{\sigma-1}e^{-t}dt \\ &\ll_b \frac{1}{\sigma} + \int_1^\infty t^{b-1}t^{-2b}dt, \\ &\ll_b \frac{1}{\sigma} + \frac{1}{b} = \frac{1}{a} + \frac{1}{b}.\end{aligned}$$

Thus we observe that in $\sigma > 0$, using integration by parts,

$$\Gamma(s+1) = s\Gamma(s).$$

Thus for any $n \in \mathbb{N}$, we have $\Gamma(n) = n!$. We have,

$$\Gamma(s) = \frac{\Gamma(s+1)}{s}, \sigma > 0.$$

Note $\Gamma(s+1)$ is analytic if $\sigma > -1$. Thus $\Gamma(s)$ is analytic in $\sigma > -1$ except simple pole at $s = 0$ with residue 1. That is

$$\lim_{s\to 0} s\Gamma(s) = 1.$$

We also have,

$$\Gamma(s) = \frac{\Gamma(s+2)}{\Gamma(s+1)}, \sigma > 0.$$

Note that in the numerator, we have $\sigma > -2$. $\Gamma(s)$ is analytic when $\sigma > -2$ except simple poles at $s = 0, -1$ with residue

$$\lim_{s\to -1}(s+1)\Gamma(s) = \frac{\Gamma(-1)}{-1} = -1.$$

Iterating the process, we have the following theorem,

**Theorem 2.31.** *$\Gamma(s)$ can be analytically continued to $\mathbb{C}$ except simple poles at $s = -k$ where $k \in \mathbb{Z}_{\geq 0}$ with residue $\frac{(-1)^k}{k!}$.*

**Remark 2.18.**

$$\Gamma(s)\Gamma(1-s) = \frac{\pi}{\sin \pi s}, s \in \mathbb{C}.$$

$$\lim_{s\to m} \Gamma(s)\Gamma(1-s)\sin \pi s = \pi.$$

$$\Gamma(s)\Gamma\left(s + \frac{1}{2}\right) = \sqrt{\pi}2^{1-2s}\Gamma(2s), \forall s \in \mathbb{C}.$$

**Exercise 2.5.** *Show*

$$\forall \alpha \in \mathbb{R}, x > 0, \sum_{n\in\mathbb{Z}} e^{-(n+\alpha)^2 \frac{\pi}{x}} = \sqrt{x} \sum_{n\in\mathbb{Z}} e^{-n^2\pi x + 2\pi i n\alpha}.$$

*Note that* $\sum_{n\in\mathbb{Z}} a_n$ *is convergent if*

$$s_N := \sum_{|n|\leq N} a_n,$$

*is convergent.*

**Theorem 2.32** (Poisson Summation Formula)**.** *Let* $\sigma > 1$ *and* $F \in L^1(\mathbb{R})$*, ie* $F : \mathbb{R} \to \mathbb{C}$ *and*

$$\int_{-\infty}^{\infty} |F(t)| dt < \infty.$$

*We have,*

$$\sum_{n\in\mathbb{Z}} F(n+u)$$

*is absolutely and uniformly convergent in* $u$*. Also we have,*

$$\sum_{n\in\mathbb{Z}} |\hat{F}(n)| < \infty,$$

*then*

$$\sum_{n\in\mathbb{Z}} F(n+u) = \sum_{n\in\mathbb{Z}} \hat{F}(n) e^{2\pi i n u}.$$

*Proof.* Use Exercise 2.5. □

**Theorem 2.33.** $\Gamma(s) \neq 0, \forall s \in \mathbb{C}$.

*Proof.* Note that the case when $s \in \mathbb{Z}$ is already shown. Thus suppose $s \notin \mathbb{Z}$. Suppose $\Gamma(s) = 0$, then it will follow that $\Gamma(1-s)$ by Remark 2.18. But this is a contradiction to Theorem 2.31 as it has a pole at non-integer point. Similarly from Remark 2.18, we see that $\frac{1}{\Gamma(s)}$ has a simple zero at $s = 0, -1, -2, \cdots$.

By definition, consider,

$$\Gamma\left(\frac{s}{2}\right) = \int_0^{\infty} e^{-t} t^{\frac{s}{2}-1} dt, \sigma > 0.$$

Replace $t = n^2\pi x, dt = n^2\pi dx$, we get,

$$\begin{aligned}
\pi^{\frac{-s}{2}}\Gamma\left(\frac{s}{2}\right)n^{-s} &= \int_0^\infty x^{\frac{s}{2}-1}e^{-n^2\pi x}dx, \sigma > 1. \\
\pi^{\frac{-s}{2}}\Gamma\left(\frac{s}{2}\right)\zeta(s) &= \sum_{n\in\mathbb{N}}\int_0^\infty x^{\frac{s}{2}-1}e^{-n^2\pi x}dx, \\
&= \int_0^\infty x^{\frac{s}{2}-1}\left(\sum_{n\in\mathbb{N}}e^{-n^2\pi x}\right)dx, \\
&= \sum_{n\in\mathbb{N}}\int_0^\infty |x^{\frac{s}{2}-1}e^{-n^2\pi x}|dx, \\
&= \sum_{n\in\mathbb{N}}\left(\int_0^\infty x^{\frac{\sigma}{2}-1}e^{-n^2\pi x}dx\right), \\
&= \sum_{n=1}^\infty \pi^{-\frac{\sigma}{2}}\Gamma\left(\frac{\sigma}{2}\right)n^{-\sigma}, \\
&= \pi^{-\frac{\sigma}{2}}\Gamma\left(\frac{\sigma}{2}\right)\zeta(s), \sigma > 1, \\
&< \infty. \\
\pi^{-\frac{s}{2}}\Gamma\left(\frac{s}{2}\right)\zeta(s) &= \int_0^\infty x^{\frac{s}{2}-1}\left(\sum_{n\in\mathbb{N}}e^{-n^2\pi x}\right).
\end{aligned}$$

Using Exercise 2.5 and set $\alpha = 0$, we have,

$$\begin{aligned}
\sum_{n\in\mathbb{Z}}e^{-n^2\frac{\pi}{s}} &= \sqrt{s}\sum_{n\in\mathbb{Z}}e^{-n^2\pi x}. \\
\sum_{n\in\mathbb{Z}}e^{-n^2\pi x} &\ll \sum_{n\in\mathbb{N}}e^{-n\pi x}, \\
&\ll \int_1^\infty e^{-\pi x t}dt, \\
&\ll \frac{e^{-\pi x}}{x}, x > 0.
\end{aligned}$$

Set $\theta(x) := \sum_{n\in\mathbb{Z}} e^{n^2\pi x}$ then,

$$\begin{aligned}
\theta(x) &= 1 + 2\sum_{n\in\mathbb{N}}e^{-n^2\pi x}. \\
\sum_{n\in\mathbb{N}}e^{-n^2\pi x} &= \frac{\theta(x)-1}{2}, x > 0.
\end{aligned}$$

Using the exercise, we have,

$$\theta\left(\frac{1}{x}\right) = \sqrt{x}\theta(x).$$

Set $w(x) := \frac{\theta(x)-1}{2}$. Thus write

$$\begin{aligned} w\left(\frac{1}{x}\right) &= \frac{\theta\left(\frac{1}{x}\right)-1}{2}, \\ &= \frac{\sqrt{x}\theta(x)-1}{2}, \\ &= \frac{\sqrt{x}\theta(x)-1}{2}. \\ w\left(\frac{1}{x}\right) &= \sqrt{x}w(x) + \frac{\sqrt{x}}{2} - \frac{1}{2}. \\ \pi^{-\frac{1}{2}}\Gamma\left(\frac{s}{2}\right)\zeta(s) &= \int_0^\infty x^{\frac{s}{2}-1}w(x)dx, \sigma > 1. \end{aligned}$$

Step 3

$$\int_0^1 x^{\frac{s}{2}-1}w(x)dx + \int_1^\infty x^{\frac{s}{2}-1}w(x)dx.$$

Taking $x = \frac{1}{y}$ and $dx = -\frac{1}{y^2}dy$ we have,

$$\begin{aligned} \int_1^\infty \left(\frac{1}{y^2}\right)^{\frac{s}{2}-1} w\left(\frac{1}{y}\right)\frac{-1}{y^2}dy &= \int_1^\infty y^{-\frac{s}{2}}w\left(\frac{1}{y}\right)\frac{dy}{y}, \\ &= \int_1^\infty y^{-\frac{1}{2}}\left(\sqrt{y}w(y) + \frac{\sqrt{y}}{2} - \frac{1}{2}\right)\frac{dy}{y}, \\ &= \int_1^\infty y^{-\frac{s}{2}+\frac{1}{2}}w(y)\frac{dy}{y} + \int_1^\infty \frac{y^{-\frac{s}{2}+\frac{1}{2}-1}}{2}dy - \frac{1}{2}\int_1^\infty y^{-\frac{s}{2}-1}dy, \\ &= \int_1^\infty y^{\frac{1-s}{2}}w(y)\frac{dy}{y} + \frac{1}{s(s-1)} + \int_1^\infty (x^{\frac{s}{2}} + x^{\frac{1-s}{2}})w(x)\frac{dx}{x}, \sigma > 1. \end{aligned}$$

Step 4,

$$\begin{aligned} \int_1^\infty |x^{\frac{s}{2}} + x^{\frac{1-s}{2}}||w(x)|\frac{dx}{x} &\leq \int_1^\infty (x^{\frac{\sigma 2}{-}1} + x^{\frac{1-\sigma}{2}-1})|w(x)|dx, \\ &\ll \int_1^\infty \frac{(x^{\frac{\sigma}{2}-1} + x^{\frac{1-\sigma}{2}-1})}{e^{\pi x}}dx, \sigma \in \mathbb{R}, \\ &\ll \int_1^\infty \frac{e^x}{e^{\pi x}}dx, \qquad \ll 1. \end{aligned}$$

$\int_1^\infty |x^{\frac{s}{2}} + x^{\frac{1-s}{2}}||w(x)|\frac{dx}{x}$ is analytic in $\mathbb{C}$.

$$s(s-1)\pi^{-\frac{s}{2}}\Gamma\left(\frac{s}{2}\right)\zeta(s) = 1 + s(s-1)\int_1^\infty (x^{\frac{s}{2}} + x^{\frac{1-s}{2}})w(x)\frac{dx}{x}.$$

Set $\xi(x) := 1 + s(s-1)\int_1^\infty (x^{\frac{s}{2}} + x^{\frac{1-s}{2}})w(x)\frac{dx}{x}$, then it is entire. and for all $s$ we have,

$$\xi(1-s) = \xi(s).$$

Thus obtain,

$$(1-s)(1-s-1)\pi^{-\frac{(1-s)}{2}}\Gamma\left(\frac{1-s}{2}\right) = s(s-1)\pi^{-\frac{s}{2}}\Gamma\left(\frac{s}{2}\right)\zeta(s).$$

Using the construction of $w(s)$ we have,

$$|w(x)| \ll |\theta(x)| \ll \frac{e^{-\pi x}}{x}, \forall x > 0.$$

Also we have,

$$\begin{aligned}
\zeta(1-s) &= \pi^{-s+\frac{1}{2}}\frac{\Gamma\left(\frac{s}{2}\right)}{\Gamma\left(\frac{1-s}{2}\right)}\zeta(s),\\
\zeta(1-s) &= \pi^{-s}2^{1-s}\cos\left(\frac{\pi s}{s}\right)\Gamma(s)\zeta(s).
\end{aligned}$$

Note that $\zeta(s)$ has an analytic continuation to $\mathbb{C}$ except simple pole at $s$.

$$\begin{aligned}
\lim_{s\to 1}\zeta(1-s) &= \pi^{-1}\lim_{s\to 1}\frac{\cos\frac{\pi s}{2}\zeta(s)(s-1)}{s-1}.\\
\lim_{s\to 1}\frac{\cos\frac{\pi s}{2}}{s-1} &= \lim_{s\to 0}\frac{\cos\left(\frac{\pi}{2}(1-s)\right)}{-s},\\
&= \lim_{s\to 0}\frac{\sin\frac{\pi}{2}s}{-s\cdot\frac{\pi}{2}}\frac{\pi}{2},\\
&= -\frac{\pi}{2}.\\
\Rightarrow \lim_{s\to 1}\zeta(1-s) &= -\frac{1}{2}.
\end{aligned}$$

$$\zeta(-2n) = \pi^{-(2n+1)}2^{1-2n-1}\cos\left(\frac{\pi}{2}(2n+1)\right)\Gamma(2n+1)\zeta(2n+1) = 0.$$

Recall

$$\begin{aligned}
\zeta(s) &= \sum_{n\in\mathbb{N}}\frac{1}{n^s},\\
&= \frac{s}{s-1} - s\int_1^\infty \frac{\{t\}}{t^{s+1}}dt, \sigma > 0, s \neq 1.
\end{aligned}$$

If $s$ is real,

$$\begin{aligned}
\left|\zeta(s) - \frac{s}{s-1}\right| &\leq |s|\int_1^\infty \frac{\{t\}}{t^{\sigma+1}}dt,\\
&< \frac{|s|}{\sigma} = \frac{\sigma}{\sigma} = 1.
\end{aligned}$$

Thus we obtain,

$$\begin{aligned}
-1+\frac{s}{s-1} &< \zeta(s) < 1+\frac{s}{s-1} \\
\frac{1}{s-1} &< \zeta(s) < \frac{2s-1}{s-1}, \\
-1 < (1-s)\zeta(s) &< 1-2s < 0 \quad \text{if } \frac{1}{2} < s < 1 \\
\Rightarrow \zeta(s) &\neq 0, \quad \text{if } \frac{1}{2} < s < 1.
\end{aligned}$$

□

## 2.7 Primitive characters

Suppose $q_0|q$. Then we have a natural map

$$\mathbb{Z}/q\mathbb{Z} \to \mathbb{Z}/q_0\mathbb{Z}$$

**Definition 2.16.** *Let $\chi \bmod q$ be a character and $q_0|q$. If there exists a character $\chi^* \bmod q_0$ making the following diagram commutative,*

$$\begin{tikzcd}
(\mathbb{Z}/q\mathbb{Z})^\times \arrow[r, "\chi"] \arrow[d, "f"'] & \mathbb{C}^\times \\
(\mathbb{Z}/q_0\mathbb{Z})^\times \arrow[ur, "\chi^*"'] &
\end{tikzcd}$$

*we say $\chi$ factors through $(\mathbb{Z}/q_0\mathbb{Z})$ or $\chi^*$ induces $\chi$. If $q_0$ is minimal among positive such numbers, we say $q_0$ is a conductor of $\chi$.*

**Definition 2.17.** *A character $\chi \bmod q$ is called primitive if its conductor is $q$ only. Otherwise the character is imprimitive.*

**Example 2.11.** *Take $\chi : (\mathbb{Z}/8\mathbb{Z})^\times \to \mathbb{C}^\times$ such that*

| $n$ | $\chi(n)$ |
|---|---|
| 1 | 1 |
| 3 | $-1$ |
| 5 | 1 |
| 7 | $-1$ |

*Then 4 is a conductor with $\chi^* : (\mathbb{Z}/4\mathbb{Z})^\times \to \mathbb{C}^\times$,*

| $n$ | $\chi^*(n)$ |
|---|---|
| 1 | 1 |
| 3 | $-1$ |

**Example 2.12.** *Take $\chi : (\mathbb{Z}/6\mathbb{Z})^\times \to \mathbb{C}^\times$ such that*

| $n$ | $\chi(n)$ |
|---|---|
| 1 | 1 |
| 5 | $-1$ |

*Then* 3 *is a conductor with* $\chi^* : (\mathbb{Z}/3\mathbb{Z})^\times \to \mathbb{C}^\times$,

| $n$ | $\chi^*(n)$ |
|---|---|
| 1 | 1 |
| 2 | $-1$ |

**Remark 2.19.** *All principal characters* $\bmod q$ *for* $q > 1$ *are imprimitive by construction.*

We set as usual $s = \sigma + it$ for $\sigma, t \in \mathbb{R}$.

**Lemma 2.14.** *Suppose* $\chi \bmod q$ *is induced by* $\chi^* \bmod q_0$. *We then have,*

$$\sigma > 1 \Rightarrow L(s,\chi) = L(s,\chi^*) \prod_{p \mid q} \left(1 - \frac{\chi^*(p)}{p^s}\right).$$

*Proof.* We know that for $\sigma > 1$,

$$L(s,\chi) = \prod_{p \nmid q} \left(1 - \frac{\chi(p)}{p^s}\right)^{-1}.$$

Also

$$\begin{aligned} L(s,\chi^*) &= \prod_{p \nmid q_0} \left(1 - \frac{\chi(p)}{p^s}\right)^{-1}, \\ &= \prod_{\substack{p \nmid q_0 \\ p \nmid q}} \left(1 - \frac{\chi^*(p)}{p^s}\right)^{-1} \prod_{\substack{p \nmid q_0 \\ p \mid q}} \left(1 - \frac{\chi^*(p)}{p^s}\right)^{-1}, \\ &= L(s,\chi) \prod_{\substack{p \nmid q_0 \\ p \mid q}} \left(1 - \frac{\chi^*(p)}{p^s}\right)^{-1}. \end{aligned}$$

□

**Remark 2.20.** *Using Identity theorem, Lemma 2.14 holds for all* $s \in \mathbb{C}$.

**Lemma 2.15.** *Suppose* $x \in \left[0, \frac{1}{2}\right]$, *then*

$$|\sin \pi x| \geq 2x.$$

*Proof.* Exercise. □

**Definition 2.18** (Gauss sum)**.** *Let $\chi \bmod q$ be a character. The Gauss sum is,*

$$\tau(\chi) := \sum_{k=1}^{q} \chi(k) e^{\frac{2\pi k i}{q}}.$$

**Lemma 2.16.** *For $n \in \mathbb{Z}$, we have,*

$$\chi(n)\tau(\overline{\chi}) = \sum_{k=1}^{q} \overline{\chi}(k) e^{\frac{2\pi i k n}{q}},$$

*where $\chi$ is a primitive character modulo $q$ and $\chi \neq \chi_0$.*

**Theorem 2.34.** *If $\chi \bmod q$ is primitive then,*

$$|\tau(\chi)| = \sqrt{q}.$$

*Proof.* By Lemma 2.16, we have,

$$\chi(n)\tau(\overline{\chi}) = \sum_{k=1}^{q} \overline{\chi}(k) e^{\frac{2\pi i k}{q}},$$
$$\overline{\chi(n)\tau(\overline{\chi})} = \sum_{l=1}^{q} \chi(l) e^{-\frac{2\pi i l}{q}}.$$

Multiplying this equation with the one in the statement we have,

$$|\chi(n)|^2 |\tau(\overline{\chi})|^2 = \sum_{k,l=1}^{q} \overline{\chi(k)} \chi(l) e^{\frac{2\pi i (k-l)}{q}}.$$

Applying $\sum_{n=1}^{q}$ to the both sides of the above equation, we have,

$$|\tau(\overline{\chi})|^2 \sum_{n=1}^{q} |\chi(n)|^2 = \sum_{k,l=1}^{q} \overline{\chi(k)} \chi(l) \left( \sum_{n=1}^{q} \left( e^{\frac{2\pi i (k-l)}{q}} \right)^n \right).$$

Note that

$$x + x^2 + \cdots + x^q = \begin{cases} \frac{x(x^q-1)}{x-1}, & (x \neq 1), \\ q, & (x = 1), \end{cases}$$
$$= \begin{cases} 0, & (x \neq 1, x^q = 1), \\ q, & (x = 1), \end{cases}$$

Take $x = e^{\frac{2\pi i (k-l)}{q}} = \cos \frac{2\pi}{q}(k-l) + i \sin \frac{2\pi}{q}(k-l) = 1$. $x = 1$ if and only if $q | k - l$ thus

$$\sum_{n=1}^{q} \left( e^{\frac{2\pi i (k-l)}{q}} \right)^n = \begin{cases} 0, & (q \nmid k-l), \\ q, & (\text{otherwise}). \end{cases}$$

Therefore, we get,

$$\begin{aligned}
|\tau(\overline{\chi})|^2 \sum_{n=1}^{q} |\chi(n)|^2 &= q \sum_{\substack{k,l=1 \\ q|k-l}}^{q} \overline{\chi}(k)\chi(l), \\
&= q \sum_{k=1}^{q} \overline{\chi}(k)\chi(k), \\
&= q \sum_{k=1}^{q} |\chi(k)|^2.
\end{aligned}$$

□

**Theorem 2.35** (Pólya-Vinogradov)**.**
*Let $\chi \bmod q$ be a primitive character. We have,*

$$\left| \sum_{n \leq x} \chi(n) \right| \ll \sqrt{q} \log q.$$

*This inequality is uniform in $q$.*

*Proof.* Consider $(n, q) = 1$, then

$$\begin{aligned}
\chi(n)\tau(\overline{\chi}) &= \chi(n) \sum_{k=1}^{q} \overline{\chi}(k) e^{\frac{2\pi i k}{q}}, \\
&= \chi(n) \sum_{\substack{k=1 \\ (k,q)=1}}^{q} \overline{\chi}(k) e^{\frac{2\pi i k}{q}},
\end{aligned}$$

Set $k = nt$, we get,

$$\begin{aligned}
\chi(n)\tau(\overline{\chi}) &= \sum_{\substack{t=1 \\ (t,q)=1}}^{q} \overline{\chi}(nt)\chi(n) e^{\frac{2\pi i n t}{q}}. \\
\overline{\chi}(nt) &= \overline{\chi}(n)\overline{\chi}(t). \\
\chi(n)\tau(\overline{\chi}) &= \sum_{\substack{t=1 \\ (t,q)=1}}^{q} \overline{\chi}(t) e^{\frac{2\pi i n t}{q}}.
\end{aligned}$$

Observe that

$$\begin{aligned}
\tau(\overline{\chi})\left(\sum_{n\le x}\chi(n)\right) &= \sum_{k=1}^{q-1}\overline{\chi}(k)\left(\sum_{n\le x}e^{\frac{2\pi ikn}{q}}\right),\\
|\tau(\overline{\chi})|\cdot\left|\sum_{n\le x}\chi(n)\right| &\le \sum_{k=1}^{q-1}\left|\sum_{n\le x}e^{\frac{2\pi ikn}{q}}\right|,\\
&= \sum_{k=1}^{q-1}\left|\frac{e^{\frac{2\pi ik}{q}}\left(e^{\frac{2\pi in[x]}{q}}-1\right)}{e^{\frac{2\pi ik}{q}}-1}\right|,\\
&\le \sum_{k=1}^{q-1}\frac{2}{|e^{\frac{2\pi k}{q}}-1|},
\end{aligned}$$

Note that for all $y\in\mathbb{R}$,

$$\begin{aligned}
2i\sin y &= e^{-iy}(e^{2iy}-1),\\
|2\sin y| &= |e^{2iy}-1|.
\end{aligned}$$

Apply this to the equation above we have,

$$\sum_{k=1}^{q-1}\frac{2}{|e^{\frac{2\pi k}{q}}-1|} = \sum_{k=1}^{q-1}\frac{1}{|\sin\frac{\pi k}{q}|}.$$

Using Lemma 2.15, we get,

$$\begin{aligned}
\sum_{k=1}^{q-1}\frac{1}{|\sin\frac{\pi k}{q}|} &= \sum_{1\le k\le\frac{q}{2}}\frac{1}{|\sin\frac{\pi k}{q}|}+\sum_{\frac{q}{2}<k}^{q-1}\frac{1}{|\sin\frac{\pi k}{q}|},\\
&= 2\sum_{1\le k\le\frac{q}{2}}\frac{1}{|\sin\frac{\pi k}{q}|},\\
&\le \sum_{1\le k\le\frac{q}{2}}\frac{k}{q}\ll q\log\frac{q}{2}\ll q\log q.
\end{aligned}$$

Recall Theorem 2.34, we have $|\tau(\chi)|=\sqrt{q}$ as $\chi$ is primitive. Therefore, for $q>1$, we have,

$$\left|\sum_{n\le x}\chi(n)\right|\ll\sqrt{q}\log q,$$

uniformly in $q$ as $x\to\infty$. □

Consider $L(q,\chi)$ functional equation, where $\chi \neq \chi_0$ and $\chi$ is primitive. Suppose $\chi(-1) = 1$ (i.e it is an even character). From the proof of Theorem 2.33, we have,

$$\pi^{-\frac{s}{2}}\Gamma\left(\frac{s}{2}\right)n^{-s} = \int_0^\infty x^{\frac{s}{2}}e^{-n^2\pi x}\frac{dx}{x}.$$

Replace $x$ with $\frac{x}{q}$ then for $\mathrm{Re}(s) = \sigma > 0$, we have,

$$\pi^{-\frac{s}{2}}q^{\frac{s}{2}}\Gamma\left(\frac{s}{2}\right)n^{-s} = \int_0^\infty x^{\frac{s}{2}}e^{-n^2\pi x}\frac{dx}{x}.$$

For $\sigma > 1$, we have,

$$\begin{aligned}\pi^{-\frac{s}{2}}q^{\frac{s}{2}}\Gamma\left(\frac{s}{2}\right)L(s,\chi) &= \sum_{n=1}^\infty \int_0^\infty \chi(n)x^{\frac{s}{2}}e^{-n^2\frac{\pi x}{q}}\frac{dx}{x},\\ &= \int_0^\infty x^{\frac{s}{2}}\left(\sum_{n=1}^\infty \chi(n)e^{-n^2\frac{\pi x}{q}}\right)\frac{dx}{x}.\end{aligned}$$

Let

$$\theta(x,\chi) = \sum_{n\in\mathbb{Z}}\chi(n)e^{-\frac{n^2\pi x}{q}} = 2\sum_{n=1}^\infty \chi(n)e^{-\frac{n^2\pi x}{q}}, (x>0).$$

Using this, we get,

$$\int_0^\infty x^{\frac{s}{2}}e^{-n^2\pi x}\frac{dx}{x} = \frac{1}{2}\int_0^\infty x^{\frac{s}{2}}\theta(x,\chi)\frac{dx}{x}, (\sigma>1).$$

Split the integral into $\int_0^1, \int_1^\infty$.

$$\begin{aligned}\tau(\overline{\chi})\theta(x,\chi) &= \left(\frac{q}{x}\right)^{\frac{1}{2}}\theta(x^{-1},\overline{\chi}),\\ &= \sum_{n\in\mathbb{Z}}(\tau(\overline{\chi})\chi(n))e^{\frac{-n^2\pi x}{q}},\\ &= \sum_{n\in\mathbb{Z}}\sum_{k=1}^q \overline{\chi}(k)e^{\frac{2\pi i n k}{q}}e^{-\frac{n^2\pi x}{q}},\\ &= \sum_{k=1}^q \overline{\chi(k)}\sum_{n\in\mathbb{Z}}e^{\frac{2\pi k n}{q}-n^2\frac{\pi x}{q}},\\ &\overset{\text{From Lecture 7 page 6}}{=} \sum_{k=1}^q \overline{\chi}(k)\left(\frac{x}{q}\right)^{-\frac{1}{2}}\sum_{n\in\mathbb{Z}}e^{-\left(m+\frac{m}{q}\right)\frac{\pi q}{x}},\\ &= \left(\frac{q}{x}\right)^{\frac{1}{2}}\sum_{k=1}^q \overline{\chi}(k)\sum_{n\in\mathbb{Z}}e^{-\left(m+\frac{m}{q}\right)\frac{\pi q}{x}}.\end{aligned}$$

Put $qn + m = t$, then

$$\overline{\chi}(qn+m) = \overline{\chi}(m) = \overline{\chi}(t).$$

Thus,

$$\left(\frac{q}{x}\right)^{\frac{1}{2}} \sum_{k=1}^{q} \overline{\chi}(k) \sum_{n\in\mathbb{Z}} e^{-\left(m+\frac{m}{q}\right)\frac{\pi q}{x}} = \left(\frac{q}{x}\right)^{\frac{1}{2}} \sum_{t\in\mathbb{Z}} \overline{\chi}(t) e^{-\frac{t^2\pi}{qx}} = \left(\frac{q}{x}\right)^{\frac{1}{2}} \theta(x^{-1}, \overline{\chi}).$$

Now for the splitted integral, we have,

$$\frac{1}{2}\int_0^1 x^{\frac{s}{2}}\theta(x,\chi)\frac{dx}{x} + \frac{1}{2}\int_1^\infty x^{\frac{s}{2}}\theta(x,\chi)\frac{dx}{x}\frac{dx}{x}.$$

Replacing $x$ with $\frac{1}{x}$, we get,

$$= \frac{\tau(\chi)}{2\sqrt{q}}\int_1^\infty x^{\frac{1-s}{2}}\theta(x,\overline{\chi})\frac{dx}{x} + \frac{1}{2}\int_1^\infty\int_0^1 x^{\frac{s}{2}}\theta(x,\chi)\frac{dx}{x}.$$

Set

$$\xi(s,\chi) := \frac{\tau(\chi)}{2\sqrt{q}}\int_1^\infty x^{\frac{1-s}{2}}\theta(x,\overline{\chi})\frac{dx}{x} + \frac{1}{2}\int_1^\infty\int_0^1 x^{\frac{s}{2}}\theta(x,\chi)\frac{dx}{x}.$$

Use that $\theta(x,\overline{\chi}) \ll e^{-\frac{\pi x}{q}}$ and $|\tau(\chi)| = \sqrt{q}$. It turns out that $\xi$ is uniformly bounded on compact subsets of $\mathbb{C}$ and in particular, this is entire.

**Lemma 2.17.**

$$\xi(1-s,\chi) = \frac{\tau(\chi)}{\sqrt{q}}\xi(s,\overline{\chi}).$$

*Proof.* Recall,

$$\tau(\overline{\chi})\theta(x,\chi) = \left(\frac{q}{x}\right)^{\frac{1}{2}}\theta(x^{-1},\overline{\chi}),$$

and

$$|\tau(\chi)|^2 = q.$$

We get,

$$\frac{\tau(\chi)}{\sqrt{q}} = \frac{\sqrt{q}}{\overline{\tau(\chi)}} = \frac{\sqrt{q}}{\tau(\overline{\chi})}.$$

Then,

$$L(1-s,\chi) = \frac{\tau(\chi)}{q^{1-s}}\pi^{-s}2^{1-s}\cos\frac{\pi s}{2}(\Gamma(s))L(s,\overline{\chi}).$$

Note that $\chi(-1) = 1$ and is primitive (in the case of odd integer replace cos with sin). Note also that

i). $L(s,\chi) = 0, s = 0, -2, -4, \cdots$ when $\chi$ is even,

ii). $L(s,\chi) = -1, -3, -5, \cdots$ when $\chi$ is odd.

When $0 < \sigma < 1$, $\square$

**Definition 2.19** (Entire function of finite order)**.** *An entire functio $nf(z)$ is saiad to be of finite order if there is $\alpha > 0$ such that*

$$|f(z)| \ll e^{|z|^\alpha},$$

*as $|z| \to \infty$. The order of $f$ is the infimum of such $\alpha$.*

**Example 2.13.** *We have the following examples,*

1. $e^z \to 1$,
2. $\sin z \to 1$,
3. $\cos z \to 1$,
4. $e^{z^n} \to n$.

*For the trigonometric functions use* $\sin z = \frac{e^{iz} - e^{-iz}}{2i}$ *and similarly for* $\cos$.

**Remark 2.21.** *Is $|\sin z| < z, z \in \mathbb{C}$? To show this consider,*

$$f(z) = \left\{ \frac{\sin z}{z}, z \neq 0, 1, z = 0. \right.$$

*And apply Liouville's theorem.*

**Theorem 2.36** (Special case of Hadamard's factorization theorem)**.** *Let $f(z)$ be an entire function of order 1 and $(z_j)_{j \in \mathbb{N}}$ be such that $z_j \in \mathbb{C} \backslash \{0\}$, and $f(z_j) = 0$.Then there are some constants $A, B$ such that*

$$f(z) = e^{A+Bz} \prod_{j \geq 1} \left(1 - \frac{z}{z_j}\right) e^{\frac{z}{z_j}}.$$

**Theorem 2.37.** *Let $f$ be an entire function of order $\alpha \in \mathbb{Z}$ without any zeros and assume $f(1) \leq 0$. Then we have,*

$$f(z) = e^{p(z)},$$

*where $p$ is some polynomial of degree $\alpha$.*

Recall the completed zeta function,

**Definition 2.20.**

$$\xi(s) := (s-1)\pi^{\frac{-s}{2}} \Gamma\left(\frac{s}{2} + 1\right) \zeta(s).$$

*We know,*

1. *$\xi(s)$ has an order 1,*
2. *$\zeta(s)$ has infinitely zeros in the critical strip.*
3. *$\xi(s) = e^{As+B} \prod_{\substack{\rho, \xi(\rho)=0 \\ 0<\mathrm{Re}(\rho)<1}} \left(1 - \frac{s}{\rho}\right) e^{\frac{z}{\rho}}$, where $A, B$ ae some constant.*

**Remark 2.22.** *Zeros of $\xi(s)$ are precisely the non-trivials zeros of $\zeta(s)$.*

**Theorem 2.38.** *There eists a constant $c > 0$ such that for $\sigma, t \in \mathbb{R}$ $s = \sigma + it$, $\sigma \geq 1 - \frac{c}{\log(|t|+2)}$, we have,*

$$\zeta(s) \neq 0.$$

Taking derivatives, we obtain,

$$\frac{\xi'(s)}{\xi(s)} = \frac{1}{s-1} - \frac{1}{2}\log\pi + \frac{\Gamma'\left(\frac{s}{2}+1\right)}{2\Gamma\left(\frac{s}{2}+1\right)} + \frac{\zeta'(s)}{\zeta(s)},$$
$$\frac{\xi'(s)}{\xi(s)} = A + \sum_{\substack{\zeta(\rho)=0 \\ 0<\mathrm{Re}(\rho)<1}} \left(\frac{1}{s-\rho} + \frac{1}{\rho}\right).$$

Note that $\frac{s}{2}\Gamma\left(\frac{s}{2}\right) = \Gamma\left(\frac{s}{2}+1\right)$. Thus we obtain,

$$-\frac{\zeta'(s)}{\zeta} = \frac{1}{s-1} - A - \frac{\log\pi}{2} + \frac{1}{2}\frac{\Gamma'\left(\frac{s}{2}+1\right)}{\Gamma\left(\frac{s}{2}+1\right)} - \sum_{\rho}\left(\frac{1}{s-\rho} + \frac{1}{\rho}\right).$$

We then get,

$$\mathrm{Re}\left(-\frac{\zeta'(s)}{\zeta}\right) = \mathrm{Re}\left(\frac{1}{s-1}\right) - A - \frac{\log\pi}{2} + \frac{1}{2}\mathrm{Re}\left(\frac{\Gamma'\left(\frac{s}{2}+1\right)}{\Gamma\left(\frac{s}{2}+1\right)}\right) - \sum_{\rho}\mathrm{Re}\left(\frac{1}{s-\rho} + \frac{1}{\rho}\right).$$

Recall that for $s = \sigma + it, \sigma, t \in \mathbb{R}$, we have,

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{n\geq 1}\frac{\Lambda(n)}{n^s}, \sigma \geq 1.$$

And also,

$$\frac{-3\zeta'(\sigma)}{\zeta(\sigma)} - \mathrm{Re}\left(\frac{\zeta'(\sigma+it)}{\zeta(\sigma+it)}\right) - \mathrm{Re}\left(\frac{\zeta'(\sigma+2it)}{\zeta(\sigma+2it)}\right).$$

Check that

$$\sum_{n=1}^{\infty}\frac{\Lambda(n)}{n^\sigma}(3 + 4\cos(t\log n) + \cos(2t\log n)) > 0, \sigma > 1.$$

For $1 < \sigma < 2, t \geq 1$, we have,

$$\mathrm{Re}\left(\frac{1}{s-1}\right) = \frac{\sigma-1}{|s-1|^2} < \frac{1}{t^2} < 1.$$

$$\mathrm{Re}\left(\frac{1}{s-\rho} + \frac{1}{\rho}\right) = \mathrm{Re}\left(\frac{\overline{s-\rho}}{|s-\rho|^2} + \frac{\overline{\rho}}{|\rho|^2}\right),$$
$$= \frac{\sigma-\beta}{|s-\rho|^2} + \frac{\beta}{|\rho|^2} > 0,$$

where $\beta = \mathrm{Re}(\rho)$. We have,

$$\frac{\Gamma'\left(\frac{s}{2}+1\right)}{\Gamma\left(\frac{s}{2}+1\right)} < B\log\left(\frac{|t|}{2}+2\right), \forall t \in \mathbb{R}.$$

We have the following equations,

$$\Gamma(z) \sim \sqrt{\frac{2\pi}{z}}\left(\frac{z}{e}\right)^z, |z| \to \infty.$$

$$\frac{1}{\Gamma(z)} = e^{\gamma z} z \prod_{n=1}^{\infty}\left(1+\frac{z}{n}\right)e^{-\frac{2}{n}},$$

where $\gamma$ is the Euler's constant.

Now consider $1 \leq \sigma \leq 2$, we have,

$$\left|\frac{\Gamma'(s)}{\Gamma(s)}\right| < B\log(|t|+2),$$

where $B$ is some constant for all $t \in \ R$. From above expressions, we obtain,

$$\mathrm{Re}\left(\frac{-\zeta'(s)}{\zeta(s)}\right) < B'\log\left(\frac{|t|}{2}+2\right), \forall t \geq 1.$$

Let $\rho = \beta + it$, be any non-trivial zero of $\zeta(s)$.

$$\mathrm{Re}\left(\frac{-\zeta'(s)}{\zeta(s)}\right) < B'\log\left(\frac{|t|}{2}+2\right) - \frac{1}{\sigma-\beta}, t \geq 1.$$

Recall for $\sigma > 0$,

$$\frac{\zeta'(s)}{\zeta(s)} = \frac{1}{s-1} + g(s),$$

where$g(s)$ is some analyti function. For $1 \leq \sigma \leq 2$,

$$\begin{aligned}\frac{-\zeta'(\sigma)}{\zeta(\sigma)} &= \frac{1}{\sigma-1} - g(\sigma), \\ &< \frac{1}{\sigma-1} + c,\end{aligned}$$

where $C$ is some constant.

$$0 < \frac{-3\zeta'(\sigma)}{\zeta(\sigma)} - 4\,\mathrm{Re}\left(\frac{\zeta'(\sigma+it)}{\zeta(\sigma+it)}\right) - \mathrm{Re}\left(\frac{\zeta'(\sigma+2it)}{\zeta(\sigma+2it)}\right).$$

We then obtain,

$$0 < 3\left(\frac{1}{\sigma-1}+C\right) + 4B'\log(|t|+2) + B'\log(|t|+2) - \frac{1}{\sigma-\beta}.$$

Note that

$$\frac{1}{\sigma-\beta} < \frac{3}{\sigma-1} + B''\log(|t|+2), t \geq 1.$$

Take $\sigma = 1 + \frac{\varepsilon}{\log(|t|+2)}$, where $\varepsilon > 0$, sufficiently small which will be chosen later.

$$\beta < 1 - \frac{\varepsilon(1-B''\varepsilon)}{\log(|t|+2)}, t \geq 1.$$

So far we have shown that for $t \geq 1$ and

$$\sigma \geq 1 - \frac{c}{\log(|t|+2)},$$

we have $\zeta(s) \neq 0$. For some constant $C > 0$.

For $|t| \leq 1$, since $\zeta 9s)$ be analytic except $s = 1$, there exists a constant $c_1 > 0$ such that

$$\sigma > 1 - c_1 \Rightarrow \zeta(s) \neq 0.$$

If

$$1 - c_1 \leq 1 - \frac{c_2}{\log(|t|+2)},$$

then we have,

$$c_1 \geq \frac{c_2}{\log(|t|+2)}, |t| < 1.$$

Choose $c_2 = c_1 \log 2$. Take $c' = \min(c_1, c_1 \log 2)$.

Above proof can be generalized to $L(s,\chi)$ where $\chi$ is a non-trivial character.

**Theorem 2.39.** *Let $\chi$ be a non-real character. Let $s = \sigma + it, \sigma, t \in \mathbb{R}$ such that $L(s,\chi) = 0$, then there is $c_1 > 0$ satisfying*

$$\sigma \geq 1 - \frac{c_1}{\log(q(|t|+2))}.$$

**Theorem 2.40** (Siegel zeros)**.** *There exists an absolute constant $c_2 > 0$ such that $L(s,\chi)$ where $\chi$ is quadratic* $\bmod q$ *has at most one zero in the region,*

$$\sigma > 1 - \frac{\delta}{\log q}, |t| < \frac{\delta}{\log q},$$

*where $0 < \delta < c_2$. If such a zero exists, it is real, simple and often called Siegel zero.*

**Remark 2.23.** *Theorem 2.39 also holds for more general characters such as a character with no Siegel zeros.*

Recall the completed Dirichlet $L$-function,

$$\xi(s,\chi) := \left(\frac{q}{\pi}\right)^{\frac{s+a}{2}} \Gamma\left(\frac{s+a}{2}\right) L(s,\chi).$$

See the lecture note from 12/16. Where

$$a = \begin{cases} 0, & \chi(-1) = 1, \\ 1, & \chi(-1) = -1. \end{cases}$$

We have $\xi(s,\chi)$ is of order 1 and

$$\xi(s,\chi) = e^{A_1 + B_1 s} \prod_{\substack{\rho, L(\rho,\chi)=0 \\ 0 \le \mathrm{Re}(\rho) < 1}} \left(1 - \frac{s}{\rho}\right) e^{\frac{s}{\rho}}.$$

Furthermore $L(s,\chi)$ has inifinitely many zero in the critical strip.

**Remark 2.24.**

$$L(s,\chi) = L(s,\chi^*) \prod_{p|q} \left(1_{\frac{\chi^*(p)}{p^s}}\right), s \in \mathbb{C}$$

*Then $L(s,\chi) = 0$ if and only if $L(s,\chi^*) = 0$ when $\sigma \neq 0$.*

Recall

$$\xi(s) = \frac{s(s-1)}{2} \pi^{\frac{-s}{2}} \Gamma\left(\frac{s}{2}\right) \zeta(s).$$

**Definition 2.21.**

$$N(T) = |\{\rho = \beta + i\gamma \mid \zeta(\rho) = 0, \beta \in (0,1), \gamma \in (0,T)\}|.$$

**Theorem 2.41.** *As $T \to \infty$, we have,*

$$N(T+1) - N(T) \ll \log T.$$

*Proof.* Enough to show that as $T \to \infty$.

$$\sum_{\substack{\rho=\beta+i\gamma \\ \zeta(\rho)=0 \\ \beta\in(0,1)}} \frac{1}{1 + (T-\gamma)^2} \ll \log T$$

Observe that,

$$N(T+1) - N(T) = \sum_{\substack{\rho=\beta+i\gamma \\ \beta\in(0,1) \\ \zeta(\rho)=0 \\ T<\gamma<T+1}} \le \sum.$$

Recall we have,

$$-\mathrm{Re}\left(\frac{\zeta'(s)}{\zeta(s)}\right) < A\log(|t|+2) - \sum_{\rho=\beta+i\gamma} \mathrm{Re}\left(\frac{1}{s-p} - \frac{1}{p}\right).$$

where $s = \sigma + it$ for $\sigma \in [1,2], |t| > 1$ and $A$ is some constant. Substitute $s = 2 + iT$ in the equation, we have,

$$\mathrm{Re}\left(\frac{1}{s-\rho}\right) = \mathrm{Re}\left(\frac{\overline{(s-\rho)}}{|s-\rho|^2}\right).$$

Put $s = 2 + iT$ and $\rho = \beta + i\gamma$, we have,

$$\mathrm{Re}\left(\frac{\overline{(s-\rho)}}{|s-\rho|^2}\right) = \frac{2-\beta}{(2-\beta)^2 + (T-\gamma)^2} \geq \frac{1}{4 + (T-\gamma)^2}.$$

Also,

$$\mathrm{Re}\left(\frac{1}{\rho}\right) = \frac{\beta}{|\rho|^2} > 0.$$

By the previous equation, we get

$$\sum_{\substack{\rho=\beta+i\gamma \\ \beta\in(0,1) \\ \zeta(\rho)=0}} \mathrm{Re}\left(\frac{1}{2+iT-\rho}\right) < A\log(T+2) + \mathrm{Re}\left(\frac{\zeta'(2+iT)}{\zeta(2+iT)}\right) \ll \log T.$$

Recall that for $\sigma > 1$,

$$\frac{\zeta'(s)}{\zeta(s)} = \sum_{n\in\mathbb{N}} \frac{\Lambda(n)}{n^s},$$

$$\left|\frac{\zeta'(2+iT)}{\zeta(2+iT)}\right| \leq \sum_{n\in\mathbb{N}} \frac{\Lambda(n)}{n^s} < \infty.$$

Therefore,

$$\sum_{\rho} \mathrm{Re}\left(\frac{1}{2+iT-\rho}\right) = \sum_{\rho} \frac{1}{4+(T-\gamma)^2},$$

and

$$4 + (T-\gamma)^2 \leq 4(1 + (T-\gamma)^2).$$

□

**Theorem 2.42.** *As $T \to \infty$,*

$$N(T) = \frac{T}{2\pi}\log\frac{T}{2\pi} - \frac{T}{2\pi} + O(\log(T)).$$

*That is, as $T \to \infty$,*

$$N(T) \sim \frac{T}{2\pi}\log T.$$

*Analogously, $\chi \bmod q$, as $T \to \infty$,*

$$N(T,\chi) = \sum_{\substack{\rho=\beta+i\gamma \\ L(\rho,\chi)=0 \\ \beta\in(0,1) \\ |\gamma|<T}} 1 \sim \frac{T}{\pi}\log qT.$$

**Exercise 2.6.**

$$N(T,\chi)' = \sum_{\substack{\rho=\beta+i\gamma \\ L(\rho,\chi)=0 \\ \beta\in(0,1) \\ 0<gamma<T}} 1 \sim \frac{T}{2\pi}\log qT.$$

Recall by Remark 2.1, we have,

$$\psi(x) := \sum_{n\leq x}\Lambda(n).$$

Also recall that Theorem 2.17, $\frac{\zeta'(s)}{\zeta(s)}$ has a simple pole at $s=1$ with residue $-1$.

Let $s=a$ be a zero of $\zeta(s)$ of order $n\in\mathbb{N}$. Then

$$\zeta(s) = (s-a)^n g(s),$$

where $g(s)$ is analytic and $g(a)\neq 0$. Taking log-derivative of both sides , we have,

$$\frac{\zeta'(s)}{\zeta(s)} = \frac{n}{s-a} + \frac{g'(a)}{g(a)}.$$

Resiue of $\frac{\zeta'(s)}{\zeta(s)}$ at 0 of $zeta(s)$ is precisely the multiplicity of that zero. Recall from the proof of Theorem **??**, we have,

$$\psi(x) = \frac{1}{2\pi i}\int_{a-iT}^{a+iT} -\frac{\zeta'(s)}{\zeta(s)}\frac{x^s}{s}ds + O\left(\sum_{n\in\mathbb{N}}\Lambda(n)\left(\frac{x}{n}\right)\min(1, T^{-1}\left|\log\frac{x}{n}\right|^{-1})\right),$$

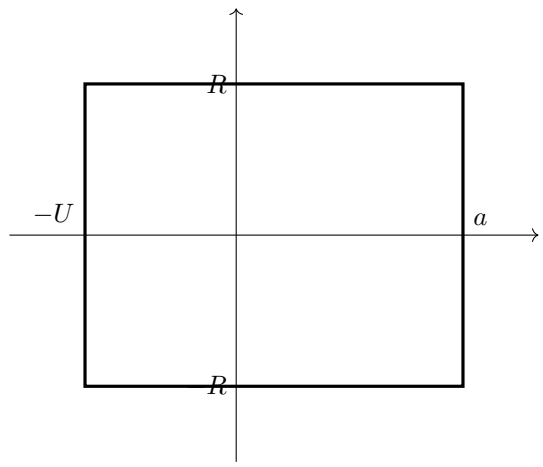
where $x$ is not an integer and $T, a\geq 1$. Similarly to the proof of Theorem **??**, take

$$a = 1 + \frac{1}{\log x}.$$

we then get, as $x\to\infty$,

$$\psi(x) = \frac{1}{2\pi i}\int_{a-iR}^{a+iR} -\frac{\zeta'(s)}{\zeta(s)}\frac{x^s}{s}dx + O\left(\frac{x\log^2 x}{R}\right).$$

Now consider the following closed path




where $U$ is some odd integer. By Cauchy's residue theorem, we have,

$$\frac{1}{2\pi i}\int_{C_R} -\frac{\zeta'(s)}{\zeta(s)}\frac{x^s}{s}ds = \text{sum of residues of } \frac{\zeta'(s)}{\zeta(s)} \text{ in } C_R,$$

$$= \overbrace{x}^{\text{contribution of poles of } \zeta(s)} - \underbrace{\frac{\zeta'(0)}{\zeta(0)}}_{\text{contribution of } s=0} + \overbrace{\sum_{0<2m<U}\frac{x^{-2m}}{2m}}^{\text{contribution of non-trivial zeros of } \zeta(s)} - \underbrace{\sum_{\substack{\rho=\beta+i\gamma\\ \zeta(\rho)=0\\ |\gamma|<R}}\frac{x^\rho}{\rho}}_{\text{contribution of trivial zeros of } \zeta(s)}.$$

Note that,

$$\int_{C_R} = \int_{a-iR}^{a+iR} + \underbrace{\int_{T_1,T_2,T_3}}_{=I}.$$

We then have

$$\psi(x) = x - \frac{\zeta'(0)}{\zeta(0)} + \sum_{0<2m<U}\frac{x^{-2m}}{2m} - \sum_{\substack{\rho=\beta+i\gamma\\ \zeta(\rho)=0\\ \beta\in(0,1)\\ |\gamma|<R}}\frac{x^\rho}{\rho} - I + O\left(\frac{x\log^2 x}{R}\right).$$

**Exercise 2.7.** *Show that*

$$I = \frac{1}{2\pi}\int_{I_1+I_2+I_3} -\frac{\zeta'(s)}{\zeta(s)}\frac{x^s}{s}ds, \qquad = O\left(\frac{x\log^2 R}{R\log x} + \frac{R\log U}{Ux^R}\right).$$

Suppose we have above equality, then we have,

$$\psi(x) = x - \frac{\zeta'(0)}{\zeta(0)} + \sum_{0<2m<U}\frac{x^{-2m}}{2m} - \sum\frac{x^\rho}{\rho} + O\left(\frac{x\log^2 R}{R\log x} + \frac{R\log U}{Ux^R} + \frac{x\log^2 x}{R}\right).$$

Recall $U$ is an odd integer. Take $U \to \infty$, we have,

$$\psi(x) = x - \frac{\zeta'(0)}{\zeta(0)} + \log(1 - x^{-2}) - \sum_{\rho} \frac{x^\rho}{\rho} + O\left(\frac{x \log^2 R}{R \log x} + \frac{x \log^2 x}{R}\right).$$

Taking $R \to \infty$, we have,

$$\psi(x) = x - \frac{\zeta'(0)}{\zeta(0)} + \log(1 - x^{-2}) - \sum_{\rho} \frac{x^\rho}{\rho}.$$

**Definition 2.22.**

$$\psi(x, \chi) := \sum_{n \leq x} \Lambda(n)\chi(n).$$

Recall we had,

$$\psi(x, q, a) := \sum_{\substack{n \leq x \\ n = a \bmod q}} \Lambda(n),$$

where $(a, q) = 1$. By Theorem **??**, we have, as $x \to \infty$,

$$\psi(x, q, a) \sim \frac{x}{\varphi(q)}.$$

**Exercise 2.8.** *Show that for $(a, q) = 1$, we have,*

$$\psi(x, q, a) = \frac{1}{\varphi(q)} \sum_{\chi \bmod q} \psi(x, \chi)\overline{\chi(a)}.$$

**Theorem 2.43.** *As $x \to \infty$, we have,*

$$\psi(x) = x + O(xe^{-c\sqrt{\log x}}),$$

*where $c > 0$ is some constant.*

*Proof.* Recalling zero free regions of $\zeta$, we have for $s = \sigma + it$,

$$\sigma > 1 - \frac{c}{\log R} \Rightarrow \zeta(s) \neq 0,$$

where $c$ is some constant and $R = \mathrm{Im}(s)$. From this we have,

$$\left| \sum_{|\gamma| < R} \frac{x^\rho}{\rho} \right| \leq x^{1 - \frac{c}{\log R}} \sum_{\substack{\rho = \beta + i\gamma \\ \zeta(\rho) = 0 \\ \beta \in (0,1) \\ |\gamma| < R}} \frac{1}{|\rho|} \ll x^{1 - \frac{c}{\log R}} \sum_{\substack{\beta \in (0,1) \\ \gamma \in (0,R) \\ \zeta(\beta + i\gamma) = 0}} \frac{1}{\sqrt{\beta^2 + \gamma^2}}.$$

Set

$$g(t) = \frac{1}{\sqrt{\beta^2 + t^2}},$$

and

$$a_t := \sum_{\substack{\beta\in(0,1)\\ \gamma\in(0,t)\\ \zeta(\beta+i\gamma)=0}} 1.$$

We now have,

$$\begin{aligned}\sum_{\substack{\beta\in(0,1)\\ \gamma\in(0,R)\\ \zeta(\beta+i\gamma)=0}} \frac{1}{\sqrt{\beta^2+\gamma^2}} &= \frac{N(R)}{\sqrt{\beta^2+\gamma^2}} + \int_0^R \frac{tN(R)}{(\beta^2+\gamma^2)^{\frac{3}{2}}}dt \\ &\ll \frac{N(R)}{R} + \int_0^R \frac{N(t)}{t^2}dt, \\ &= \log R + \int_0^2 \frac{\log t}{t} + \int_2^R \cdots, \\ &= \log^2 R.\end{aligned}$$

Combining all we get,

$$\left|\sum_{|\gamma|<R} \frac{x^\rho}{\rho}\right| \ll x^{1-\frac{c}{\log R}} \log^2 R.$$

Choose $\log R = c_2(\log \overline{x})^{\frac{1}{2}}$ we have,

$$\psi(x) = x + O(xe^{-c\sqrt{\log x}}).$$

Using Riemann hypotehsis, choose $R = \sqrt{x}$, we then have as $x \to \infty$,

$$\psi(x) = x + O(\sqrt{x}\log^2 x) + \frac{R\log U}{Ux^R} + \frac{x\log^2 x}{R}.$$

□

**Theorem 2.44** (Siegel-Walfisz)**.** *For any $N \in \mathbb{R}$ there is $c_N > 0$ such that as $x \to \infty$,*

$$\psi(x,q,a) = \frac{x}{\varphi(q)} + O(xe^{-c_N\sqrt{\log x}}),$$

*where $q \le (\log x)^N$. This is uniform in $q$.*

**Exercise 2.9.** *Show that Rieman hypothesis is equivalent to that as $x \to \infty$,*

$$\psi(x) = x + O(\sqrt{x}\log^2 x).$$

**Theorem 2.45.** *Riemann hypothesis implies that for $(a,q) = 1$, we have, as $x \to \infty$,*

$$\psi(x;q,a) = \frac{x}{\varphi(q)} + O(\sqrt{x}\log^2 qx).$$

**Remark 2.25.** *For $(a,q) = 1$, fixed we have as $x \to \infty$.*

$$\pi(x,q,a) < \frac{2x}{\varphi \log \frac{x}{q}}, q < x.$$

# 3 Exercise Solutions

## 3.1 Lecture 1 $\sim$ 4

**Exercise 3.1** (1.2.7)**.** *Show that*

$$\sum_{n=1}^{\infty} \frac{|\mu(n)|}{n^s} = \frac{\zeta(s)}{\zeta(2s)}.$$

*Solution.* From Theorem 2.5, we have,

$$\frac{1}{\zeta(2s)} = \sum_{n=1}^{\infty} \frac{\mu(n)}{n^{2s}}.$$

Therefore, we examine,

$$\left(\sum_{n=1}^{\infty} \frac{|\mu(n)|}{n^s}\right)\left(\sum_{n=1}^{\infty} \frac{\mu(n)}{n^s}\right) = \sum_{n=1}^{\infty} \frac{1}{n^s} \sum_{d|n} \left(|\mu(n)|\mu\left(\frac{n}{d}\right)\right)$$

Note that if $n \neq 1$ is not square free,

$$|\mu(n)|\mu\left(\frac{n}{d}\right) = 0,$$

unless $n = k^2$ and $d = k$. If $n$ is square free then by Theorem 2.1, we have

$$\left(\sum_{n=1}^{\infty} \frac{|\mu(n)|}{n^s}\right)\left(\sum_{n=1}^{\infty} \frac{\mu(n)}{n^s}\right) = \sum_{n=1}^{\infty} \frac{\mu(n)}{n^{2s}}.$$

□

**Exercise 3.2** (1.5.10)**.** *Let $\pi(x; z)$ be the number of $n \leq x$ such that it is coprime to all $p \leq z$. Then we have,*

$$\pi(x; z) = x \prod_{p \leq z} \left(1 - \frac{1}{p}\right) + O(2^z).$$

*Solution.* We have,

$$\begin{aligned}
\pi(x; z) &= \sum_{k=0,\cdots} \sum_{p_1, p_2, \cdots, p_k \leq z} (-1)^{k+1} \left[\frac{x}{p_1 p_2 \cdots p_k}\right], \\
&= \sum_{p_1, p_2, \cdots, p_k \leq z} (-1)^k \left(\frac{x}{p_1 p_2 \cdots p_k} - \left\{\frac{x}{p_1 p_2 \cdots p_k}\right\}\right), \\
&= x \prod_{p \leq z} \left(1 - \frac{1}{p}\right) + \sum_{p_1, \cdots, p_k \leq z} (-1)^{k+1} \left\{\frac{x}{p_1 p_2 \cdots p_k}\right\},
\end{aligned}$$

We have at most $[z]$ primes and there are $\binom{[z]}{k}$ many ways to pick $k$-many primes. Thus,

$$\left| \sum_{p_1, \cdots, p_k \leq z} (-1)^k \left\{ \frac{x}{p_1 p_2 \cdots p_k} \right\} \right| \leq \sum_{k=0}^{[z]} \binom{[z]}{k} = 2^{[z]} \leq 2^z.$$

□

**Exercise 3.3** (1.5.11)**.** *Show that*

$$\sum_{p \leq x} \frac{1}{p} \geq \log \log x + c,$$

*for some constant c.*

*Solution.* Let $\{p_1, \cdots, p_k\}$ be the primes less than $x$, then

$$\sum_{l \leq x} \frac{1}{l} \leq \sum_{\alpha \in \mathbb{Z}^k_{\geq 0}} \frac{1}{\prod p_i^{\alpha_i}}.$$

$$\log[x] \leq \sum_{l \leq x} \frac{1}{l}.$$

$$\Rightarrow \sum_{i=1}^{k} \log \left( \frac{1}{1 - \frac{1}{p}} \right) \geq \log \log [x].$$

$$\log p - \log(p-1) \leq \frac{1}{p}.$$

We then conclude that

$$\sum_{p \leq x} \frac{1}{p} \geq \log \log x - 1.$$

□

**Exercise 3.4** (1.5.12)**.** *Show that*

$$\pi(x) = O\left( \frac{x}{\log \log x} \right).$$

*Solution.*

**Claim 1.** *We have,*

$$-\sum_{p \leq z} \log \left( 1 - \frac{1}{p} \right) = \sum_{p \leq z} \frac{1}{p} + O(1).$$

<u>Proof:</u> We have,

$$\log(1-x) = -\sum_{n=0}^{\infty} \frac{x^n}{n}.$$

Thus substituting $x = \frac{1}{p}$, we obtain,

$$-\log\left(1-\frac{1}{p}\right) = \frac{1}{p} + \sum_{n=2}^{\infty} \frac{1}{np^n} \leq \frac{1}{p} + \frac{\frac{1}{p^2}}{1-\frac{1}{p}} \leq \frac{1}{p} + \frac{2}{p^2}.$$

Therefore,

$$-\sum_{p\leq z} \log\left(1-\frac{1}{p}\right) \leq \sum_{p\leq z} \frac{1}{p} + 2\zeta(2).$$

■

Using $\pi(x,z)$, we have,

$$\pi(x) \leq \pi(x,z) + z = x\prod_{p\leq z}\left(1-\frac{1}{p}\right) + O(2^z) + z.$$

Setting $z = \log x$ and using Claim 1, we have,

$$\pi(x) \leq x\exp\left(-\sum_{x\leq \log x}\frac{1}{p} + O(1)\right) + O(x^{\log 2}).$$

Using Exercise 3.3, we get,

$$\pi(x) \leq x\exp\left(-\log\log\log(x) + O(1)\right) + O(x^{\log 2}).$$

This completes the proof. □

**Exercise 3.5** (3.1.2)**.** *Prove that* $\theta(2m+1) - \theta(m) \leq 2m\log 2$*. Deduce that* $\theta(n) \leq 2n\log 2$*.*

*Solution.* Observe that

$$\theta(2) = \log 2.$$

And by definition, we may assume $n$ is odd. Note that $\binom{2m+1}{m}$ is divisible by all primes between $m+1$ and $2m+1$.

$$\binom{2m+1}{m} + \binom{2m+1}{m+1} \leq 2^{2m+1} \Rightarrow 2\binom{2m+1}{m} \leq 2^{2m+1}.$$

Thus we hae,

$$\theta(2m+1) - \theta(m) \leq 2m\log 2.$$

Suppose $\theta(m) \leq 2m\log 2$. Then

$$\theta(2m+1) \leq 4m\log 2 \leq 2(2m+1)\log 2.$$

Thus by induction, we proved the statement. □

**Exercise 3.6** (3.1.4)**.** *Show that*

$$e^{\psi(2n+1)} \int_0^1 x^n (1-x)^n dx,$$

*is a positive integer and deduce that*

$$\psi(2n+1) \geq 2n \log 2.$$

*Solution.*

**Claim 1.** *For any $n \in \mathbb{N}$, we have,*

$$e^{\psi(n)} = \operatorname{lcm}(2, \cdots, n).$$

Using binomial coefficient, we have,

$$(1-x)^n = \sum_{k=0}^{n} \binom{n}{k} (-1)^k x^k.$$

Thus we have,

$$I = \int_0^1 x^n (1-x)^n = \int_0^1 \sum_{k=0}^{n} \binom{n}{k} (-1)^k x^{n+k} = \sum_{k=0}^{n} (-1)^k \binom{n}{k} \frac{1}{n+k+1} x^{n+k+1}$$

Therefore, the first statement follows. For the latter observe that

$$x(1-x) \leq \frac{1}{4}.$$

Therefore,

$$I \leq 2^{-2n}.$$

Finally,

$$e^{\psi(2n+1)} I \geq 1 \Rightarrow e^{\psi(2n+1)} \geq 2^{2n}.$$

□

**Exercise 3.7** (3.1.5)**.** *Show that for all sufficiently large $x$, there are $A, B > 0$, such that*

$$\frac{Ax}{\log x} \leq \pi(x) \leq \frac{Bx}{\log x}.$$

*Solution.* As the smallest prime between $\sqrt{x}$ and $x$ is at least $\sqrt{x}$, we have,

$$(\pi(x) - \pi(\sqrt{x})) \log \sqrt{x} \leq 2x \log 2.$$

Thus we get,

$$\pi(x) \leq \frac{4x \log 2}{\log x} + \pi(\sqrt{x}).$$

Note that $\pi(\sqrt{x}) \leq \sqrt{x}$. Therefore,

$$\pi(x) = O\left(\frac{x}{\log x}\right).$$

Using $\psi \gg x$ and Remark 2.9

$$\psi(x) - \theta(x) = O(\sqrt{x}\log x).$$

Hence $\theta \gg x$. As before,

$$\sum_{\sqrt{x}<p\leq x} \log p + O(\sqrt{x}\log x) \gg x.$$

So that sufficiently large $x$, we have,

$$\pi(x)\log x \gg x.$$

□

**Exercise 3.8** (3.1.8)**.** *Prove that*

$$\sum_{p\leq x} \frac{1}{p} = \log\log x + O(1).$$

*Solution.* By Remark 1.4, we have

$$\psi(x) \sim x.$$

Also by Remark 2.1, we have,

$$\psi(x) = \sum_{n\leq x} \Lambda(n).$$

Using Proposition 2.7, we get,

$$\begin{aligned}\sum_{n\leq x} \frac{\Lambda(n)}{n} &= \frac{\psi(x)}{x} + \int_1^x \frac{\psi(t)}{t^2}dt,\\ &= \log x + O(1).\end{aligned}$$

Apply Corollary 2.1 to $\frac{\log n}{n}$, we get,

$$\begin{aligned}\sum_{p\leq x} \frac{\log p}{p}\frac{1}{\log p} &= O(1) + \int_2^x \frac{(\log t + O(1))}{(\log t)^2}dt,\\ &= \int_2^x \frac{(\log t + O(1))}{(\log t)^2}\frac{dt}{t},\\ &= \log\log x + O(1).\end{aligned}$$

□

**Exercise 3.9** (3.1.12)**.** *Show that under the assumption,*

$$\lim_{x\to\infty} \frac{\pi(x)}{\frac{x}{\log x}} = \alpha,$$

*we have,*

$$\sum_{p\le x} \frac{1}{p} = \alpha \log\log x + o(\log\log x).$$

*Furthermore, in this case, we have* $\alpha = 1$.

*Solution.* Set

$$\chi(n) = \begin{cases} 1, & (n = p), \\ 0, & (\text{otherwise}). \end{cases}$$

Then $\pi(n) = \sum_{m\le x} \chi(m)$. Therefore,

$$\sum_{p\le x} \frac{1}{p} = \frac{\pi(x)}{x} + \int_1^x \frac{\pi(t)}{t^2} dt.$$

Using Exercise 3.8, we have

$$\begin{aligned} \sum_{p\le x} \frac{1}{p} &= \frac{\pi(x)}{x} + \int_2^x \frac{\pi(t)}{t^2} dt, \\ &= \frac{\alpha}{\log x} + \int_1^x \frac{\alpha}{t\log t} dt, \\ &= \frac{\alpha}{\log x} + \alpha \log\log(x) - \alpha\log\log(2). \\ &\Rightarrow \alpha = 1. \end{aligned}$$

□

**Exercise 3.10** (2.4.6)**.** *Let* $\chi \bmod q$ *be a non-principal character. Then we have,*

$$\sum_{n>x} \frac{\chi(n)}{n} = O\left(\frac{1}{x}\right).$$

*Proof.* Using Proposition 2.7, we have,

$$\sum_{n\le x} \frac{\chi(n)}{n} \le \frac{\varphi(q)}{x} - \int_1^x \frac{\varphi(q)}{t}^2 dt.$$

Recall $L(s,\chi)$ is analytic for $\mathrm{Re}(s) > 0$, in particular,

$$L(1,\chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n}$$

is bounded. Thus we have,

$$\sum_{n>x} \frac{\chi(n)}{n} \leq \int_x^\infty \frac{\varphi(q)}{t^2} dt = O\left(\frac{1}{x}\right).$$

□

**Exercise 3.11** (2.4.7)**.** *Show that for a non-principal character* $\chi \bmod q$*, we have,*

$$\sum_{n\leq x} \sum_{d|n} \chi(d) = xL(1,\chi) + O(\sqrt{x}).$$

*Solution.*

$$\begin{aligned}
\sum_{n\leq x} \sum_{d|n} \chi(d) &= \sum_{de\leq x} \chi(d), \\
&= \sum_{d\leq\sqrt{x}} \chi(d) \left[\frac{x}{d}\right] + \sum_{e\leq\sqrt{x}} \sum_{\sqrt{x}<d\leq\frac{x}{e}} \chi(d). \\
\sum_{\sqrt{x}<d\leq\frac{x}{e}} \chi(d) &= \sum_{d\leq\frac{x}{e}} \chi(d) - \sum_{d\leq\sqrt{x}} \chi(d). \\
\sum_{e\leq\sqrt{x}} \sum_{\sqrt{x}<d\leq\frac{x}{e}} \chi(d) &\leq 2\varphi(q)[\sqrt{x}] = O(\sqrt{x}). \\
\sum_{d\leq\sqrt{x}} \chi(d) \left[\frac{x}{d}\right] &= x \sum_{d\leq\sqrt{x}} \frac{\chi(d)}{d} + O(1), \\
&= xL(1,\chi) - \sum_{n>x} \frac{\chi(n)}{n} + O(1).
\end{aligned}$$

By the previous exercise. We conclude the solution. □

**Exercise 3.12** (2.4.8)**.** *For non-principal real character* $\chi \bmod q$*, show that* $L(1,\chi) \neq 0$*.*

*Solution.* By Lemma 2.10, we have

$$a_n := \sum_{d|n} \chi(d) \geq 0.$$

Set

$$F(x) = \sum_{n\leq x} a_n.$$

Then by the previous exercise, we have $F(x) = O(\sqrt{x})$. Let

$$\sum_{n=1}^\infty \frac{a_n}{n^s} = s\int_1^\infty \frac{F(t)}{t^{s+1}} dt.$$

Note that by the definition of Dirichlet series, we have,

$$\sum_{n=1}^{\infty} \frac{a_n}{n^s} = L(s,\chi)\zeta(s).$$

$\zeta(s)$ has a simple pole at $s = 1$ and $L$ has zero of order at least one at $s = 1$, however,

$$\lim_{s\to 1} s \int_1^{\infty} \frac{F(t)}{t^{s+1}} dt = \infty.$$

This is a contradiction. □

**Exercise 3.13** (2.5.11)**.** *Let $\chi \bmod q$ be a non-principal character then,*

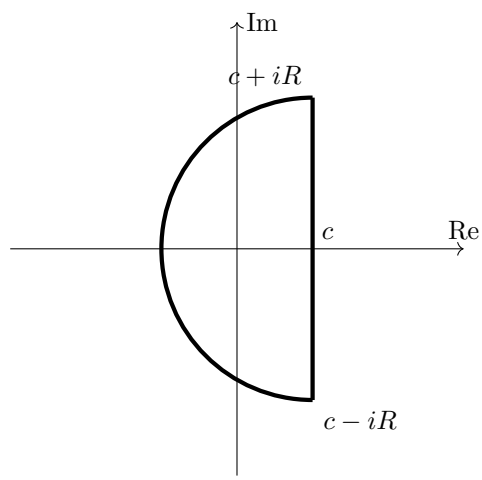
$$\sum_{n\le x} \sum_{d|n} \chi(d) = xL(1,\chi) + O(q\sqrt{x}).$$

*Solution.* Previous exercise and $\varphi(q) \le q$. □

**Exercise 3.14** (4.1.1)**.** *Show that for $x > 1$ and $c > 0$, we have*

$$\frac{1}{2\pi i} \int_{(c)} \frac{x^s}{s} ds = 1.$$

*Solution.* Let $\zeta_c$ be the contour




going the bold closed path in counterclockwise. Then

$$\frac{1}{2\pi i}\left(\int_{c-iR}^{c+iR} \frac{x^s}{s} ds + x^c \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \frac{x^{e^{i\theta}}}{c + e^{i\theta}} d\theta\right) = \mathrm{Res}_s\left(\frac{x^s}{s}, 0\right) = 1.$$

Let us examine the second term, for some small $\delta > 0$,

$$\left|\int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \frac{x^{Re^{i\theta}}}{c+Re^{i\theta}} d\theta\right| \leq \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} x^{R\cos\theta}\theta,$$

$$= \left(\int_{\frac{\pi}{2}}^{\frac{\pi}{2}+\delta} + \int_{\frac{\pi}{2}+\delta}^{\frac{3\pi}{2}-\delta} + \int_{\frac{3\pi}{2}-\delta}^{\frac{3\pi}{2}}\right) x^{R\cos\theta} d\theta.$$

$$\delta \geq \int_{\frac{\pi}{2}}^{\frac{\pi}{2}+\delta} x^{R\cos\theta} d\theta, \int_{\frac{3\pi}{2}-\delta}^{\frac{3\pi}{2}+\delta} x^{R\cos\theta} d\theta.$$

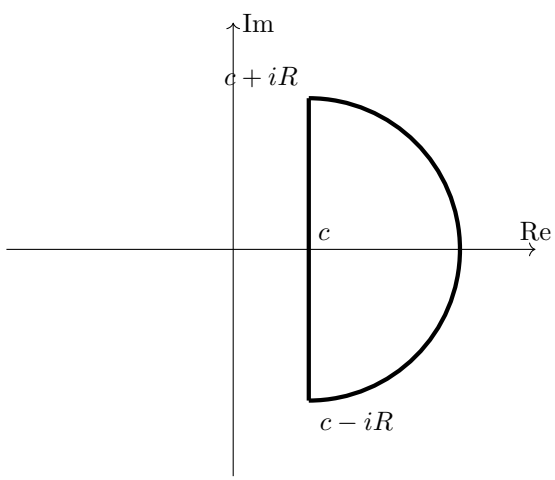
$$\int_{\frac{\pi}{2}+\delta}^{\frac{3\pi}{2}-\delta} x^{R\cos\theta} d\theta \leq \pi x^{R\cos\left(\frac{\pi}{2}+\delta\right)} = \pi x^{-R\sin\delta} \overset{R\to\infty}{\to} 0.$$

As $\delta$ is arbitrary, we conclude the proof. □

**Exercise 3.15** (4.1.2)**.** *Show that for $x \in (0,1)$ and $c > 0$, we have*

$$\frac{1}{2\pi i}\int_{(c)} \frac{x^s}{s} ds = 0.$$

*Solution.* As in the previous exercise, consider the following contour,




Then,

$$\frac{1}{2\pi i}\left(\int_{c-iR}^{c+iR} \frac{x^s}{s} ds + x^c \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{x^{e^{i\theta}}}{c+e^{i\theta}} d\theta\right) = 0.$$

Similarly, we observe that

$$\int_{-\frac{\pi}{2}+\delta}^{\frac{\pi}{2}-\delta} x^{R\cos\theta} d\theta \leq \pi x^{R\cos\left(\frac{\pi}{2}-\delta\right)} = \pi x^{R\sin\delta} \overset{R\to\infty}{\to} 0.$$

□

**Exercise 3.16** (4.1.3)**.** *Show that for $c > 0$, we have*

$$\frac{1}{2\pi i}\int_{(c)} \frac{1}{s} ds = \frac{1}{2}.$$

*Solution.*

$$\begin{aligned}\frac{1}{2\pi i}\int_{c-iR}^{c+iR} \frac{1}{s} ds &= \frac{1}{2\pi i}\int_{-R}^{R} \frac{i}{c+it} dt, \\ &= \frac{1}{2\pi}\int_{-R}^{R} \frac{c-it}{c^2+t^2} dt,\end{aligned}$$

Since $\frac{t}{c^2+t^2}$ is odd, the imaginary part of the integral vanishes. As $\frac{1}{c^2+t^2}$ is even, we consider,

$$\frac{1}{\pi}\int_0^R \frac{c}{c^2+t^2} dt = \frac{1}{\pi}\int_0^{\frac{R}{c}} \frac{1}{1+u^2} du.$$

As $R$ goes to infinity, the left hand side equals to $\frac{1}{\pi}\lim_{u\to\infty}\arctan u = \frac{1}{2}$. □

**Exercise 3.17** (4.1.4)**.**

*Solution.* □

**Exercise 3.18** (4.3.4)**.** *Show that*

$$M(x) := \sum_{n\le x} \mu(n) = O(x\exp(-c(\log x)^{\frac{1}{10}})),$$

*for some constant $c > 0$.*

*Proof.* Using Lemma 2.13, and recall that

$$\frac{1}{\zeta(s)} = \sum_{n=1}^{\infty} \frac{\mu(n)}{n^s},$$

we get,

$$M(x) = \frac{1}{2\pi i}\int_{(c)} \frac{x^s}{s\zeta(s)} ds.$$

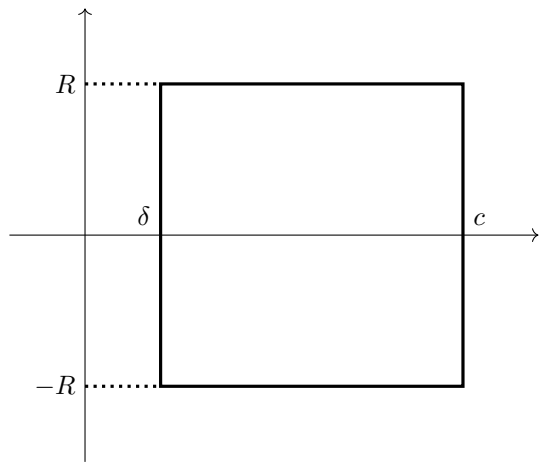
Thus we examine

$$\frac{1}{2\pi i}\int_{c-iR}^{c+iR} \frac{x^s}{s\zeta(s)} ds,$$

Take $c_1, c_2$ from Theorem 2.28, thus

$$\delta = 1 - \frac{c_1}{(\log R)^9} \le \sigma \le 2, 1 \le |t| \le R \Rightarrow .$$

Thus consider the counterclockwise contour integral $\mathcal{C}$,




By Cauchy's theorem, we have,

$$\frac{1}{2\pi i}\int_{\mathcal{C}} \frac{x^s}{s\zeta(s)}ds.$$

Therefore,

$$\frac{1}{2\pi i}\int_{c-iR}^{c+iR} \frac{x^s}{s\zeta(s)}ds = -\frac{1}{2\pi i}\left(\int_{\delta-iR}^{c-iR} + \int_{c+iR}^{\delta+iR} + \int_{\delta+iR}^{\delta-iR}\right)\frac{x^s}{s\zeta(s)}ds,$$

By Theorem 2.28, we have,

$$-\frac{1}{2\pi i}\left(\int_{\delta-iR}^{c-iR} + \int_{c+iR}^{\delta+iR}\right)\frac{x^s}{s\zeta(s)}ds \ll \frac{x^c(\log R)^7}{R\log x}.$$

On the vertical line, we have,

$$\left|\frac{1}{2\pi i}\int_{\delta+iR}^{\delta-iR} \frac{x^s}{s\zeta(s)}ds\right| \ll x^\delta(\log R)^8.$$

□

**Exercise 3.19** (4.3.5)**.** *Let $E(x)$ be the number of square free integers less than or equal to $x$ with even number of prime factors. Show that*

$$E(x) = \frac{3}{\pi^2}x + O(x\exp(-c\log^{\frac{1}{10}} x)).$$

*Solution.* Consider,

$$a_n = \frac{\mu(n)^2(1+\mu(n))}{2}.$$

$a_n = 1$ if $n$ is square-free and has even number of prime factors otherwise 0. Let $Q(x)$ be the number of square free integers less than or equal to $x$. Then

$$\begin{aligned} E(x) &= \sum_{n \le x} a_n, \\ &= \frac{1}{2} \sum_{n \le x} \mu(n)^2 + \frac{1}{2} \sum_{n \le x} \mu(n), \\ &= \frac{Q(x)}{2} + \frac{M(x)}{2}. \end{aligned}$$

The second term is shown by Exercise 3.18. Remains to show the first term. By Exercise 3.1, we have,

$$Q(x) = O\left(\frac{6}{\pi^2} \log x\right).$$

We then conclude the solution. □

**Exercise 3.20** (4.4.2)**.** *Show that for any $s$ with* $\mathrm{Re}(s) = 1$*, we have,*

$$\sum_{n=1}^{\infty} \frac{\mu(n)}{n^s}$$

*converges.*

*Solution.* By the assumption, we set $s = 1 + it$ for $t \in \mathbb{R}$. Using Proposition 2.7, we have,

$$\sum_{n \le N} \frac{\mu(n)}{n^{1+it}} = \frac{O(N \exp(-c(\log N)^{\frac{1}{10}}))}{N^{1+it}} + (1+it) \int_1^x \frac{M(w)}{w^{2+it}} dw.$$

Examine the splitted integral,

$$\int_1^{\infty} \frac{M(w)}{w^{2+it}} dw - \int_N^{\infty} \frac{M(w)}{w^{2+it}} dw.$$

The first term conevrges as $\frac{M(w)}{w^{2+it}} = O(w^{-1-\delta})$ for some $\delta > 0$. Note that

$$\exp\left(-\frac{c}{2}(\log w)^{\frac{1}{10}}\right) \le \exp\left(-\frac{c}{2}(\log w)^{\frac{1}{10}}\right).$$

Thus,

$$\begin{aligned} \left| \int_N^{\infty} \frac{M(w)}{w^{2+it}} dw \right| &\le \int_N^{\infty} \exp\left(-\frac{c}{2}(\log w)^{\frac{1}{10}}\right) \exp\left(-\frac{c}{2}(\log w)^{\frac{1}{10}}\right) \frac{dw}{w}, \\ &\le c' \exp\left(-\frac{c}{2}(\log N)^{\frac{1}{10}}\right) \overset{N \to \infty}{\to} 0. \end{aligned}$$

□

**Exercise 3.21** (4.4.12)**.** *Show that*

$$\sum_{n\le x}\frac{n}{\varphi(n)}\ll x.$$

*Proof.*

$$\begin{aligned}\sum_{n\le x}\frac{n}{\varphi(n)}&=\sum_{n\le x}\frac{1}{\prod_{p|n}\left(1-\frac{1}{p}\right)},\\&=\sum_{n\le x}\sum_{\gamma(d)|n}\frac{1}{d},\\&\le\sum_{\gamma(d)\le x}\frac{1}{d}\frac{x}{\gamma(d)},\\&\le x\prod_{p\le x}\frac{1}{\left(1-\frac{1}{p}\right)}.\end{aligned}$$

□

**Exercise 3.22** (4.4.13)**.** *Show that*

$$\sum_{n\le x}\frac{1}{\varphi(n)}\ll\log x.$$

*Proof.* Using the previous problem and Proposition 2.7, we get,

$$\begin{aligned}\sum_{n\le x}\frac{n}{\varphi(n)}\frac{1}{n}&\le\frac{Cx}{x}+C\int_1^x\frac{t}{t^2}dt,\\&=C+C\log x.\end{aligned}$$

□

**Exercise 3.23** (4.4.14)**.** *Prove that*

$$\sum_{n\le x}\frac{1}{\varphi(n)}\sim c\log x.$$

*Proof.* From Exercise 3.22, it remains to prove that

$$c\log x\le\sum_{n\le x}\frac{1}{\varphi(n)}.$$

We have $\varphi(n)\le n$ for any $n\in\mathbb{N}$, in the argument of Lemma 2.2,

$$\sum_{n\le x}\frac{1}{n}\ge\log x.$$

This finishes the proof. □

**Exercise 3.24** (5.3.2)**.** *Show that for a character* $\chi \bmod q$ *and* $(n,q)=1$*, we have,*

$$\chi(n)\tau(\overline{\chi}) = \sum_{k=1}^{q} \overline{\chi}(n) e^{\frac{2\pi i k n}{q}}.$$

*Solution.* Note that we have $\chi(n) = \overline{\chi}(n)^{-1}$. And use

$$kn \equiv l \mod q \Rightarrow k \equiv ln^{-1} \mod q.$$

□

**Exercise 3.25.** *Let* $\chi \bmod q$ *be a primitive character, then we have,*

$$L(1,\chi) = \sum_{n \le x} \frac{\chi(n)}{n} + O\left(\frac{\sqrt{q}\log q}{x}\right).$$

*Proof.* By Proposition 2.7,

$$L(1,\chi) = \int_1^\infty \frac{A(t)}{t^2} dt \Rightarrow \sum_{n>x} \frac{\chi(n)}{n} \le \int_x^\infty \frac{A(t)}{t^2} dt,$$

where

$$A(x) = \sum_{n \le x} \chi(n).$$

By Theorem 2.11, we have,

$$|A(x)| \le \sqrt{q}\log q.$$

Thus we obtain,

$$\int_x^\infty \frac{A(t)}{t^2} dt \le \frac{\sqrt{q}\log q}{x}.$$

□

**Exercise 3.26** (5.4.3)**.** *Show that*

$$\pi^{-\left(\frac{s+1}{2}\right)} q^{\left(\frac{s+1}{2}\right)} \Gamma\left(\frac{s+1}{2}\right) n^{-s} = \int_0^\infty n e^{-\frac{\pi n^2 x}{q}} x^{\frac{s+1}{2}} \frac{dx}{x}.$$

*Suppose* $\chi \bmod q$ *is odd and set,*

$$\theta_1(x,\chi) = \sum_{n \in \mathbb{Z}} n\chi(n) e^{-\frac{\pi n^2 x}{q}}.$$

*Then we have,*

$$\pi^{-\left(\frac{s+1}{2}\right)} q^{\left(\frac{s+1}{2}\right)} \Gamma\left(\frac{s+1}{2}\right) L(s,\chi) = \frac{1}{2}\int_0^\infty \theta_1(x,\chi) x^{\frac{s+1}{2}} \frac{dx}{x}.$$

*Solution.* By setting $s \to s-1$, we turn the first equation into

$$\pi^{-\left(\frac{s}{2}\right)} q^{\left(\frac{s}{2}\right)} \Gamma\left(\frac{s}{2}\right) n^{-s} = \int_0^\infty e^{-\frac{\pi n^2 x}{q}} x^{\frac{s+1}{2}} \frac{dx}{x}.$$

This is derived from the definition of $\Gamma$ and change of variables by $t \to \frac{\pi n^2 x}{q}$. The second equation is due to the parity of $\chi$ that is,

$$\theta_1(x,\chi) = \sum_{n\in\mathbb{Z}} n\chi(n) e^{-\frac{\pi n^2 x}{q}} = \sum_{n\in\mathbb{Z}} (n\chi(n) - n\chi(-n)) e^{-\frac{\pi n^2 x}{q}}.$$

□

**Exercise 3.27** (5.4.4)**.**

$$\sum_{n\in\mathbb{Z}} \left(n + \frac{m}{q}\right) e^{-\frac{q\left(n+\frac{m}{q}\right)^2 \pi}{x}} = i x^{\frac{3}{2}} \sum_{n\in\mathbb{Z}} n e^{\frac{-n^2 \pi x}{q} + 2\pi i \frac{mn}{q}}.$$

*Solution.* Recall that for $F(t) = e^{-\pi t^2}$, we have $\hat{F}(t) = F(t)$. Therefore, let

$$f(t) = e^{-\frac{t^2}{\alpha}\pi},$$

we obtain,

$$\begin{aligned} \hat{f}(n) &= \sqrt{\alpha} \int_{-\infty}^{\infty} e^{-\pi t^2} e^{-2\pi i \sqrt{\alpha} n t} dt, \\ &= \sqrt{\alpha} \hat{F}(\sqrt{\alpha} n) = \sqrt{\alpha} e^{-\alpha \pi n^2}. \end{aligned}$$

Using Theorem 2.32, we obtain,

$$\sum_{n\in\mathbb{Z}} e^{-\frac{(n+a)^2 \pi}{x}} = \sqrt{x} \sum_{n\in\mathbb{Z}} e^{-n^2 \pi x + 2\pi i n a}.$$

Differentiate with respect to $a$, we get,

$$\frac{2\pi}{x} \sum_{n\in\mathbb{Z}} (n+a) e^{-\frac{(n+a)^2 \pi}{x}} = 2\pi i \sqrt{x} \sum_{n\in\mathbb{Z}} n e^{-n^2 \pi x + 2\pi i n a}.$$

Put $x \to \frac{x}{q}$, and $a = \frac{m}{q}$, we obtain the result. □

**Exercise 3.28** (5.4.5)**.**

$$\xi(s,\chi) := \pi^{-\left(\frac{s}{2}\right)} q^{\left(\frac{s}{2}\right)} \Gamma\left(\frac{s+1}{2}\right) L(s,\chi).$$

*Then show that $\xi$ is entire and*

$$\xi(s,\chi) = \frac{\tau(\chi)}{i\sqrt{q}} \xi(1-s,\overline{\chi}).$$

*Solution.* Note that $\chi(1) = e^{2\pi i \frac{m}{q}}$. Therefore,

$$\theta(s,\chi) =$$

$$\begin{aligned}
&\pi^{-\left(\frac{s+1}{2}\right)} q^{\left(\frac{s+1}{2}\right)} \Gamma\left(\frac{s+1}{2}\right) L(s,\chi), \\
&\qquad = \frac{1}{2}\int_0^\infty \theta_1(x,\chi) x^{\frac{s+1}{2}} \frac{dx}{x}, \\
&\qquad = \frac{1}{2}\int_1^\infty \theta_1(x,\chi) x^{\frac{s+1}{2}} \frac{dx}{x} + \frac{1}{2}\int_1^\infty \theta_1(x^{-1},\chi) x^{-\frac{s+1}{2}} \frac{dx}{x},
\end{aligned}$$

□

**Exercise 3.29** (5.5.7)**.** *Suppose $\chi$ is a primitive character then we have,*

$$L(1,\chi) = \sum_{n \le x} \frac{\chi(n)}{n} + O\left(\frac{\sqrt{q}\log q}{x}\right).$$

*Solution.* We have,

$$L(1,\chi) = \sum_{n \le x} \frac{\chi(n)}{n} + \sum_{n > x} \frac{\chi(n)}{n}.$$

Also by Proposition 2.7, we have,

$$L(1,\chi) \le \int_1^\infty \left(\frac{\sqrt{q}\log q}{x^2}\right).$$

□

**Exercise 3.30** (5.5.8)**.** *Show that*

$$\sum_{\chi \ne \chi_0} L(1,\chi) = \varphi(q) + O(\sqrt{q}\log q).$$

*Proof.* If $\chi$ is imprimitive, then use Theorem 2.11 to the primitive conductor of it, we obtain the same inequality. Thus

$$\sum_{\chi \ne \chi_0} L(1,\chi) = \sum_{n \le x} \left(\frac{\sum_{\chi \ne \chi_0} \chi(n)}{n} + O\left(\frac{\sqrt{q}\log q}{x}\right)\right).$$

Using Remark 2.12 and Theorem 2.10, furthermore choosing $x = q$ we obtain,

$$\sum_{\chi \ne \chi_0} L(1,\chi) = \varphi(q) \sum_{(n,q)=1} \frac{1}{n} - \sum_{n=1}^{q} \frac{1}{n} + O(\sqrt{q}\log q).$$

□

**Exercise 3.31** (5.5.9)**.** *Let $s \in \mathbb{C}$ be such that* $\mathrm{Re}(s) = \sigma > 0$*. Then for any $x$ and a non-trivial character $\chi$, show that*

$$L(s,\chi) = \sum_{n \le x} \frac{\chi(n)}{n^s} + O\left(\frac{|s|\sqrt{q}\log q}{\sigma x^\sigma}\right).$$

*Solution.* Using Proposition 2.7 and Theorem 2.11, we have,

$$\sum_{n>x} \frac{\chi(n)}{n^s} \le |s| \int_x^\infty \frac{\sqrt{q}\log q}{t^{\sigma+1}} dt.$$

□

**Exercise 3.32** (5.5.10)**.** *For $\sigma > \frac{1}{2}$, show that*

$$\sum_{\chi \ne \chi_0} L(\sigma,\chi) = \varphi(q) + O(q^{\frac{3}{2}-\sigma}).$$

*Solution.* By previous exercises, by letting $x = q$, we have

$$\sum_{\chi \ne \chi_0} L(\sigma,\chi) = \varphi(q) + O(q^{1-\sigma}) + O(q^{\frac{3}{2}-\sigma}).$$

□

**Exercise 3.33** (5.5.14)**.** *For a prime $p$ and a quadratic character $\chi$, we have,*

$$\tau(\chi) = \begin{cases} \sqrt{p}, & (p \equiv 1 \mod 4), \\ i\sqrt{p}, & (p \equiv 3 \mod 4). \end{cases}$$

*Proof.* As $(\mathbb{Z}/p\mathbb{Z})^\times$ is cyclic, there is a unique quadratic character $\chi$ for each $p$. Let $\zeta_p = e^{\frac{2\pi i}{p}}$. Then

$$\sum_{k=1}^{p-1} \zeta^k = -1.$$

Therefore,

$$\tau(\chi) - 1 = \sum_{k=1}^{p-1} (\chi(k) + 1)\zeta^k.$$

As $\mathbb{Z}/p\mathbb{Z}$ is a field, if an element has square roots then there are exactly two besides 0. And if $k$ is a square $\chi(k) - 1 = 2$ otherwise 0. Therefore,

$$\tau(\chi) = 1 + \sum_{k=1}^{p-1} \zeta_p^{k^2}$$

Consider,

$$|\tau(\chi)|^2 = \sum_{k,l=0}^{p-1} \zeta_p^{k^2-l^2} = \sum_{a,b=0}^{p-1} \zeta_p^{ab} = p.$$

Remains to prove the parity of this. But note that

$$\overline{\tau(\chi)} = \chi(-1)\tau(\chi),$$

therefore,

$$\chi(-1)\tau(\chi)^2 = p.$$

Note that $\chi(-1) = \pm 1$ and for a generator $g \in (\mathbb{Z}/p\mathbb{Z})^\times$ we have

$$g^{\frac{p-1}{2}} \equiv -1 \mod p.$$

Thus we conclude the proof. $\square$